



2018. Fall.

1. Let  $A \in M_{n \times n}(\mathbb{R})$  such that  $A^2 = 0$ . Prove

$$2 \cdot \text{rank } A \leq n.$$

Proof. Given  $x \in \mathbb{R}^n$ ,  $A(Ax) = A^2x = 0$ , so  $\text{Im } A \subseteq \ker A$ . So,  $\dim \text{Im } A \leq \dim \ker A$ .  
By the dimension theorem,

$$n = \text{nullity } A + \text{rank } A \geq 2 \text{rank } A \quad \square$$

2. Determine over  $\mathbb{R}$ ,

$$A = \begin{pmatrix} 0 & 0 & -2 \\ 1 & 2 & 1 \\ 1 & 0 & 3 \end{pmatrix}$$

is diagonalizable.

Proof. The characteristic polynomial of  $A$  is

$$\det \begin{pmatrix} t & 0 & 2 \\ -1 & t-2 & -1 \\ -1 & 0 & t-3 \end{pmatrix} = t(t-2)(t-3) + 2 \cdot (-1)(t-2) = (t-2)(t^2 - 3t + 2) = (t-1)(t-2)(t-3).$$

By the Cayley-Hamilton theorem, the minimal polynomial has same roots to characteristic polynomial, so we need to check that  $(A-I)(A-2I) = 0$ ,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = 0$$

So, the minimal polynomial of  $A$  is  $(t-1)(t-2)$ , it implies  $A$  is diagonalizable, because the minimal polynomial splits over  $\mathbb{R}$  and separable. Explicitly,

$$\begin{pmatrix} -1 & 0 & -2 \\ 1 & 1 & 1 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x = -2z \\ x+y+z=0 \end{cases} \Rightarrow \ker(A-I) = \text{span} \left\{ \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} \right\}$$
$$\begin{pmatrix} -2 & 0 & -2 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Leftrightarrow \begin{cases} x+z=0 \end{cases} \Rightarrow \ker(A-2I) = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} \right\}$$

Therefore,  $A = UDU^{-1}$  where  $U = \begin{pmatrix} -2 & 0 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & -1 \end{pmatrix}$  and  $D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ .

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3. Let  $R$  be a commutative ring and let  $x \in R$  such that  $x^{2018} = 0$ .  
prove that  $1+x$  is unit.

Proof. observe that

$$\begin{aligned} & (1+x) \cdot (-x^{2017} + x^{2016} - x^{2015} + \dots - x + 1) \\ &= 1 - x + x^2 - \dots + x^{2016} - x^{2017} \\ & \quad + x - x^2 + \dots - x^{2016} + x^{2017} - x^{2018} \\ &= 1, \quad \text{since } x^{2018} = 0. \quad \text{So, } 1-x+\dots+x^{2016}-x^{2017} \text{ is inverse of } 1+x. \end{aligned}$$

4. prove that  $A_4$  does not have a subgroup of order 6.

Proof. Since  $|S_4:A_4| = 2$ ,  $|A_4| = 4 \cdot 3 = 12$ .

Suppose that  $A_4$  has a subgroup  $H$  of order 6.

Then,  $|A_4:H| = 12/6 = 2$ , so  $gH = Hg$  for any  $g \in A_4$ , it means  $H \trianglelefteq A_4$  is normal.  
Consider the afforded Homomorphism of  $A_4$  acts  $A_4/H$  as left multiplication:

$$\mathcal{Q}: A_4 \rightarrow S_{A_4/H}: g \mapsto (x) \mapsto gxH$$

And, since every 3-cycle  $\sigma \in A_4$  satisfies

$$(\mathcal{Q}(\sigma))^3 = \mathcal{Q}(\sigma^3) = \mathcal{Q}(\text{id}) = \text{id}$$

and  $S_{A_4/H} \cong \mathbb{Z}/2\mathbb{Z}$  so  $\mathcal{Q}(\sigma) = \text{id}$ . It implies all 3-cycles are contained in  $\ker \mathcal{Q}$ . Meanwhile, observe that  $(1\ 2\ 3), (1\ 2\ 4) \in A_4$  satisfies

$$(1\ 2\ 3)^{-1}(1\ 2\ 4)(1\ 2\ 3) = (1\ 3\ 2)(1\ 2\ 4)(1\ 2\ 3) = (1\ 4\ 3)$$

$(1\ 2\ 3)(1\ 4\ 3) = (1\ 4)(2\ 3)$ , ... in this way,  $A_4$  is generated by all 3-cycles in  $A_4$ , so  $\ker \mathcal{Q} = A_4$ . By the first isomorphism theorem,  $A_4/\ker \mathcal{Q} \cong \text{Im } \mathcal{Q}$  is trivial, so the action is trivial.

But, the left coset action is transitive, it cannot be trivial, contradiction.  $\square$

Similar Problems,

1. Show that  $A_5$  has no subgroup of order 15.

Proof.  $|A_5| = |S_5|/2 = 60 = 2^2 \cdot 3 \cdot 5$ .

Suppose that there is a subgroup  $H < A_5$  of order 15.

By the Sylow Theorem,  $\# \text{Syl}_3(H) \in \{1\}$  and  $\# \text{Syl}_5(H) \in \{1\}$ , that is,

there are order 3 subgroup  $P < H$  and order 5 subgroup  $Q < H$ ,

and both are normal subgroup of  $H$ . Since  $P \cong C_3$  and  $Q \cong C_5$  both are

cyclic, let  $x \in P$  and  $y \in Q$  such that  $P = \langle x \rangle$  and  $Q = \langle y \rangle$ .

Consider the afforded Homomorphism by  $H$  acts  $P$  by conjugation

$$\varphi: H \rightarrow \text{Sym}(P) : h \mapsto (k \mapsto hkh^{-1})$$

Then,  $\text{Ker } \varphi = \{h \in H : hkh^{-1} = k \text{ for all } k \in P\} = C_H(P)$  and

$H/\text{Ker } \varphi \cong \text{Im } \varphi \leq \text{Aut}(P)$ . But, since  $\text{Aut}(P) \cong \mathbb{Z}/2\mathbb{Z}$ ,

and  $|H|$  has no factor 2, odd number, so  $|\text{Ker } \varphi| = |H|$ , that is,

$C_H(P) = H$ . Now,  $x$  and  $y$  commute, therefore  $|xy| = 15$ .

But,  $A_5$  cannot have order 15 element, because:

order of permutation in  $S_5$  is determined by the least common multiplier of disjoint cycle decompositions, but  $\text{lcm } 5 = 5$ ,

$\text{lcm}(4, 1) = 4$ ,  $\text{lcm}(3, 2) = 6$ ,  $\text{lcm}(2, 1, 1, 1) = 2$ , ...

the lcm cannot be 15, so  $H$  contradicts.

[Similar problem

Let  $G = S_3 \times C_2$ .

1. Find all normal subgroups of  $G$ .

2. Prove that  $G$  has no normal subgroup of order 4.

Proof. Define three maps:

$$\mathcal{Q}_1: G \rightarrow S_3: (\sigma, \alpha) \mapsto \sigma$$

$$\mathcal{Q}_2: G \rightarrow C_2: (\sigma, \alpha) \mapsto \alpha$$

$$\mathcal{Q}_3: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma \cdot \pi(\alpha) \quad \text{where } C_2 \stackrel{\pi}{\cong} \{\pm 1\}.$$

$$\mathcal{Q}_4: G \rightarrow \{\pm 1\} \times C_2: (\sigma, \alpha) \mapsto (\text{sgn } \sigma, \alpha).$$

$$\mathcal{Q}_5: G \rightarrow \{\pm 1\}: (\sigma, \alpha) \mapsto \text{sgn } \sigma$$

Then,  $\mathcal{Q}_1$  and  $\mathcal{Q}_2$  are Homomorphisms, being the Coordinatively Projection, and  $\mathcal{Q}_3$  is also Homomorphism, because  $\text{sgn}$ ,  $\pi$  are Homomorphism and  $\{\pm 1\}$  is abelian. Precisely,

$$\begin{aligned} \mathcal{Q}_3((\sigma_1, \alpha_1) \cdot (\sigma_2, \alpha_2)) &= \mathcal{Q}_3(\sigma_1 \sigma_2, \alpha_1 \alpha_2) = \text{sgn } \sigma_1 \text{sgn } \sigma_2 \pi(\alpha_1) \pi(\alpha_2) \\ &= \text{sgn } \sigma_1 \pi(\alpha_1) \text{sgn } \sigma_2 \pi(\alpha_2) = \mathcal{Q}_3(\sigma_1, \alpha_1) \cdot \mathcal{Q}_3(\sigma_2, \alpha_2). \end{aligned}$$

And,  $\mathcal{Q}_4$  is Homomorphism, being entrywise Homomorphism, and  $\mathcal{Q}_5$  is Homomorphism, being Composition of Projection and  $\text{sgn}$ .

Now, the kernel

$$\ker \mathcal{Q}_1 = \{\text{id}\} \times C_2, \quad \ker \mathcal{Q}_2 = S_3 \times \{x^0\}, \quad \ker \mathcal{Q}_3 = A_3 \times \{x^0\} \cup (S_3 \setminus A_3) \times \{x^1\}$$

$$\ker \mathcal{Q}_4 = A_3 \times \{x^0\}, \quad \ker \mathcal{Q}_5 = A_3 \times C_2.$$

are we desired.

Are these all possibilities?

Observe that: if  $N \subseteq G_1 \times G_2$  is normal, then the image of projections  $\pi_i: G_1 \times G_2 \rightarrow G_i: (g_1, g_2) \mapsto g_i$ ,  $\pi_i[N]$  is also normal of  $\pi_i[G_1 \times G_2] = G_i$ , because image of normal subgroup is normal in the Range, and the projections are onto.

But, as we know, normal subgroup of  $S_3$  is only  $\{\text{id}\}$ ,  $A_3$ ,  $S_3$

and normal subgroup of  $\{\pm 1\}$  is  $\{1\}$  and  $\{\pm 1\}$ .

2018. Fall.

7. Consider the identity  $\frac{e^{-x}}{1-x} = \sum_{n=0}^{\infty} C_n x^n$ .

a). Determine the sequence  $(C_n) \subseteq \mathbb{R}$  and the range of  $x \in \mathbb{R}$  for the identity holds.

b). Show that for each  $n \geq 0$ ,

$$\sum_{k=0}^n \frac{C_k}{(n-k)!} = 1.$$

Proof. a),

Notice that  $e^{-x} = \sum_{n=0}^{\infty} \frac{(-x)^n}{n!}$  for all  $x \in \mathbb{R}$  and

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \text{ for all } |x| < 1.$$

Therefore, the coefficient  $C_n$  is determined by the Cauchy product

$$C_n = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!} \cdot 1^k = \sum_{k=0}^n \frac{(-1)^{n-k}}{(n-k)!}.$$

And, the identity valid in  $\mathbb{R} \cap (-1, 1) = (-1, 1)$ .  $\perp 5m$ .

b). Since

$$\frac{1}{1-x} = e^{-x} \cdot \sum_{n=0}^{\infty} C_n x^n, \text{ so we can write for } x \in (-1, 1)$$

$$\sum_{n=0}^{\infty} x^n = \sum_{n=0}^{\infty} \sum_{k=0}^n \frac{C_k}{(n-k)!} x^n.$$

And, the coefficients equals for each  $n \geq 0$ , because these are infinitely differentiable, so the coefficients of power series are determined uniquely by the Taylor Theorem. Thus,

$$1 = \sum_{k=0}^n \frac{C_k}{(n-k)!} \text{ for all } n \geq 0. \perp 4m$$

2019, Spring

1. (10 pts.) Let  $M$  be an  $n \times n$  matrix over a field. The  $(i, j)$ -minor of  $M$ , denoted by  $\widehat{M}_{ij}$ , is the determinant of the  $(n-1) \times (n-1)$  submatrix of  $M$  that is given by removing the  $i$ -th row and the  $j$ -th column from  $M$ . The adjoint of  $M$ , denoted by  $\text{adj}(M)$ , is the  $n \times n$  matrix whose  $(i, j)$ -entry is given by  $(-1)^{i+j} \widehat{M}_{ji}$ . Prove that if  $A$  and  $B$  are invertible  $n \times n$  matrices over a field, then  $\text{adj}(AB) = \text{adj}(B) \text{adj}(A)$ . (Hint:  $M \text{adj}(M) = \det(M) I_n$ .)

Proof. Observe that

$$\begin{aligned} \text{adj}(AB) &= B^{-1} A^{-1} A B \cdot \text{adj}(AB) = B^{-1} A^{-1} \cdot \det(AB) \cdot I_n = B^{-1} A^{-1} \cdot \det A \cdot \det B \\ &= B^{-1} \cdot \det B \cdot A^{-1} \cdot \det A = \text{adj}(B) \cdot \text{adj}(A) \end{aligned}$$

So, proved.

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2. (10 pts.) Let  $V = \mathcal{P}_2(\mathbb{R}) = \{f(x) \in \mathbb{R}[x] \mid \deg f(x) \leq 2\}$  and let  $T$  be the linear operator on  $V$  defined by  $T(f(x)) = f(0) + f(1)(x + x^2)$ . Determine whether  $T$  is diagonalizable, and, if  $T$  is diagonalizable, find a basis  $\mathcal{B}$  for  $V$  such that  $[T]_{\mathcal{B}}$  is a diagonal matrix.

Proof. Let  $A = \{1, x, x^2\}$  be the standard basis for  $V$ . Then,

$$[T]_A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}, \text{ so that the characteristic is } \det([T]_A + tI) \\ = t(t-1)(t-2)$$

Since the characteristic polynomial is separable and splits over  $\mathbb{R}$ , so the minimal polynomial equals to the characteristic, and so the  $T$  is diagonalizable. Moreover, observe that

$$\ker T = \text{Span}\{x^2 - x\}$$

$$\ker(T - \text{id}) = \text{Span}\{-1 + x + x^2\}$$

$$\ker(T - 2 \cdot \text{id}) = \text{Span}\{x^2 + x\}$$

Therefore, for  $\mathcal{B} = \{x^2 - x, x^2 + x - 1, x^2 + x\}$ ,

$$[T]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}. \quad \square$$

2019 Spring

3. Rationalize the denominator of  $\frac{1}{3+2\sqrt[3]{2}+\sqrt[3]{4}}$

Proof Denote that  $\alpha = \sqrt[3]{2}$ . Then,  $\alpha^3 - 2 = 0$ .

Meanwhile, we want to find that the  $\mathbb{Q}$ -linear combination for

$$(3+2\sqrt[3]{2}+\sqrt[3]{4})^{-1} = (\alpha^2+2\alpha+3)^{-1}.$$

By the division algorithm,

$$\alpha^3 - 2 = (\alpha^2 + 2\alpha + 3) \cdot (\alpha - 2) + \alpha + 4$$

$$\alpha^2 + 2\alpha + 3 = (\alpha + 4)(\alpha - 2) + 11.$$

$$\Rightarrow (\alpha^2 + 2\alpha + 3) + (\alpha^2 + 2\alpha + 3)(\alpha - 2)^2 = 11 \quad (\alpha^3 - 2 = 0)$$

$$\Rightarrow (\alpha^2 + 2\alpha + 3)(\alpha^2 - 4\alpha + 5) = 11.$$

Therefore,

$$(\alpha^2 + 2\alpha + 3)^{-1} = \frac{1}{11} \cdot (\alpha^2 - 4\alpha + 5) = \frac{\sqrt[3]{4} - 4\sqrt[3]{2} + 5}{11}.$$

□

Tzola, Spring

6. For each  $n \in \mathbb{N}$ , let

$$f_n(x) = \frac{|x|^n}{1+|x|^n}, \quad x \in \mathbb{R}.$$

a). Show that  $(f_n)_{n \geq 1}$  does not converge uniformly.

b) For each fixed  $q > 1$ , Show that  $(f_n)$  converges uniformly on  $\{x \in \mathbb{R} \mid |x| \geq q\}$  and  $\{x \in \mathbb{R} \mid |x| \leq \frac{1}{q}\}$ .

a). Given  $x \in \mathbb{R}$ ,

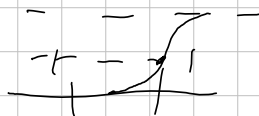
if  $|x| < 1$ , then  $f_n(x) \rightarrow 0$  as  $n \rightarrow \infty$ , since  $|x|^n \rightarrow 0$ .

if  $|x| = 1$ , then  $f_n(x) = \frac{1}{2}$  for all  $n \in \mathbb{N}$ , since  $|x|^n = 1$ .

if  $|x| > 1$ , then  $f_n(x) = \frac{1}{1/|x|^n + 1} \rightarrow 1$  as  $n \rightarrow \infty$ , since  $1/|x|^n \rightarrow 0$  as  $n \rightarrow \infty$ .

Therefore,  $f_n \rightarrow f$  pointwise where

$$f(x) = \begin{cases} 0 & |x| < 1 \\ \frac{1}{2} & |x| = 1 \\ 1 & |x| > 1 \end{cases}$$



But, for  $\epsilon = 1/4$ , and for all  $n \in \mathbb{N}$ ,

$$\sup_{0 \leq x < 1} |f_n(x) - f(x)| = \sup_{0 \leq x < 1} |f_n(x)| = \sup_{0 \leq x < 1} \frac{|x|^n}{1+|x|^n} \geq \frac{1}{4}$$

because given  $n \in \mathbb{N}$ , we can take  $x \in [0, 1)$  such that  $\frac{x^n}{1+x^n} \geq \frac{1}{4}$

Since  $\lim_{x \rightarrow 1^-} \frac{|x|^n}{1+|x|^n} = \frac{1}{2}$ , so,  $\sup_{x \in \mathbb{R}} |f_n - f| \geq \sup_{0 \leq x < 1} |f_n - f|$  implies

$f_n \rightarrow f$  but not uniformly on  $\mathbb{R}$ .

b). If  $|x| \geq q > 1$ , then  $f(x) = 1$ . So, enough to show that  $\sup |f_n(x) - f(x)| \rightarrow 0$ .

Since  $\sup_{|x| \geq q} |f_n(x) - f(x)| = \sup_{|x| \geq q} \left| \frac{1}{1+|x|^n} \right| \leq \frac{1}{1+q^n} \rightarrow 0$  as  $n \rightarrow \infty$ , so  $f_n \rightarrow f$  uniformly on  $|x| \geq q$ . Similarly, for  $|x| \leq \frac{1}{q}$ ,  $f(x) = 0$  and so

$$\sup_{|x| \leq \frac{1}{q}} |f_n(x) - f(x)| = \sup_{|x| \leq \frac{1}{q}} \left| \frac{|x|^n}{1+|x|^n} \right| \leq \frac{1}{q^n + 1} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

so proved all cases.  $\square$

2019, Spring

9. Evaluate the integral  $\int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx$ .

Proof. Let  $f(z) = 1/(1+z^2)^2$ , and a Curve  $C_R: t \mapsto Re^{it}$  ( $0 \leq t \leq \pi$ ), and  $C_L: t \mapsto -R \leq t \leq \pi$ . Then the composition  $C_R \times C_L$  is a closed, and  $f(z)$  has only one singularity at  $z=i$ , as a pole.

To obtain the Residue at  $z=i$ , observe that

$$f(z) = \frac{1}{(1+z^2)^2} = \frac{1}{(z+i)^2 \cdot (z-i)^2}$$

So, therefore,  $z=i$  is a pole of order 2, so

$$\begin{aligned} \text{Res}(f, i) &= \lim_{z \rightarrow i} \frac{d}{dz} \left( (z-i)^2 \cdot f(z) \right) \\ &= \lim_{z \rightarrow i} \frac{1}{(z+i)^2} = \lim_{z \rightarrow i} -2 \cdot \frac{1}{(z+i)^3} = \frac{1}{4i}. \end{aligned}$$

By the Residue theorem, for any  $R > 1$ ,

$$\int_{C_R \times C_L} f = \text{Res}(f, i) \cdot 2\pi i = \frac{\pi}{2}.$$

Meanwhile,  $\int_{C_R} f(z) dz = \int_0^\pi \frac{i R e^{it}}{(1+R^2 e^{2it})^2} dt$ , so, for  $R > 2$ ,

$$\left| \int_{C_R} f \right| \leq \int_0^\pi \frac{R}{(R^2-1)^2} dt \rightarrow 0 \text{ as } R \rightarrow \infty, \quad \text{nam,}$$

$$\lim_{R \rightarrow \infty} \int_{C_R \times C_L} f = \int_{-\infty}^{\infty} \frac{1}{(1+x^2)^2} dx = \frac{\pi}{2}, \quad \square$$

201a, Spring.

12. For  $a, b \in \mathbb{Z}$ , let  $A_{a,b} = \{a + n \cdot b \mid n \in \mathbb{Z}\}$ .

a). Prove that  $\mathcal{A} = \{A_{a,b} \mid a, b \in \mathbb{Z}\}$  is a basis for a topology on  $\mathbb{Z}$ .

b). Is  $\mathbb{Z}$  compact under the topology generated by  $\mathcal{A}$ ?

Proof. Given  $a \in \mathbb{Z}$ ,  $a \in A_{a,0} = \{a\} \subseteq \mathbb{Z}$ , so  $\bigcup A_{a,b} = \mathbb{Z}$ .

And, let  $A_{a,b}$  and  $A_{c,d}$  be given and let  $k \in A_{a,b} \cap A_{c,d}$  be given.

If  $A_{a,b} \cap A_{c,d} \neq \emptyset$ , then for some  $n_1, n_2 \in \mathbb{Z}$  such that

$$k = a + n_1 b = c + n_2 d$$

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17, Let  $S = \{(x, y, z) \in \mathbb{R}^3 \mid z = x \cdot f(\frac{y}{x}), x \neq 0\}$  where  $f: \mathbb{R} \rightarrow \mathbb{R}$  is differentiable. Show that the tangent plane at any point of  $S$  passes through the origin. Prove by the definition of  $S$ , given  $(x_0, y_0, z_0) \in S$ , a map

$X: \mathcal{O} \rightarrow S: (u, v) \mapsto (u, v, u \cdot f(\frac{v}{u}))$  where  $\mathcal{O}$  is an open set containing  $(x_0, y_0)$  is a parametrization of  $S$  at  $(x_0, y_0, z_0)$ . The differential of  $X$  at  $(x_0, y_0)$  is:

$$dX = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ f(\frac{v}{u}) - \frac{v}{u} \cdot f'(\frac{v}{u}) & f'(\frac{v}{u}) \end{bmatrix} \bigg|_{(u,v)=(x_0,y_0)}$$

So, the tangent plane of  $S$  at  $(x_0, y_0)$  is generated by

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ f(\frac{y_0}{x_0}) - \frac{y_0}{x_0} \cdot f'(\frac{y_0}{x_0}) \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ f'(\frac{y_0}{x_0}) \end{pmatrix} \right\}.$$

Now, for  $t_1, t_2 \in \mathbb{R}$ , the coordinate of a point of the tangent plane can be described by

$$(x_0, y_0, x_0 \cdot f(\frac{y_0}{x_0})) + (t_1, 0, t_1 \cdot f(\frac{y_0}{x_0}) - t_1 \cdot \frac{y_0}{x_0} f'(\frac{y_0}{x_0})) \\ + (0, t_2, t_2 \cdot f'(\frac{y_0}{x_0}))$$

is zero when  $t_1 = -x_0$  and  $t_2 = -y_0$ , so proved  $\square$

2019, Fall.

1. Classify the following groups according to the FTFGA:

$$\frac{\mathbb{Z} \times \mathbb{Z}}{\langle (n, n) \rangle}$$

Proof. Define a map for each  $n \in \mathbb{N}$ :

$$\mathcal{Q}: \mathbb{Z} \times \mathbb{Z} \rightarrow \mathbb{Z}_n \times \mathbb{Z} : (a, b) \mapsto (a \bmod n, a - b).$$

Since

$$\begin{aligned} \mathcal{Q}((a_1, b_1) + (a_2, b_2)) &= \mathcal{Q}(a_1 + a_2, b_1 + b_2) \\ &= (a_1 + a_2 \bmod n, a_1 + a_2 - b_1 - b_2) \\ &= (a_1 \bmod n, a_1 - b_1) + (a_2 \bmod n, a_2 - b_2) \\ &= \mathcal{Q}(a_1, b_1) + \mathcal{Q}(a_2, b_2) \end{aligned}$$

So  $\mathcal{Q}$  is Homomorphism, and given  $(c, d) \in \mathbb{Z}_n \times \mathbb{Z}$ ,  
take  $a = c$  and  $b = c - d$ , then

$$\mathcal{Q}(a, b) = (a, a - b) = (c, d)$$

So  $\mathcal{Q}$  is onto. And,

$$(a, b) \in \ker \mathcal{Q} \Leftrightarrow a \equiv 0 \bmod n \text{ and } a - b = 0$$

So,  $\ker \mathcal{Q} = \langle (n, n) \rangle$ . Now, by the first 'isomorphism' theorem

$$\mathbb{Z} \times \mathbb{Z} / \ker \mathcal{Q} \cong \text{Im } \mathcal{Q} = \mathbb{Z}_n \times \mathbb{Z}.$$

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2. Let  $p$  be the prime number and  $\alpha \in \mathbb{F}_p$ . Show that  $x^p + \alpha$  is reducible over  $\mathbb{F}_p$ .

Proof. If  $\alpha = 0$ , then  $x^p$  is clearly reducible. Assume that  $\alpha \neq 0$ .

Since  $\mathbb{F}_p^\times$  is a cyclic group of order of  $p-1$ , so  $\alpha^{p-1} = 1$ , so  $\alpha^p = \alpha$ .

Therefore, observe that

$$(x + \alpha)^p = \sum_{n=0}^p \binom{p}{n} x^{p-n} \cdot \alpha^n = x^p + \alpha^p = x^p + \alpha \quad \text{and } x + \alpha \in \mathbb{F}_p[x],$$

So,  $x^p + \alpha$  is reducible

Note:  $\mathbb{F}_p^\times \cong C_{p-1}$ .

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3. Show that  $\alpha = 3^{\frac{1}{3}} + 5^{\frac{1}{5}}$  is algebraic over  $\mathbb{Q}$  and find  $[\mathbb{Q}(\alpha) : \mathbb{Q}]$ .  
Proof. Since  $3^{\frac{1}{3}}$  and  $5^{\frac{1}{5}}$  both are roots of  $x^3 - 3$  and  $x^5 - 5$ , respectively, therefore both are algebraic over  $\mathbb{Q}$ . And, since sum of algebraic numbers is algebraic, so  $3^{\frac{1}{3}} + 5^{\frac{1}{5}}$  is algebraic.

By the Eisenstein criterion, both  $x^3 - 3$  and  $x^5 - 5$  are irreducible over  $\mathbb{Q}$ , so that  $[\mathbb{Q}(3^{\frac{1}{3}}) : \mathbb{Q}] = 3$  and  $[\mathbb{Q}(5^{\frac{1}{5}}) : \mathbb{Q}] = 5$ . Therefore,  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}]$  is divided by 3 and 5, so  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}] \geq 15$ , and  $x^3 - 3$  has a root as  $3^{\frac{1}{3}}$  over  $\mathbb{Q}(5^{\frac{1}{5}})$ , so  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}(5^{\frac{1}{5}})] \leq 3$ , so that  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}] \leq 15$ , so the degree is 15.

Since  $3^{\frac{1}{3}} + 5^{\frac{1}{5}} \in \mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}})$ ,  $\mathbb{Q}(3^{\frac{1}{3}} + 5^{\frac{1}{5}}) \subseteq \mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}})$ . And, since  $\mathbb{Q}(3^{\frac{1}{3}} + 5^{\frac{1}{5}}, 3^{\frac{1}{3}}) = \mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}})$ , so, since  $(3^{\frac{1}{3}})^3 - 3 = 0$ ,  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}(3^{\frac{1}{3}} + 5^{\frac{1}{5}})] \leq 3$ , so,  $= 1$  or  $3$ .

Conti.

To find the degree, notice that  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}(5^{\frac{1}{5}})] = 3$ , because  $x^3 - 3$  has no root in  $\mathbb{Q}(5^{\frac{1}{5}})$ , so the tower lemma gives that  $[\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) : \mathbb{Q}] = [\mathbb{Q}(\alpha) : \mathbb{Q}(5^{\frac{1}{5}})] \cdot [\mathbb{Q}(5^{\frac{1}{5}}) : \mathbb{Q}] = 3 \cdot 5 = 15$ .  
( $x^5 - 5$  is also irreducible over  $\mathbb{Q}$ , by the Eisenstein criterion)  
Meanwhile,  $\mathbb{Q}(3^{\frac{1}{3}}, 5^{\frac{1}{5}}) = \mathbb{Q}(3^{\frac{1}{3}} + 5^{\frac{1}{5}})$  is trivial

$\alpha^p - 5^p$

Similar problem.

4. Determine the degree over  $\mathbb{Q}$  of  $2 + \sqrt{3}$  and of  $1 + \sqrt[3]{2} + \sqrt[3]{4}$ .
5. Let  $F = \mathbb{Q}(i)$ . Prove that  $x^3 - 2$  and  $x^3 - 3$  are irreducible over  $F$ .
6. Prove directly from the definitions that the field  $F(\alpha_1, \alpha_2, \dots, \alpha_n)$  is the composite of the fields  $F(\alpha_1), F(\alpha_2), \dots, F(\alpha_n)$ .
7. Prove that  $\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3})$  [one inclusion is obvious, for the other consider  $(\sqrt{2} + \sqrt{3})^2$ , etc.]. Conclude that  $[\mathbb{Q}(\sqrt{2} + \sqrt{3}) : \mathbb{Q}] = 4$ . Find an irreducible polynomial satisfied by  $\sqrt{2} + \sqrt{3}$ .

4.  $2 + \sqrt{3} = x \Rightarrow x^2 - 4x + 1 = 0$ , it has no root in  $\mathbb{Q}$ , so that  $[\mathbb{Q}(2 + \sqrt{3}) : \mathbb{Q}] = 2$ .  
Since  $(\sqrt[3]{2} - 1)(\sqrt[3]{4} + \sqrt[3]{2} + 1) = 2 - 1 = 1$ , so that  
$$\sqrt[3]{4} + \sqrt[3]{2} + 1 = \frac{1}{\sqrt[3]{2} - 1}$$

Therefore

$$\mathbb{Q}(\sqrt[3]{4} + \sqrt[3]{2} + 1) = \mathbb{Q}\left(\frac{1}{\sqrt[3]{2} - 1}\right) = \mathbb{Q}(\sqrt[3]{2} - 1) = \mathbb{Q}(\sqrt[3]{2})$$

So, the degree is 3, since  $x^3 - 2$  is irreducible, by the Eisenstein.

2010, Fall.

5. Show that a symmetric  $n \times n$  matrix over  $\mathbb{R}$  has a real eigenvalue.

Proof. Let  $f(x)$  be the characteristic polynomial of the given matrix.

By the Fundamental Theorem of Algebra,  $f$  has a root  $\lambda$  in  $\mathbb{C}$ .

Meanwhile, observe that: for the given matrix  $A$ ,  $Ax = \lambda x$  for some  $x \in \mathbb{C}^n \setminus \{0\}$ .

Now,  $\lambda \|x\|^2 = \langle \lambda x, x \rangle = \langle Ax, x \rangle = \langle x, A^* x \rangle = \langle x, A^T x \rangle = \langle x, Ax \rangle = \langle Ax, x \rangle = \lambda \|x\|^2$ .

Since  $\|x\| > 0$ , so  $\lambda = \bar{\lambda}$ , it proves the  $\lambda$  is real root of  $f(x)$ , so

$A$  has a real eigenvalue.  $\square$

Note:  $f(\lambda) = 0 \Leftrightarrow \det(A - \lambda I) = 0$

$\Rightarrow A - \lambda I$  is non-singular, not invertible,

$\Rightarrow \ker(A - \lambda I) \neq \{0\}$ .

※ 가장 간단한 소리가 제일 시원한 것은 늘 그렇다.

2019, Fall.

6. Find the general solution

$$y' + xy = xy^2.$$

201a, Fall.

8. Let  $(b_n)_{n \in \mathbb{N}}$  be a decreasing sequence of positive real numbers with  $b_n \rightarrow 0$ . Let  $(a_n)_{n \in \mathbb{N}}$  be a sequence of complex numbers such that  $(\sum_{n=1}^N a_n)_{N \in \mathbb{N}}$  is a bounded sequence.

a). Show that  $\sum_{n=1}^{\infty} a_n b_n$  converges.

b). Show  $\sum_{n=2}^{\infty} \frac{\sin(n\pi x)}{\log n}$  converges for all  $x \in [-\pi, \pi]$ .

Proof. Denote  $A_n = \sum_{k=1}^n a_k$  and  $A_0 = 0$ . And let  $M \geq 0$  be a bound for  $\sum a_n$ .

$$\begin{aligned} \sum_{n=1}^N a_n b_n &= \sum_{n=1}^N [A_n - A_{n+1}] \cdot b_n = \sum_{n=1}^N A_n b_n - \sum_{n=1}^N A_{n+1} b_n \left( \sum_{n=1}^N A_n b_n = \sum_{n=0}^{N-1} A_n b_{n+1} \right) \\ &= \sum_{n=1}^N A_n b_n - \left( \sum_{n=1}^N A_n b_{n+1} + A_0 b_1 - A_N b_{N+1} \right) \\ &= \sum_{n=1}^N A_n (b_n - b_{n+1}) + A_N b_{N+1}. \end{aligned}$$

Therefore, for any  $m > n \geq 1$ ,

$$\sum_{k=n}^m a_k b_k = \sum_{k=n}^m A_k (b_k - b_{k+1}) + A_m b_{m+1} - A_n b_n$$

Now, using the hypothesis,

$$\begin{aligned} \left| \sum_{k=n}^m a_k b_k \right| &\leq \sum_{k=n}^m |A_k| \cdot |b_k - b_{k+1}| + |A_m| |b_{m+1}| + |A_n| |b_n| \\ &\leq M \cdot \left( \sum_{k=n}^m |b_k - b_{k+1}| + |b_{m+1}| + |b_n| \right) \\ &= M \cdot \left( \sum_{k=n}^m (b_k - b_{k+1}) + b_{m+1} + b_n \right) \end{aligned}$$

$$= M \cdot 2 b_n \rightarrow 0 \text{ as } n \rightarrow \infty, \text{ so proved.}$$

b). Since  $1/\log n$  is decreasing and positive real numbers, so by a), enough to show that  $|\sum \sin(n\pi x)|$  is bounded for all  $x \in [-\pi, \pi]$ .

observe that:  $2 \sin x \cdot \sin nx = \cos(nx - x) - \cos(nx + x)$

Therefore,  $2 \sin x \cdot \sum_{n=1}^N \sin(n\pi x) = \sum_{n=1}^N [\cos((n-1)x) - \cos((n+1)x)] \leq |\cos 0| + |\cos x| + |\cos Nx| + |\cos(N+1)x| \leq 4$

so for  $x \in [-\pi, \pi] \setminus \{0, \pm\pi\}$ ,  $\left| \sum_{n=2}^N \sin n\pi x \right| \leq \frac{2}{|\sin x|}$  for any  $N \in \mathbb{N}$ , bounded.  
for  $x = 0$  or  $\pm\pi$ , then the sum is just zero.  $\square$

2019, Fall,

1. Let  $\varphi: [0, 1] \rightarrow \mathbb{R}$  be a continuous map. Evaluate

$$\lim_{\varepsilon \rightarrow 0^+} \int_0^1 \left[ \frac{\varphi(x)}{x - i\varepsilon} - \frac{\varphi(x)}{x + i\varepsilon} \right] dx.$$

Proof. Observe that

$$\frac{1}{x - i\varepsilon} - \frac{1}{x + i\varepsilon} = \frac{2i\varepsilon}{x^2 + \varepsilon^2}$$

Therefore, the given integral is

$$2i\varepsilon \int_0^1 \frac{\varphi(x)}{x^2 + \varepsilon^2} dx.$$

Let  $x = \varepsilon t$ . Then,  $dx = \varepsilon dt$ , so that

$$2i\varepsilon \int_0^1 \frac{\varphi(x)}{x^2 + \varepsilon^2} dx = 2i\varepsilon \cdot \int_0^{\frac{1}{\varepsilon}} \frac{\varphi(\varepsilon t)}{\varepsilon^2 t^2 + \varepsilon^2} \varepsilon dt = 2i \cdot \int_0^{\frac{1}{\varepsilon}} \frac{\varphi(\varepsilon t)}{t^2 + 1} dt$$
$$= 2i \cdot \int_0^{\infty} \frac{\varphi(\varepsilon t)}{t^2 + 1} \cdot \chi_{[0, \frac{1}{\varepsilon}]}(t) dt$$

where  $\chi_{[0, \frac{1}{\varepsilon}]}$  is the characteristic function. By the dominated convergence theorem,

$$\lim_{\varepsilon \rightarrow 0^+} \left( 2i \cdot \int_0^{\infty} \frac{\varphi(\varepsilon t)}{t^2 + 1} \cdot \chi_{[0, \frac{1}{\varepsilon}]}(t) dt \right)$$
$$= 2i \cdot \int_0^{\infty} \lim_{\varepsilon \rightarrow 0^+} \frac{\varphi(\varepsilon t)}{t^2 + 1} \cdot \chi_{[0, \frac{1}{\varepsilon}]}(t) dt$$
$$= 2i \cdot \int_0^{\infty} \frac{\varphi(0)}{t^2 + 1} dt = i\pi \cdot \varphi(0).$$

2019, Fall.

10. (10 pts.) Let  $I \subseteq \mathbb{R}$  be a non-empty open interval. Let  $f : I \rightarrow \mathbb{R}$  be a real analytic function, i.e. for any  $x_0 \in I$  there is an open interval  $J \subseteq I$  containing  $x_0$  and  $(a_n)_{n=1}^{\infty} \subseteq \mathbb{R}$  such that  $f(x) = \sum_{n=1}^{\infty} a_n(x-x_0)^n, \forall x \in J$ . Let  $y_0 \in I$  be a point such that  $f'(y_0) \neq 0$ .

(a) Show that there is an open interval  $J \subseteq I$  containing  $y_0$  such that  $f'(x) \neq 0, \forall x \in J$ .

(b) Show that there is an open interval  $J' \subseteq I$  containing  $y_0$  and a constant  $C > 0$  such that

$$|f(x) - f(y_0)|^{\frac{1}{2}} \leq C|f'(x)|, \forall x \in J'.$$

Proof. a). Since every analytic function is  $C^{\infty}$ , and the derivative at  $y_0$  is

$$f'(y_0) = \sum_{n=1}^{\infty} n a_n (x - y_0)^{n-1} = a_1 + \sum_{n=2}^{\infty} n a_n (x - y_0)^{n-1}. \quad (x \in J)$$

By the assumption,  $f'(y_0) = a_1 \neq 0$ , and since  $\sum n a_n (x - y_0)^{n-1}$  converges uniformly,

$$\lim_{x \rightarrow y_0} \sum_{n=2}^{\infty} n a_n (x - y_0)^{n-1} = \sum_{n=2}^{\infty} \lim_{x \rightarrow y_0} n a_n (x - y_0)^{n-1} = 0$$

therefore for  $\epsilon = |a_1|/2$ , there is  $\delta > 0$  such that

$$|x - y_0| < \delta \Rightarrow |f'(x) - a_1| < \frac{|a_1|}{2}.$$

So,  $J_0 = (y_0 - \epsilon, y_0 + \epsilon)$  is the open interval we desired.

b) Since  $f(x) \rightarrow f(y_0)$  as  $x \rightarrow y_0$  by the continuity and  $|f'(x)| \geq |a_1|/2$  for all  $x \in J_0$  by a), so we can take  $\delta' > 0$  such that

$$|x - y_0| < \delta' \Rightarrow |f(x) - f(y_0)|^{\frac{1}{2}} \leq \frac{|a_1|}{2} \leq f'(x),$$

so proved.

Practice in the Next.

Debris.

Since  $f$  is analytic, for given  $y_0 \in I$ , there is an open interval  $J_0$  containing  $y_0$  such that  $f(x) = \sum_{n=0}^{\infty} a_n (x - y_0)^n$ , for some  $\{a_n\}$ .

Moreover,  $f \in C^{\infty}$  in  $J_0$ , so the derivative  $f'$  is continuous on  $J_0$ . Since the set

$$K = \{x \in J_0 \mid f'(x) = 0\} = (f')^{-1}[\{0\}] \cap J_0$$

is closed set of  $J_0$ , therefore  $J_0 \setminus (f')^{-1}[\{0\}]$  is an open set containing  $y_0$ , thus for some open interval  $J \subset J_0$ ,  $f'(x) \neq 0$  for all  $x \in J$ .

b). By the mean-value theorem, for any  $x \in I$ , there is  $\xi \in I$  such that

$$|f(x) - f(y_0)| = |f'(\xi)| \cdot |x - y_0|.$$

Meanwhile, the given condition equals to

$$|f(x) - f(y_0)| \leq C^2 |f'(x)|^2,$$

so combining the above condition then

$$|f'(x) - f'(y_0)|$$

By the assumption, for some open interval  $J$  containing  $y_0$ , for any  $x \in J_0$ ,

$$f(x) = \sum_{n=0}^{\infty} a_n (x - y_0)^n.$$

$$\text{So, } |f(x) - f(y_0)| = |f(x) - a_0| = \left| \sum_{n=1}^{\infty} a_n (x - y_0)^n \right| \quad \text{and}$$

$$|f'(x)| = \left| \sum_{n=1}^{\infty} n a_n (x - y_0)^{n-1} \right|.$$

Problem. Let  $I \subset \mathbb{R}$  be an open interval, and let  $f: I \rightarrow \mathbb{R}$  be real analytic. Suppose that for some  $m \geq 2$ , and  $a \in I$ ,

$$f(a) = f'(a) = \dots = f^{(m-1)}(a) = 0 \quad \text{and} \quad f^{(m)}(a) \neq 0.$$

1). Show that there exist an open interval  $J \subset I$  containing  $a \in I$  and real analytic  $g: J \rightarrow \mathbb{R}$  such that

$$f(x) = (x-a)^m \cdot g(x) \quad (x \in J) \quad \text{and} \quad g(x) \neq 0 \quad (x \in J).$$

2). Show that there exist  $C_1, C_2 > 0$  and open interval  $J' \subset J$  containing  $a$  such that  $C_1 |x-a|^m \leq |f(x)| \leq C_2 |x-a|^m \quad (x \in J')$

3). Show that there exists  $J'' \subset J'$  and  $C > 0$  such that  $|f(x)|^{\frac{m-1}{m}} \leq C \cdot |f'(x)| \quad (x \in J'').$

2019. Fall.

11. (9 pts.) Consider topological spaces  $X$  and  $Y$ , and suppose that  $f : X \rightarrow Y$  is a homotopy equivalence.

(a) The mapping cylinder of  $f$  is defined by

$$M(f) = (X \times [0, 1]) \amalg Y / \sim,$$

where  $\sim$  is given by  $(x, 1) \sim f(x)$  for  $(x, 1) \in X \times [0, 1]$ . Show that  $M(f)$  deformation retracts onto  $Y$ .

(b) Show that  $M(f)$  is homotopy equivalent to  $X$ .

(c) Is the homotopy equivalence in (b) also a deformation retract? Prove or disprove.

Proof. a). Let  $\pi : (X \times [0, 1]) \amalg Y \rightarrow M(f)$  be the natural projection. Define a retraction

$$r : M(f) \rightarrow Y : \begin{cases} [(x, t)] \mapsto f(x) & \text{if } [(x, t)] \in \pi[X \times [0, 1]] \\ [y] \mapsto y & \text{if } [y] \in \pi[Y]. \end{cases}$$

Since  $(x, 1) \sim f(x)$  for all  $x \in X$ , so that  $[(x, 1)] = [f(x)]$ , thus

$$r([(x, 1)]) = f(x) = r([f(x)]), \text{ so this } r \text{ is well-defined.}$$

Then, for the natural inclusion  $i : Y \rightarrow M(f)$ ,  $r \circ i = \text{id}_Y$  and  $i \circ r \simeq \text{id}_{M(f)}$  by the deformation

$$H : M(f) \times I \rightarrow M(f) : \begin{cases} [(x, t)] \times s \mapsto [(x, (1-s) \cdot t + s)] \\ [y] \times s \mapsto [y] \end{cases}$$

Then,  $H(\cdot, 0) = \text{id}_{M(f)}(\cdot)$ , and  $H(\cdot, 1) = i \circ r(\cdot)$

To show  $M(f)$  deformation retracts onto  $Y$ , we want to find a retraction  $r : M(f) \rightarrow Y$  such that  $i \circ r \simeq \text{id}_{M(f)}$  where  $i : Y \rightarrow M(f)$  is the inclusion.

Define  $r : M(f) \rightarrow Y : \begin{cases} f(x) & \text{if } (x, t) \in X \times I, \\ y & \text{if } y \in Y \end{cases}$

Consider the map  $\tilde{f} : X \times \{1\} \rightarrow Y : (x, t) \mapsto f(x)$ .

Then, we can consider the quotient map

$$\pi : (X \times [0, 1]) \amalg Y \rightarrow M(f).$$

Since  $f$  is homotopy equivalence, by definition, there is a continuous map  $g : Y \rightarrow X$  such that  $f \circ g \simeq \text{id}_Y$  and  $g \circ f \simeq \text{id}_X$ .

a).

2020 spring

2. Let  $A = \begin{pmatrix} 0 & I_n \\ 2I_n & 0 \end{pmatrix} \in M_{2n}(\mathbb{R})$

a). Determine if  $A$  is diagonalizable over  $\mathbb{R}$ .

b). Find the characteristic and the minimal polynomial of  $A$ .

Proof a). Observe that

$$A^2 = \begin{pmatrix} 2I_n & 0 \\ 0 & 2I_n \end{pmatrix} = 2 \cdot I_{2n}, \text{ so } A^2 - 2I = 0, \text{ with } A \neq \pm \sqrt{2} \cdot I_{2n}.$$

therefore, the minimal polynomial of  $A$  is  $t^2 - 2 = (t - \sqrt{2})(t + \sqrt{2})$ , which is separable and splits over  $\mathbb{R}$ , so  $A$  is diagonalizable.

b). The characteristic polynomial is:

$$\det(tI - A) = \det \begin{pmatrix} tI_n & -I_n \\ -2I_n & tI_n \end{pmatrix} = \det(t^2 I_n - 2I_n) = (t^2 - 2)^n$$

And the minimal is already obtained in a), so it is done.  $\square$

2020 spring.

3. Show that no group of order 48 is simple.

25

Proof. Notice that  $48 = 2^4 \cdot 3$ . By the Sylow theorem, for a group  $G$  of order 48,  $n_2 = \# \text{Syl}_2(G) \in \{1, 3\}$  and  $n_3 = \# \text{Syl}_3(G) \in \{1, 4, 16\}$ .

If either  $n_2 = 1$  or  $n_3 = 1$ , then  $G$  has a non-trivial proper normal subgroup, so it is non-simple. Now, suppose that  $n_2 = 3$  and  $n_3 \in \{4, 16\}$ .

Consider the afforded Homomorphism

$$\mathcal{Q}: G \rightarrow \text{Sym}(\text{Syl}_2(G)): g \mapsto (p \mapsto gPg^{-1})$$

By the Sylow theorem, this action is transitive on  $\text{Syl}_2(G)$ , so  $\ker \mathcal{Q} \neq G$ .

And,  $\text{Sym}(\text{Syl}_2(G)) \cong S_3$ , so  $|G| = 48 > 6 = |S_3|$  implies that

$\mathcal{Q}$  cannot be injective, so  $\ker \mathcal{Q} \neq \{e\}$ . Now,  $\ker \mathcal{Q} \triangleleft G$  is the non-trivial normal subgroup, so  $G$  is non-simple.  $\square$

Note: By the Sylow theorem, all elements in  $\text{Syl}_p(G)$  are conjugate.

Thus, the map  $\mathcal{Q}: G \rightarrow \text{Sym}(\text{Syl}_p(G)): g \mapsto (p \mapsto gPg^{-1})$  is not trivial,

because: assume  $n_p > 1$ , then

$\mathcal{Q}$  is trivial  $\stackrel{\text{def}}{\Leftrightarrow} \ker \mathcal{Q} = G \Leftrightarrow$  for all  $g \in G$  and  $P \in \text{Syl}_p(G)$ ,  $gPg^{-1} = P$

And it contradicts to the Sylow Theorem. So  $\ker \mathcal{Q} < G$ .

Meanwhile,

$$\ker \mathcal{Q} = \{e\} \Leftrightarrow \mathcal{Q} \text{ is injective} \Rightarrow G \cong \mathcal{Q}[G] \leq \text{Sym}(\text{Syl}_p(G)) \cong S_{n_p}$$

Therefore, by the Contraposition,

$$|G| \nmid n_p! \Rightarrow \ker \mathcal{Q} > \{e\}.$$

pebr15

$$\begin{aligned} |P_1 \cup P_2 \cup P_3| &= |P_1| + |P_2| + |P_3| - |P_1 \cap P_2| - |P_2 \cap P_3| - |P_1 \cap P_3| + |P_1 \cap P_2 \cap P_3| \\ &= 24 + |P_1 \cap P_2 \cap P_3| \end{aligned}$$

$$\text{And, since } 1 \leq |P_1 \cap P_2 \cap P_3|, \quad 25 \leq |P_1 \cup P_2 \cup P_3| \leq 32$$

If  $n_3 = 16$ , then there are 16 numbers distinct order 3 subgroups of  $G$ , so there are  $16 \cdot 2 = 32$  distinct order 3 elements, by the Lagrange theorem. Meanwhile, since  $25 \leq |P_1 \cup P_2 \cup P_3|$ , and  $P_1 \cup P_2 \cup P_3$  does not contain the order 3 elements, so 25 + 32 elements are lay in  $G$ , it is contradiction.

Since  $n_2 = 3$ , there are three Sylow 2-subgroups  $P_1, P_2, P_3$ .

$$\text{For } i \neq j, \quad |P_i P_j| = \frac{|P_i| \cdot |P_j|}{|P_i \cap P_j|} = \frac{256}{|P_i \cap P_j|} \leq 48,$$

so  $4 < 256/48 \leq |P_i \cap P_j|$ . Meanwhile, by the Lagrange's Theorem,  $|P_i \cap P_j|$  is a multiple of 2, so  $|P_i \cap P_j| = 8$ . Now, let  $\mathcal{Q} = P_1 \cap P_2$ .

Since  $|P_i : \mathcal{Q}| = 2$ , by the Lagrange's theorem, so  $\mathcal{Q} \triangleleft P_1$  and  $\mathcal{Q} \triangleleft P_2$ .

Therefore,  $N_G(\mathcal{Q})$  contains  $P_1$  and  $P_2$ , so

$$|P_1 \cup P_2| = |P_1| + |P_2| - |P_1 \cap P_2| = 32 - 8 = 24 \text{ implies that}$$

$|N_G(\mathcal{Q})| \geq 24$ . So, either  $|N_G(\mathcal{Q})| = 24$  or 48.

If  $|N_G(\mathcal{Q})| = 24$ , then  $|G : N_G(\mathcal{Q})| = 2$ , so  $N_G(\mathcal{Q}) \triangleleft G$ .

If  $|N_G(\mathcal{Q})| = 48$ , then  $N_G(\mathcal{Q}) = G$  and so  $\mathcal{Q} \triangleleft G$ , so  $G$  is non-simple.

So proved  $\square$

2020. Spring.

6. Find the general solution of the differential equation

$$y'' - 3y' + 2y = \frac{1}{1+e^{-t}}.$$

Proof. Since the characteristic of  $y'' - 3y' + 2y = 0$  is  $r^2 - 3r + 2 = (r-1)(r-2)$ , therefore the general solution of the homogeneous part is:

$$y_h(t) = C_1 e^t + C_2 e^{2t} \quad (C_1, C_2 \in \mathbb{R}).$$

And, to find a solution of the given equation, we need to find the particular solution  $y_p(t)$ . First, calculate the Wronskian,

$$\begin{vmatrix} e^t & e^{2t} \\ e^t & 2e^{2t} \end{vmatrix} = 2e^{3t} - e^{3t} = e^{3t}.$$

Therefore, the particular solution is

$$\begin{aligned} y_p(t) &= \int_{x_0}^x \begin{vmatrix} e^t & e^{2t} \\ e^{2t} & e^{2t} \end{vmatrix} \cdot \frac{1}{1+e^{-t}} \cdot \frac{1}{e^{3t}} dt \\ &= \int_{x_0}^x \frac{e^{t+2t} - e^{2t+t}}{e^{3t} + e^{2t}} dt \\ &= e^{2x} \int_{x_0}^x \frac{1}{e^{2t} + e^t} dt - e^x \int_{x_0}^x \frac{1}{e^t + 1} dt \end{aligned}$$

Meanwhile, since  $\int \frac{1}{e^t + 1} dt = \int \frac{e^{-t}}{1 + e^{-t}} dt = -\log(1 + e^{-x})$

and  $\int \frac{1}{e^{2t} + e^t} dt = \int \frac{e^{-t}}{e^t + 1} dt = \int \frac{-u}{\frac{1}{u} + 1} \cdot \frac{1}{u} du = \int \frac{-u}{1+u} du$

$$= \int -1 + \frac{1}{1+u} du = -u + \log(1+u) = -e^{-x} + \log(1+e^{-x})$$

Finally, the general solution is:

$$\begin{aligned} y_p + C_1 e_1 + C_2 e_2 &= \left[ e^x + e^{2x} \log(1+e^{-x}) + e^x \cdot \log(1+e^x) \right] \\ &+ C_1 e^x + C_2 e^{2x}, \quad C_1, C_2, C_3 \in \mathbb{R}, \end{aligned}$$

2020 Spring.

?, Let  $(a_n)_{n \in \mathbb{N}}$  be a sequence of non-negative real numbers and consider  
$$f(x) = \sum_{n=1}^{\infty} a_n \cdot \max\{|x-2n|, 0\}, \quad x \in [0, \infty)$$

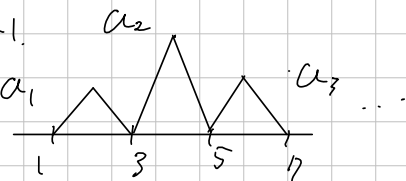
Give a necessary and sufficient condition on  $(a_n)$  such that  
a), the series to converge uniformly on  $[0, \infty)$ .

b).  $f$  is continuous uniformly.

Proof. Notice that: Given  $x \geq 0$ ,

$$|-|x-2n|| \geq 0 \Leftrightarrow 2n-1 \leq x \leq 2n+1.$$

Therefore,  $f$  can be described as



a), The given series converges uniformly if and only if

$$\sup_{x \geq 0} \left| f(x) - \sum_{n=1}^N a_n \cdot \max\{|x-2n|, 0\} \right| \rightarrow 0 \quad \text{as } N \rightarrow \infty.$$

Meanwhile, the supremum term above equals to

$$\sup_{k \geq N+1} a_k$$

Therefore,  $\lim_{n \rightarrow \infty} \sup a_n = 0$  is the condition we desired.

(Since  $a_n$  is non-negative,  $\lim_{n \rightarrow \infty} \sup a_n = \lim_{n \rightarrow \infty} a_n = 0$ )

b) If  $(a_n)$  is bounded by  $M > 0$ , then, given  $\varepsilon > 0$ , take  $\delta = \frac{\varepsilon}{M}$ . Then

$$|x-y| < \delta = \frac{1}{M} \Rightarrow |f(x) - f(y)| \leq M \cdot |x-y| < \varepsilon, \quad \text{so continuous uniformly.}$$

Conversely, if  $(a_n)$  is unbounded, then

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ such that } |x-y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon$$

'is false, because for  $\varepsilon=1$ , for any  $\delta > 0$ , there exists  $x_n, y_n$  such that  
 $|f(x_n) - f(y_n)| = a_n \cdot |x_n - y_n| = a_n \cdot \frac{\delta}{2} > 1$ , we take  $n \in \mathbb{N}$  such that  $a_n > \frac{2}{\delta}$ .

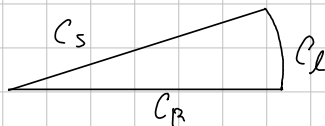
So, the condition that we desired is  $(a_n)$  is bounded.  $\square$

(Technically, we can take  $x_n = 2n$ ,  $y_n = 2n + \delta$ .)

2020 Spring.

8. For each  $n=2,3,\dots$  evaluate

$$\int_0^\infty \frac{1}{1+x^n} dx.$$



Proof. Let  $f(z) = \frac{1}{1+z^n}$ . Take a Contour  $C$  as a Composition of:

$$C_R: t \mapsto t \quad (0 \leq t \leq R), \quad C_\theta: t \mapsto R \cdot e^{i\frac{\pi t}{n}} \quad (0 \leq t \leq 2\pi),$$

$$C_s: t \mapsto t \cdot e^{\frac{2\pi i}{n}} \quad (0 \leq t \leq R), \text{ that is, } C = C_R * C_\theta * (-C_s).$$

Inside  $C$ ,  $1+z^n$  has only one zero,  $\exp(i\pi/n)$ , so  $f$  has only one singularity inside  $C$ , the simple pole  $\exp(i\pi/n)$ , by the Residue theorem,

$$\begin{aligned} \int_C f &= 2\pi i \cdot \text{Res}(f, \exp(i\pi/n)) = \lim_{z \rightarrow \exp(i\pi/n)} (z - \exp(i\pi/n)) \cdot \frac{1}{1+z^n} \cdot 2\pi i \\ &= \frac{2\pi i}{n \cdot \exp(i\pi/n)} = -\frac{\exp(i\pi/n)}{n} \cdot 2\pi i \end{aligned}$$

Meanwhile,

$$\begin{aligned} \left| \int_{C_\theta} f \right| &= \int_0^{2\pi} \left| \frac{1}{1+R^n \cdot e^{i\pi t}} \cdot \frac{Ri}{n} \cdot e^{i\frac{\pi t}{n}} \right| dt \\ &= \int_0^{2\pi} \frac{R/n}{|1+R^n \cdot e^{i\pi t}|} dt \leq \int_0^{2\pi} \frac{R/n}{|R^n - 1|} dt \rightarrow 0 \text{ as } R \rightarrow \infty. \end{aligned}$$

and

$$\begin{aligned} \int_{C_s} f &= \int_0^R \frac{e^{\frac{2\pi i t}{n}}}{1+t^n e^{2\pi i}} dt = e^{\frac{2\pi i}{n}} \int_0^R \frac{1}{1+t^n} dt. \text{ So that} \\ -\frac{2\pi i}{n} \cdot \exp(i\pi/n) &= \lim_{R \rightarrow \infty} \int_C f = \int_0^\infty \frac{1}{1+x^n} dx - \int_0^\infty \frac{1}{1+x^n} dx \cdot e^{\frac{2\pi i}{n}} \\ &= (1 - e^{\frac{2\pi i}{n}}) \cdot \int_0^\infty \frac{1}{1+x^n} dx \end{aligned}$$

$$\Rightarrow \int_0^\infty \frac{1}{1+x^n} dx = -\frac{2\pi i}{n} \cdot \exp(i\pi/n) / (1 - e^{\frac{2\pi i}{n}}).$$

□

2020. Spring

15. Let  $r=r(\theta)$  be a curve in  $\mathbb{R}^2$  defined in polar coordinates  $(r, \theta)$  with  $r > 0$ ,  $0 \leq \theta < 2\pi$ .

a. Express the arc length of the curve using  $r(\theta), r'(\theta)$ .

b. Express the curvature at  $r(\theta)$  using  $r(\theta), r'(\theta), r''(\theta)$ .

Proof. Denote  $x = r \cdot \cos \theta$  and  $y = r \cdot \sin \theta$ ,  $\alpha(\theta) = (x, y)$ .

$$a), \frac{dx}{d\theta} = r' \cos \theta - r \sin \theta, \quad \frac{dy}{d\theta} = r' \sin \theta + r \cos \theta.$$

$$\text{therefore, } \int_0^{2\pi} \left| \frac{d\alpha}{d\theta} \right| d\theta = \int_0^{2\pi} \sqrt{r'^2 + r^2} d\theta.$$

b. Since the tangent vector with respect to the arc length  $s$

$$t(\theta) = \frac{d\alpha}{ds} = \frac{1}{\frac{ds}{d\theta}} \cdot \frac{d\alpha}{d\theta} = \left| \frac{d\alpha}{d\theta} \right|^{-1} \cdot \frac{d\alpha}{d\theta} = \frac{1}{\sqrt{r'^2 + r^2}} \cdot (r' \cos \theta - r \sin \theta, r' \sin \theta + r \cos \theta)$$

and since the  $\alpha$  is plane curve, the normal vector is

$$n(\theta) = \frac{1}{\sqrt{r'^2 + r^2}} \cdot (-r' \sin \theta - r \cos \theta, r' \cos \theta - r \sin \theta)$$

Since the curvature is

$$K(\theta) = \left\langle \frac{dt}{ds}, n \right\rangle = \left\langle \frac{d\theta}{ds} \cdot \frac{dt}{d\theta}, n \right\rangle$$

$$= \left| \frac{d\theta}{d\alpha} \right| \cdot \frac{1}{(\sqrt{r'^2 + r^2})^2} \cdot \left\langle (r' \cos \theta - r \sin \theta - r' \sin \theta - r \cos \theta, r' \sin \theta + r \cos \theta + r \cos \theta - r \sin \theta), \right. \\ \left. (-r' \sin \theta - r \cos \theta, r' \cos \theta - r \sin \theta) \right\rangle$$

$$= \frac{1}{(r'^2 + r^2)^{\frac{3}{2}}} \cdot \left\langle ((r'' - r) \cos \theta - 2r' \sin \theta, (r'' - r) \sin \theta + 2r' \cos \theta), \right. \\ \left. (-r' \sin \theta - r \cos \theta, r' \cos \theta - r \sin \theta) \right\rangle$$

$$= \frac{1}{(r'^2 + r^2)^{\frac{3}{2}}} \cdot (r''r + r^2 + 2r'^2)$$

Comment. 밑줄표는 사다리꼴이긴 하다.

Pebriis

observe that

$$\frac{d\alpha}{d\theta} = \left( \frac{dx}{d\theta}, \frac{dy}{d\theta} \right), \text{ so } t(\theta) = \frac{1}{\sqrt{r'^2 + r^2}} \cdot (r' \cos \theta - r \sin \theta, r' \sin \theta + r \cos \theta)$$

$$n(\theta) = \frac{1}{\sqrt{r'^2 + r^2}} \cdot (-r' \sin \theta - r \cos \theta, r' \cos \theta - r \sin \theta)$$

$$t'(\theta) = \left( \frac{1}{2} \cdot \frac{2r'r'' + 2rr'}{\sqrt{r'^2 + r^2}} \right) \cdot \sqrt{r'^2 + r^2} \cdot t(\theta)$$

$$+ \frac{1}{\sqrt{r'^2 + r^2}} \cdot (r'' \cos \theta - r' \sin \theta - r' \sin \theta - r \cos \theta, r'' \sin \theta + r' \cos \theta + r' \cos \theta - r \sin \theta)$$

Now, the curvature

$$K(\theta) = \langle t', n \rangle = \frac{1}{(r'^2 + r^2)^{\frac{3}{2}}} \left\langle ((r'' - r) \cos \theta - 2r' \sin \theta, (r'' - r) \sin \theta + 2r' \cos \theta), \right. \\ \left. (-r' \sin \theta - r \cos \theta, r' \cos \theta - r \sin \theta) \right\rangle$$

$$= \frac{1}{(r'^2 + r^2)^{\frac{3}{2}}} \cdot (r^2 - rr'' + 2r'^2)$$

12/20, Fall.

1. Let  $p$  be a prime. Find the number of subgroups  $H$  of  $\mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$ ,  $|H| = p^2$ .  
Proof. By the fundamental theorem of finitely generated abelian group,  
either  $|H| \cong \mathbb{Z}_{p^2}$  or  $\mathbb{Z}_p \times \mathbb{Z}_p$ .

If  $|H| \cong \mathbb{Z}_p \times \mathbb{Z}_p$ , then for all  $x \in H$ ,  $p \cdot x = 0$ . Therefore,  
 $H \subseteq \{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\}$ .

Meanwhile,  $pa = 0$  and  $pb = 0$  if and only if  $a \in \langle p \rangle \subseteq \mathbb{Z}_{p^2}$  and  $b \in \langle p^2 \rangle \subseteq \mathbb{Z}_{p^3}$ ,  
so  $\{(a, b) \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3} \mid p \cdot (a, b) = 0\} = \langle p \rangle \times \langle p^2 \rangle$ .

Since  $\langle p \rangle \times \langle p^2 \rangle$  has already order  $p^2$ , so  $H = \langle p \rangle \times \langle p^2 \rangle$ .

If  $|H| \cong \mathbb{Z}_{p^2}$ , that is,  $H$  is generated by order  $p^2$  element,  
Since  $x \in \mathbb{Z}_{p^2} \times \mathbb{Z}_{p^3}$  has order  $p^2 \iff p^2 \cdot x = 0$  and  $px \neq 0$ .

Every element  $a \in \mathbb{Z}_{p^2}$  satisfies  $p^2 \cdot a = 0$ , and  $b \in \mathbb{Z}_{p^3}$  satisfies  $p^2 \cdot b = 0$   
if and only if  $b \in p\mathbb{Z}_{p^3}$  and  $\mathbb{Z}_{p^3}$  has number of  $p^2$  of  $p$ -multiples,  
so there are  $p^2 \cdot p^2 = p^4$  elements satisfying  $p^2 \cdot x = 0$ .

And,  $px = 0$  if and only if  $a \in p\mathbb{Z}_{p^2}$  and  $b \in p^2\mathbb{Z}_{p^3}$ , where  $x = (a, b)$ .  
So, there are  $p^2$  elements satisfying  $px = 0$ .

Consequently, there are  $p^4 - p^2$  elements  $x$  such that  $|H| = \langle x \rangle$ .

Moreover, the cyclic group of order  $p^2$  has  $\phi(p^2) = p^2 - p$  generators,

so the possible  $|H| \cong \mathbb{Z}_{p^2}$  are

$$(p^4 - p^2) / (p^2 - p)$$

numbers.

Finally, the number of  $|H|$  is  $1 + (p^4 - p^2) / (p^2 - p)$ .

Note: for a prime  $p$ ,  $\phi(p^n) = p^n - p^{n-1}$ .

2020 Fall.

2. For each  $n \in \mathbb{N}$ , let  $a_n = \cos(2\pi/n)$ .

a). Prove that  $a_n$  is algebraic over  $\mathbb{Q}$ .

b). Calculate  $\deg_{\mathbb{Q}}(a_5)$ .

Proof. a). Let  $\zeta = \exp(2\pi i/n)$ . Then,  $\zeta^{-1} = \exp(-2\pi i/n) = \cos(2\pi/n) - i\sin(2\pi/n)$  gives that  $a_n = \cos(2\pi/n) = (\zeta + \zeta^{-1})/2$ .

Meanwhile, since  $\zeta^n = \exp 2\pi i = 1$ , so  $\zeta^n - 1$  gives  $\zeta$  is algebraic over  $\mathbb{Q}$ .

Therefore,  $\zeta^{-1}$  is also algebraic, thus  $a_n = \frac{\zeta + \zeta^{-1}}{2}$  is algebraic over  $\mathbb{Q}$ .

b). Let  $w = \exp(2\pi i/5)$ . Since  $w$  is the primitive 5th root of unity,  $w^4 + w^3 + w^2 + w + 1 = 0$ . And, dividing by  $w^2$  then  $2a_5 = w + w^{-1}$  implies

$$w^4 + w^3 + w^2 + w + 1 = 0 \Rightarrow w^2 + w^{-2} + w + w^{-1} + 1 = 0$$

$$\Rightarrow 4a_5^2 - 2 + 2a_5 + 1 = 0$$

$$\Rightarrow 4a_5^2 + 2a_5 - 1 = 0.$$

Since  $4x^2 + 2x - 1$  is irreducible over  $\mathbb{Q}$  by the discriminant  $2^2 + 16 = 20$  but  $\sqrt{20} \notin \mathbb{Q}$ , so  $\deg_{\mathbb{Q}}(a_5) = 2$ .  $\square$

Fall.

a) Notice that De Moivre's formula gives  $e^{i \cdot n z} = (\cos n z + i \cdot \sin n z) = (\cos z + i \cdot \sin z)^n$

Therefore

$$\begin{aligned} \cos n \cdot \frac{2\pi}{n} = 1 &= \operatorname{Re} \left( \cos \frac{2\pi}{n} + i \cdot \sin \frac{2\pi}{n} \right)^n \\ &= \operatorname{Re} \left( \sum_{k=0}^n \binom{n}{k} \left( \cos \frac{2\pi}{n} \right)^k \cdot \left( i \cdot \sin \frac{2\pi}{n} \right)^{n-k} \cdot \binom{n}{k} \right) \end{aligned}$$

Meanwhile,  $\sin^2(\frac{2\pi}{n}) = 1 - \cos^2(\frac{2\pi}{n})$  and

the real part of  $\sum_{k=0}^n \binom{n}{k} \cdot \left( \cos \frac{2\pi}{n} \right)^k \cdot \left( i \cdot \sin \frac{2\pi}{n} \right)^{n-k}$  yield only  $n-k$  is even,

since  $i^{n-k}$  is real  $\Leftrightarrow n-k$ , so  $\operatorname{Re} \left( \sum_{k=0}^n \binom{n}{k} \cdot \left( \cos \frac{2\pi}{n} \right)^k \cdot \left( i \cdot \sin \frac{2\pi}{n} \right)^{n-k} \right) - 1 = 0$  is evaluated  $\cos(\frac{2\pi}{n})$  for some polynomial over  $\mathbb{Q}$ , so  $a_n$  is algebraic over  $\mathbb{Q}$ .

b). From above argument,

$$\begin{aligned} 1 &= \underbrace{1 \cdot a_5^0}_{k=0} + \underbrace{5 \cdot a_5^1 \cdot (1 - a_5^2)}_{k=1} - \underbrace{10 \cdot a_5^3 \cdot (1 - a_5^2)^2}_{k=3} + \underbrace{a_5^5}_{k=5} \\ &= 5 \cdot a_5 - 5 \cdot a_5^3 - 10 a_5^3 \cdot (1 - 2a_5^2 + a_5^4) + a_5^5 \\ &= 5 \cdot a_5 - 15 a_5^3 + 11 a_5^5 - 10 \cdot a_5^7. \end{aligned}$$

$$\text{So, } 10 a_5^7 - 11 a_5^5 + 15 a_5^3 - 5 a_5 + 1 = 0.$$

2020, Fall

3. Let  $R$  be a field. Show that  $(x^3 - y^5) \in R[x, y]$  is prime ideal.

Proof. Since  $R[x, y]$  is U.F.D., every irreducible element is prime.

Thus, enough to show that  $x^3 - y^5$  is irreducible.

Notice that:  $R[x, y] = R[y][x]$  by definition, so  $x^3 - y^5 \in R[y][x]$ .

Since  $\gcd(1, y^5) = 1$ ,  $x^3 - y^5$  is primitive, by the Gauss Lemma,

$x^3 - y^5$  is irreducible over  $R[y] \Leftrightarrow x^3 - y^5$  is irreducible over  $R(y)$ .

But, since  $x^3 - y^5$  has degree 3,  $x^3 - y^5$  is reducible only if it has a linear factor, in other say,  $x^3 - y^5$  has a root in  $R(y)$ .

Suppose that for some  $f(y)/g(y) \in R(y)$ ,

$$\frac{f(y)^3}{g(y)^3} - y^5 = 0.$$

Then,  $f(y)^3 = y^5 g(y)^3$ , But  $\deg f(y)^3 \equiv 0 \pmod{3}$ , and

$$\deg y^5 g(y)^3 = 5 + \deg g(y)^3 \equiv 2 \pmod{3}.$$

Therefore, there is no such a root, so  $x^3 - y^5$  is irreducible over  $R(y)$ ,  
so it is irreducible over  $R[x, y]$ .

Note:

$$K[x, y] = K[y][x] \xrightarrow{\text{Gauss}} K(y)[x] \xrightarrow{\text{irr. Test.}}$$

Modified

Prove that  $\gcd(m,n)=1$  implies  $x^m - y^n \in F[x,y]$  is irreducible.

2020. Fall.

4. Let  $W \subseteq V = \mathbb{R}^n$  be a subspace. Show that there exists a homogeneous system of linear equations whose solution space is  $W$ .

Proof. Let  $\beta = \{\alpha_1, \dots, \alpha_k\}$  be a basis for  $W$ .

Then, we can extend the basis  $\beta$  to  $\overline{\beta} = \beta \cup \{\alpha_{k+1}, \dots, \alpha_n\}$  for  $V$ .

Now, consider the map:

$$f_{\beta}: V \rightarrow \mathbb{R}^k; \sum c_i \alpha_i \mapsto C_{\beta}$$

Then,  $W = \ker f_{\alpha_{k+1}} \cap \dots \cap \ker f_{\alpha_n}$ , so for the standard basis  $\beta$  for  $V$ ,

$$A = \begin{pmatrix} [f_{\alpha_{k+1}}]_{\beta} \\ \vdots \\ [f_{\alpha_n}]_{\beta} \end{pmatrix} \text{ satisfies } \ker A = W, \text{ so}$$

([ $f$ ] $_{\beta}$  is row vector representation).

$$A \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ is the system we desired.}$$

2021. Spring.

1. Show that no group of 2020 is simple.

Proof  $2020 = 2^2 \cdot 5 \cdot 101$ . Denote  $n_p$  as the number of Sylow  $p$ -subgroups.

By the Sylow theorem,  $n_{101} \equiv 1 \pmod{101}$  and  $n_{101} \mid 20$ , so  $n_{101}$  must be 1. Therefore, the Sylow 101-subgroup is normal so proved.

Without using Sylow theorem.

By the Cauchy's theorem,  $G$  has cyclic subgroup  $H \leq G$  of order 101.

Denote  $A = \{g \mid |g| = 2 \in G\}$ . Then,  $A$  is orbit of  $H$  from the action

$$G \times \{S \leq G\} \rightarrow \{S \leq G\} : (g, S) \mapsto gSg^{-1}.$$

Since the stabilizer of this action of  $H$  is  $G_H = N_G(H)$ , we obtain

$$\#A = |G : N_G(H)|.$$

And since  $H \leq N_G(H)$ ,

$$|G : N_G(H)| \leq |G : H| = 2020/101 = 20, \text{ so } \#A \leq 20.$$

Now, consider the action

$$H \times A \rightarrow A : (h, K) \mapsto hKh^{-1}.$$

Given  $K \in A$ , the  $H$ -orbit of this action is  $\text{Orb}_H(K) = \{hKh^{-1} \mid h \in H\}$  and

$$\#\text{Orb}_H(K) = |H : \text{Stab}_H(K)|, \text{Stab}_H(K) \leq H \Rightarrow |\text{Stab}_H(K)| \in \{1, 101\}.$$

So,  $\#\text{Orb}_H(K) \in \{1, 101\}$  and  $\text{Orb}_H(K) \leq A$ , so  $\#\text{Orb}_H(K) = 1$ .

Therefore, every element in  $H$  normalizes  $K \in A$ , that is,  $H \leq N_G(K)$ .

Now, for some  $K \in A$ , if  $H \neq K$  then  $HK \leq G$  and

$$|HK| = \frac{|H| \cdot |K|}{|H \cap K|} = 101^2$$

but, since  $101^2 \nmid 2020$ , so  $A = \{1\}$ . That is,  $H \leq G$   $\square$

2021, Spring.

3. Let  $p$  be a prime number, and  $L$  be the Galois extension  $\mathbb{Q}(\zeta)$  where  $\zeta$  is a primitive  $p^2$ th root of unity. 55.

a). Show that all  $\mathbb{Q}$ -automorphisms of  $L$  are of the form

$$\zeta \mapsto \zeta^k, \quad p \nmid k, \quad k \in \mathbb{Z}.$$

b). Show that  $L$  contains a subfield whose Galois group is  $\mathbb{Z}/p\mathbb{Z}$ .  
Proof.

a). By definition,  $\{\zeta^0, \zeta^1, \dots, \zeta^{p^2-1}\}$  is a set of roots of the polynomial  $x^{p^2}-1 = (x-1)(x^{p^2-1} + x^{p^2-2} + \dots + x + 1)$  'irreducible'

Meanwhile,  $x^{p^2-1} + x^{p^2-2} + \dots + x + 1$  has a factor  $\prod_{\substack{1 \leq a \leq p^2 \\ \gcd(a, p^2)=1}} (x - \zeta^a) = p(x)$

So, since the automorphisms maps a root to the conjugate, so every automorphism of  $L/\mathbb{Q}$  maps  $\zeta \mapsto \zeta^k$ ,  $\gcd(k, p^2)=1$ , so,  $p \nmid k$ .  
Moreover, since  $\mathbb{Q}(\zeta)$  is a finite dimensional vector space generated by  $\zeta^k$  ( $k=0, 1, \dots$ ), so if  $\zeta \mapsto \zeta^k$  chosen, then the automorphism is determined. it proves a).

b). Since  $\varphi(p^2) = p^2 - p$ , where  $\varphi$  is the Euler-Phi function, the Galois extension of  $p(x)$  over  $\mathbb{Q}$  is contained in  $\mathbb{Q}(\zeta)$  and it has the degree of  $p^2 - p$ , so, this extension  $K$  satisfies

$$|\text{Gal}(K/\mathbb{Q})| = p^2 - p. \text{ And, by the Cauchy's theorem,}$$

it has a subgroup of order  $p$ ,  $G \leq \text{Gal}(K/\mathbb{Q})$ . Now the fixed field  $\mathbb{Q}(\zeta)^G$  is we desired.

Revised in the Next

2021. Spring

7. Let  $f$  be a polynomial given by

$$f(z) = a_0 + a_1 z + \dots + a_d z^d \quad \text{over } \mathbb{C}, \quad a_d \neq 0$$

8m.

Evaluate the value  $\lim_{R \rightarrow \infty} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz$

Proof. By the fundamental theorem of Algebra,  $f$  has  $d$ -roots  $z_1, \dots, z_d \in \mathbb{C}$ .

That is,  $f(z) = a_d (z - z_1) \dots (z - z_d)$

Moreover,  $\frac{f'(z)}{f(z)} = \frac{1}{z - z_1} + \dots + \frac{1}{z - z_d}$ , just calculation.

Therefore, for  $R > \max\{|z_1|, \dots, |z_d|\}$ , by the residue theorem,

$$\frac{1}{2\pi i} \int_{|z|=R} \frac{z \cdot f'(z)}{f(z)} dz = \sum_{n=1}^d \operatorname{Res} \left( \frac{z}{z - z_n}, z_n \right) = \sum_{n=1}^d \lim_{z \rightarrow z_n} z = z_1 + \dots + z_d$$

Consequently, the value is

$$2\pi i \cdot (z_1 + \dots + z_d) = 2\pi i \cdot \left( -\frac{a_{d-1}}{a_d} \right),$$

□

2021. Fall.

8. Let  $w \in \mathbb{R}$ . Prove that

$$\int_{-\infty}^{\infty} \cos(wt) \cdot e^{-t^2} dt = \sqrt{\pi} \cdot e^{-\frac{w^2}{4}}.$$

Proof. Since  $\cos(\cdot)$  and  $e^{-t^2}$  are even functions, so

$$\int_{-\infty}^{\infty} \cos(wt) \cdot e^{-t^2} dt = 2 \cdot \int_0^{\infty} \cos(wt) \cdot e^{-t^2} dt$$

Meanwhile, for  $R > 0$ ,

$$\int_0^R \cos(wt) e^{-t^2} dt = \frac{1}{w} \cdot \sin(wt) e^{-t^2} \Big|_0^R - \int_0^R \frac{1}{w} \cdot$$

If  $w=0$ , then the integral

$$\int_{-\infty}^{\infty} e^{-t^2} dt = \sqrt{\pi}, \text{ so we are done. Since } \cos(wt) = \cos(-wt),$$

assume that  $w > 0$ . Let  $f(z) = \cos(wz) \cdot e^{-z^2}$ .

Consider the half-circle curve, composition of

$$C_R: t \mapsto t \quad (-R \leq t \leq R) \text{ and } C_R: t \mapsto Re^{it} \quad (0 \leq t \leq \pi).$$

By the Cauchy's theorem, the entire  $f$  satisfies

$$\int_{C_R \cup C_R} f = 0, \text{ meanwhile,}$$

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Q. Suppose that  $f$  is entire and  $\operatorname{Re} f(x+iy) \geq x$  for all  $x, y \in \mathbb{R}$ . Show that  $f$  is a polynomial of degree one.

Proof. Notice that the given condition can be rewritten as

for all  $z \in \mathbb{C}$ ,  $\operatorname{Re} f(z) \geq \operatorname{Re} z$ .

Define  $g(z) = \exp(z - f(z))$ . Then,

$$\begin{aligned} |g(z)| &= |\exp(\operatorname{Re} f(z) - i \operatorname{Im} f(z) + (\operatorname{Re} z + i \operatorname{Im} z))| \\ &= \exp(-\operatorname{Re} f(z) + \operatorname{Re} z) \leq \exp(0) = 1 \end{aligned}$$

Since  $g$  is entire, being composite of entire functions,

so, the Liouville's theorem gives that  $g$  is a constant.

Therefore,  $g(z) = \exp(z - f(z)) = C$  for some  $C \in \mathbb{C} \setminus \{0\}$ , so that

$$0 = g'(z) = (1 - f'(z)) \cdot \exp(z - f(z)) \Rightarrow f'(z) = 1$$

Consequently,  $f(z) = z + \alpha$  for some  $\alpha \in \mathbb{C}$ , so proved.  $\square$

2021 Fall.

12. (10 pts) Define an equivalence relation  $\sim$  on  $X := \mathbb{C}^2 \setminus \{(0,0)\}$  by

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow \exists t \in \mathbb{C} \setminus \{0\} \text{ such that } (x_1, y_1) = (tx_2, ty_2).$$

(a) Show that  $X/\sim$  is homeomorphic to the 2-sphere  $S^2$ .

(b) Show that the 2-sphere  $S^2$  cannot be a union of countably many circles embedded in  $S^2$ .

Proof. a). Define a map

$$f: X \rightarrow S^2: (x, y) \mapsto (2 \cdot \operatorname{Re}(x\bar{y}), 2 \cdot \operatorname{Im}(x\bar{y}), |x\bar{y}|^2 - 1) \cdot \frac{1}{1 + |x\bar{y}|^2}$$

$\downarrow \quad \uparrow \quad \searrow \quad \swarrow$   
 $\mathbb{C} \cup \{\infty\} \quad \frac{x}{y}$

2021. Fall.

15. (10 pts) For a given smooth surface  $S$ , let us denote the Gaussian curvature at  $p$  by  $K(p)$  and the mean curvature at  $p$  by  $H(p)$ .

(a) Compute the Gaussian curvature  $K(p)$  of the surface  $z = y^2 - x^2$  at  $p = (0, 0, 0)$ .

(b) Describe the surface  $S$  for which  $K(p) = H(p)^2$  for all  $p \in S$ . (Justify your answer.)

Proof (a). Let  $X: B_1(0) \rightarrow \mathbb{R}^3: (u, v) \mapsto (u, v, v^2 - u^2)$ , the Monge patch.

observe  $X_u = (1, 0, -2u)$ ,  $X_v = (0, 1, 2v)$

$X_{uu} = (0, 0, -2)$ ,  $X_{uv} = (0, 0, 0)$ ,  $X_{vv} = (0, 0, 2)$

and the normal  $n = (2u, -2v, 1) / \sqrt{1 + 4u^2 + 4v^2}$

the coefficients of the first-fundamental form is

$$E = 1 + 4u^2, \quad F = -4uv, \quad G = 1 + 4v^2$$

and

$$e = -2 / \sqrt{1 + 4u^2 + 4v^2}, \quad f = 0, \quad g = 2 / \sqrt{1 + 4u^2 + 4v^2}$$

Now, the Gaussian curvature is

$$K(p) = \left( -4 / (1 + 4u^2 + 4v^2) \right) / (1 + 4u^2 + 4v^2)$$

$$= \frac{-4}{(1 + 4u^2 + 4v^2)^2}, \quad \text{so that at } p = (0, 0, 0), \quad K(p) = -4.$$

b).

2022, Spring

1. Let  $G$  be a group of order 2021.

a). Prove that  $G$  is not simple.

b) Prove that  $G$  is cyclic

Proof. Notice that:  $2021 = 43 \cdot 47$ .

a). By the Sylow theorem, the number  $n_{43}$  of Sylow 43-subgroup satisfies  $n_{43} \equiv 1 \pmod{43}$  and  $n_{43} \mid 47$ , so  $n_{43} = 1$ .

Therefore,  $\text{Syl}_{43}(G) = \{P\}$ , and  $gPg^{-1} \in \text{Syl}_{43}(P)$  for all  $g \in G$  implies  $gPg^{-1} = P$ , so  $P$  is normal. Thus  $G$  is not simple.

b). Similarly,  $n_{47} = 1$ , so there is a <sup>unique</sup> Sylow 47-subgroup  $Q$ .

By the Lagrange's theorem, all non-zero elements in  $P$  and  $Q$  have order of 43 and 47, respectively, so  $P \cap Q = \{e\}$ .

Since both 43 and 47 are prime,  $P$  and  $Q$  are cyclic, denote

$P = \langle x \rangle$  and  $Q = \langle y \rangle$ . Since  $[x, y] = x^{-1}y^{-1}xy \in P \cap Q$ , because

$P$  and  $Q$  both are normal, so  $x^{-1}y^{-1}xy \in Q$  and  $y^{-1}xy \in P$ , so

$[x, y] = e$ . Thus  $x$  and  $y$  commute. Now,  $xy$  is order of  $43 \cdot 47$ , so  $G = \langle xy \rangle$  is cyclic.

2022 Spring

2. Prove or disprove that each of following is PID:

a)  $\mathbb{R}[x]$

b)  $\mathbb{R}[x, y]$

c)  $\mathbb{Z}[\sqrt{-1}]$

d)  $\mathbb{Z}[x]$

Proof.

a)  $\mathbb{R}[x]$  is PID, because since  $\mathbb{R}$  is a field, so  $\mathbb{R}[x]$  is Euclidean domain, so for any ideal  $I \subseteq \mathbb{R}[x]$ , we can find a generator of  $I$

b)  $\mathbb{R}[x, y]$  is not P.I.D., because: let  $(x, y) \subseteq \mathbb{R}[x, y]$ .

If  $(x, y) = (p(x, y))$  for some  $p \in \mathbb{R}[x, y]$ ,

then  $p(0, 0) = 0$ , that is,  $p$  has zero constant term.

But,  $x, y \in (x, y) = (p(x, y))$  implies  $\deg_x p, \deg_y p \leq 1$ ,

2022. Spring.

3. Find the Galois group of the polynomial  $x^5 + 3x^3 - 2x^2 - 6$  over  $\mathbb{Q}$ .

Proof. Notice that

$$x^5 + 3x^3 - 2x^2 - 6 = x^2 \cdot (x^3 - 2) + 3(x^3 - 2) = (x^3 - 2) \cdot (x^2 + 3).$$

Since the irreducible factor  $x^3 - 2$  has three roots  $\sqrt[3]{2}$ ,  $\sqrt[3]{2} \cdot e^{\frac{2\pi i}{3}}$ ,  $\sqrt[3]{2} \cdot e^{\frac{4\pi i}{3}}$  and the irreducible factor  $x^2 + 3$  has two roots  $\pm\sqrt{-3}$ ,

The splitting field is

$$K = \mathbb{Q}(\sqrt[3]{2}, \sqrt[3]{2} \cdot e^{\frac{2\pi i}{3}}, \sqrt[3]{2} \cdot e^{\frac{4\pi i}{3}}, \sqrt{-3}, -\sqrt{-3}) = \mathbb{Q}(\sqrt[3]{2}, e^{\frac{2\pi i}{3}}, \sqrt{-3})$$

Moreover, since  $\mathbb{Q}$  is perfect, so  $K/\mathbb{Q}$  is Galois.

Meanwhile, notice that

$$e^{\frac{2\pi i}{3}} = -\frac{1}{2} + \frac{\sqrt{-3}}{2}$$

Therefore,  $K = \mathbb{Q}(\sqrt[3]{2}, e^{\frac{2\pi i}{3}})$ . Finally,

$$[\mathbb{Q}(\sqrt[3]{2}) : \mathbb{Q}] = 3$$

and since  $w = e^{\frac{2\pi i}{3}}$  satisfies  $w^2 + w + 1$ , so

$$[\mathbb{Q}(\sqrt[3]{2}, w) : \mathbb{Q}(\sqrt[3]{2})] \leq 2$$

But since  $w \in \mathbb{C} \setminus \mathbb{R}$  and  $\mathbb{Q}(\sqrt[3]{2}) \subseteq \mathbb{R}$ , so the degree is actually 2.

Since  $\text{Gal}(K/\mathbb{Q}) \subseteq \text{Sym}(\{\sqrt[3]{2}, \sqrt[3]{2}w, \sqrt[3]{2}w^2\})$ , the tower lemma gives

$[\mathbb{Q}(\sqrt[3]{2}, w) : \mathbb{Q}] = 6$ , so  $|\text{Gal}(K/\mathbb{Q})| = 6$ , that is,

$$\text{Gal}(K/\mathbb{Q}) \cong S_3.$$

2022 Spring.

8. Define  $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$

Then,  $f$  is differentiable on  $\mathbb{R}$ .

Proof. For each  $m \in \mathbb{N}$ , define

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

At first,  $f_m(x) \rightarrow f(x)$  as  $m \rightarrow \infty$ , because that

$$\sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1 \text{ as } m \rightarrow \infty,$$

so the increasing bounded sequence  $f_m(x)$  converges. Now,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

if  $|x| \leq 1$ , then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{2}{n^4} \text{ and } \sum \frac{1}{n^4} < \infty,$$

if  $|x| > 1$ , then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq 2 \cdot \frac{|x|}{x^2 + n^2} \cdot \frac{1}{x^2 + n^2} \leq 2 \cdot \frac{1}{n^2} \text{ and } \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives  $f'_m$  converges uniformly.

Now,  $f_m$  converges at some point and  $f'_m$  converges uniformly on  $\mathbb{R}$ , the limit function  $f$  is differentiable.  $\square$  sm.

~~First, for any  $x \in \mathbb{R}$ ,  $|x^2 + n^2| \geq n^2$ , so~~

~~$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2 + n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1,$$~~

~~as  $m \rightarrow \infty$ . Therefore, given  $\varepsilon_0$ , for some  $N \in \mathbb{N}$ ,  $m > n \geq N$  implies~~

~~$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^m \frac{1}{x^2 + k^2} \right| \leq \sup_{x \in \mathbb{R}} \sum_{k=n}^m \left| \frac{1}{x^2 + k^2} \right| < \sum_{k=n}^m \frac{1}{k^2} < \varepsilon.$$~~

~~Now, define for each  $m \in \mathbb{N}$ ,~~

~~$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$~~

~~We already shown  $f_m \rightarrow f$  uniformly. And,~~

~~$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$~~

~~converges uniformly, because~~

2022 Spring.

11. Prove or disprove that:

a). A Compact subset of any topological space is closed.

b). A Connected metric space having more than one point is uncountable.  
Proof.

a). Let  $\mathbb{Z}_3 = \{0, 1, 2\}$  with indiscrete topology  $\mathcal{C} = \{\mathbb{Z}_3, \emptyset\}$ .

Then, a singleton  $\{1\}$  is not closed, but it is compact set, because every open cover of  $\mathbb{Z}_3$  is finite. So, false.

b). Let  $a, b \in X$  be distinct elements of a connected metric space  $X$ .  
Given  $\lambda \in (0, 1)$ , if there is no point  $x \in X$  such that  $d(a, x) = \lambda \cdot d(a, b)$ ,  
then  $U = \{x \in X \mid d(a, x) < \lambda \cdot d(a, b)\}$  and  $V = \{x \in X \mid d(a, x) > \lambda \cdot d(a, b)\}$

forms a separation of  $X$ , so it contradicts to the connectedness.

( $U \cap V = \emptyset$  and  $U, V$  are open is clear, and

$U \cup V = X$  is obtained by the assumption).

So, given  $\lambda \in (0, 1)$ , there is a point  $x_\lambda \in X$  such that  $d(a, x_\lambda) = \lambda \cdot d(a, b)$ .

Thus, if  $\lambda, \lambda' \in (0, 1)$  are different, then

$$d(a, x_\lambda) \neq d(a, x_{\lambda'})$$

So the map  $(0, 1) \rightarrow X: \lambda \mapsto x_\lambda$  is well-defined injective map,  
thus  $X$  is uncountable.  $\square$

2022. Fall.

1. Let  $G$  be a group and  $H$  and  $K$  are normal subgroup of  $G$ .

If  $H \cap K = \{e\}$ , then  $H/K \cong H \times K$ .

Proof. Define a map

$$\varphi: H \times K \rightarrow H/K: (h, k) \mapsto hK.$$

It is well-defined by the definition of  $H/K$ , and it is homomorphism; given  $h_1, h_2 \in H$  and  $k_1, k_2 \in K$ ,

$$\varphi((h_1, k_1) \cdot (h_2, k_2)) = \varphi((h_1 h_2, k_1 k_2)) = h_1 h_2 K_1 k_2$$

meanwhile, given  $h \in H$  and  $k \in K$ ,  $hkh^{-1}k^{-1} \in K$  and  $\in H$ , because  $hkh^{-1} \in K$  and  $kh^{-1}k^{-1} \in H$ , by the normality. But,  $H \cap K = \{e\}$  implies  $hkh^{-1}k^{-1} = e$ , so  $hk = kh$ , these commute so,

$$h_1 h_2 k_1 k_2 = h_1 k_1 h_2 k_2 = \varphi((h_1, k_1)) \varphi((h_2, k_2)), \text{ so } \varphi \text{ preserves the operation.}$$

And,  $\varphi[H \times K] = H/K$  by just definition,  $\varphi$  is onto. Now, the first isomorphism theorem gives that  $H \times K / \ker \varphi \cong H/K$ .

$$\text{And, } \ker \varphi = \{(h, k) \in H \times K \mid hK = e\} = \{e\},$$

so  $H \times K \cong H \times K / \{e\}$ , it proves the assertion.  $\square$

2022 Fall.

8. For continuous functions  $g: [0,1] \rightarrow \mathbb{R}$  and  $h: [-1,1] \rightarrow \mathbb{R}$ , define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \int_0^1 h(x-y)g(y) dy$$

For each  $n \in \mathbb{N}$ , define

$$f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sum_{i=0}^{n-1} h(x - \frac{i}{n}) \cdot g(\frac{i}{n})$$

Prove that  $f_n \rightarrow f$  uniformly on  $[0,1]$

Proof. Define  $\ell: [0,1] \times [0,1]; (x,y) \mapsto h(x-y) \cdot g(y)$ .

Since  $h$  and  $g$  both are continuous, therefore  $\ell$  is continuous and since  $[0,1]^2$  is compact set, so  $\ell$  is continuous uniformly.

Given  $\varepsilon > 0$ , there exists  $\delta > 0$  such that for any  $(x_1, y_1), (x_2, y_2) \in [0,1]^2$ ,

$$\|(x_1, y_1) - (x_2, y_2)\| < \delta \Rightarrow \|\ell(x_1, y_1) - \ell(x_2, y_2)\| < \varepsilon.$$

Now, given  $x \in [0,1]$ , and  $n > \frac{1}{\delta}$ , ( $n > \frac{1}{\delta}$  gives  $\frac{1}{n} < \delta$ )

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \int_0^1 \ell(x, y) dy - f_n(x) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \ell(x, y) dy - \frac{1}{n} \cdot \sum_{i=0}^{n-1} \ell(x, \frac{i}{n}) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \ell(x, y) dy - \sum_{i=1}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \ell(x, \frac{i}{n}) dy \right| \\ &\leq \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \left| \ell(x, y) - \ell(x, \frac{i}{n}) \right| dy \\ &< \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \varepsilon dy = \varepsilon. \end{aligned}$$

Therefore,  $n > \frac{1}{\delta} \Rightarrow \sup_{x \in [0,1]} |f_n(x) - f(x)| < \varepsilon$ , So, proved.  $\square$

2022 Fall.

11. (10 pts) Consider the set  $\mathbb{R}$  of real numbers with the co-countable topology  $\mathcal{T}$  given by

$$\mathcal{T} = \{\emptyset\} \cup \{S \subset \mathbb{R} : \mathbb{R} \setminus S \text{ is countable}\}.$$

(a) Show that  $\mathcal{T}$  defines a topology.

(b) Determine if the interval  $[0, 1]$  is compact with the co-countable topology. (Explain your answer.)

Proof. (a).

Since  $\mathbb{R} \setminus \mathbb{R} = \emptyset$  is countable,  $\emptyset, \mathbb{R} \in \mathcal{T}$ .

Let  $U_\alpha$  ( $\alpha \in I$ ) be sets in  $\mathcal{T}$ . Then, (if one of  $U_\alpha = \emptyset$ , the trivial, let  $U_\alpha \neq \emptyset$ )

$$\mathbb{R} \setminus \left( \bigcup_{\alpha \in I} U_\alpha \right) = \mathbb{R} \cap \left( \bigcap_{\alpha \in I} U_\alpha^c \right) = \bigcap_{\alpha \in I} \mathbb{R} \setminus U_\alpha$$

is 'intersection of countable sets' so it is also countable. And,

for any finite sets  $U_1, \dots, U_n$  in  $\mathcal{T}$ ,

$$\mathbb{R} \setminus \left( \bigcap_{k=1}^n U_k \right) = \mathbb{R} \cap \left( \bigcup_{k=1}^n U_k^c \right) = \bigcup_{k=1}^n \mathbb{R} \setminus U_k.$$

is finite union of countable sets, so it is countable.

Now, the set  $\mathcal{T}$  satisfies the Axiom of Topology, it is a Topology.

b). For each  $n \in \mathbb{N}$ , define  $U_n = ([0, 1] \setminus \{1/n\}) \cup \{1/n\}$  where  $K = \{1/n \mid n \in \mathbb{N}\}$ .

Then,  $[0, 1] \setminus U_n = ([0, 1] \cap K) \setminus \{1/n\}$ , countable, so each  $U_n$  is open in  $[0, 1]$ .

Moreover,  $\bigcup_{n=1}^{\infty} U_n = [0, 1]$ , but  $\{U_n \mid n \in \mathbb{N}\}$  has no finite subcover,

because: if there is a finite subcover

$$\{U_{n_1}, \dots, U_{n_k}\} \text{ for } [0, 1],$$

then for  $m = \max\{n_1, \dots, n_k\}$ ,

$$U_{n_1} \cup \dots \cup U_{n_k} \subseteq \bigcup_{n=1}^m U_n \subsetneq [0, 1],$$

so  $[0, 1]$  is not compact.  $\square$

2022. Fall.

12. (10 pts)

- (a) Let  $X$  be a topological space and let  $Z \subset X$  be a subspace with the induced topology. Suppose that  $Z$  is connected. Determine if its closure  $\bar{Z}$  is connected or not. (Explain your answer.)
- (b) Show that, for  $n \geq 1$ , the standard sphere  $S^n \subset \mathbb{R}^{n+1}$  is connected.

Proof.

a). Suppose that the closure  $\bar{Z}$  is disconnected, with separation  $\{U, V\}$ , such that  $U, V \subset X$  are open sets such that  $U \cap \bar{Z} \neq \emptyset$ ,  $V \cap \bar{Z} \neq \emptyset$ ,  $U \cap V \cap \bar{Z} = \emptyset$ ,  $\bar{Z} \subseteq U \cup V$ .

Want to show that:  $\{U, V\}$  is also separation for  $Z$

First, if  $U \cap Z = \emptyset$ , then  $Z \subseteq U^c$ . But, since  $U^c$  is closed and  $\bar{Z}$  is the smallest closed set containing  $Z$ , so  $\bar{Z} \subseteq U^c$ . But it equals to  $\bar{Z} \cap U = \emptyset$ , it contradicts to the assumption so  $U \cap Z \neq \emptyset$ , and  $V \cap Z \neq \emptyset$ . Similarly, Now, since  $Z \subseteq \bar{Z}$ ,

$$U \cap V \cap Z \subseteq U \cap V \cap \bar{Z} = \emptyset \Rightarrow U \cap V \cap Z = \emptyset$$

and  $Z \subseteq \bar{Z} \subseteq U \cup V$ , as a result,  $\{U, V\}$  is the separation for  $Z$ , it is contradiction to the hypothesis. So  $\bar{Z}$  is connected.

b). Since  $S^n$  is one-point compactification of  $\mathbb{R}^n$ , so,  $S^n \setminus \{*\} \cong \mathbb{R}^n$ . Since  $\mathbb{R}^n$  is connected, so  $S^n = \overline{S^n \setminus \{*\}}$  is also connected by a).

12022. Fall.

18. Let  $S$  be the subset of  $\mathbb{R}^3$  given by  $\{(x, y, z) \mid z^2 = x^2 + y^2, z \geq 0\}$ .

Show that  $S$  is not regular surface.

Proof. Suppose that  $S$  is a regular surface. Then, for  $(0, 0, 0) \in S$ , there exists a neighbor  $V$  containing  $(0, 0, 0)$  such that  $V \cap S$  is a graph of

$$z = f(x, y) \quad \text{or} \quad y = g(x, z) \quad \text{or} \quad z = h(y, z)$$

where  $f, g, h$  are differentiable. Since projection of  $S$  into  $xz$ ,  $yz$ -plane are not 1-1,  $g, h$  cannot be such a map.

Meanwhile,  $f(x, y) = \sqrt{x^2 + y^2}$  at locally  $(0, 0, 0)$ , but  $f$  is not differentiable at  $(x, y) = (0, 0)$ , so it cannot be possible. Therefore  $S$  is not regular.

(More specifically,  $(0, \pm \epsilon, \sqrt{1 - \epsilon^2}) \in S$  for any  $\epsilon > 0$ , so there is no 1-1  $yz$ -projection, similarly  $xz$ .)

2023 spring

- 1). Let  $A = \begin{bmatrix} 0 & -3 & 7 \\ -1 & -1 & 5 \\ -1 & -2 & 6 \end{bmatrix}$
- a). Find the minimal polynomial of  $A$ .  
b). " Jordan form "  
c). " rational form "

proof.

a). First, the characteristic polynomial  $\chi(t)$

\*:  $\text{tr} = \text{sum eigen.}$

$$\begin{aligned} \det \begin{bmatrix} t & 3 & -7 \\ 1 & t+1 & -5 \\ 1 & 2 & t-6 \end{bmatrix} &= t \cdot [(t+1)(t-6) + 10] - 3 \cdot [t-6+5] - 7 \cdot [2-1-t] \\ &= t \cdot [t^2 - 5t + 4] - 3 \cdot [t-1] + 7[t-1] \\ &= t \cdot [t-4] - [t-1] + 4[t-1] \\ &= (t-1)(t-2)^2. \end{aligned}$$

Since the minimal polynomial have same roots to  $\chi$ , so we need to check that  $(A-I)(A-2I) = 0$ . (If not, then Cayley-Hamilton theorem guarantees that  $\chi$  is the minimal polynomial.)

$$(A-I) \cdot (A-2I) = \begin{bmatrix} -1 & -3 & 7 \\ -1 & -2 & 5 \\ -1 & -2 & 5 \end{bmatrix} \cdot \begin{bmatrix} -2 & -3 & 7 \\ -1 & -3 & 5 \\ -1 & -2 & 4 \end{bmatrix} \neq 0$$

$$\text{So, } \chi = (t-1)(t-2)^2.$$

b).  $\dim \ker(A-I) = 1$ , since  $0 < \dim \ker(A-I) \leq \deg(t-1) = 1$ ,

and  $\dim \ker(A-2I) \leq 2 = \deg(t-2)^2$ . Meanwhile,

$\begin{bmatrix} 2 & -3 & 7 \\ -1 & -3 & 5 \\ -1 & -2 & 4 \end{bmatrix}$  has more than one linearly independent row vectors, so the nullity is 1. So, the Jordan form is

$$J_A = \begin{bmatrix} 1 & & \\ & J_2 & \\ & & 2 \end{bmatrix}$$

c). The minimal polynomial is equal to the characteristic, so  $A$  is cyclic, thus the rational form is

$$P_A = \begin{bmatrix} 0 & 0 & 4 \\ 1 & 0 & -8 \\ 0 & 1 & 5 \end{bmatrix}.$$

2023. Spring

4. Let  $V = \{\text{id}, (12)(34), (13)(24), (14)(23)\} \subseteq S_4$ , 57,

a). Show that  $V$  is normal subgroup.

b).  $S_4/V \cong S_3$ ,

a). Since conjugate cannot change the order of elements, and cannot change the parity: since the sign function is homomorphism,

$$\text{sgn}(\sigma \tau \sigma^{-1}) = \text{sgn}(\sigma) \cdot \text{sgn}(\sigma^{-1}) \cdot \text{sgn}(\tau) = \text{sgn}(\tau), \text{ for any } \sigma, \tau \in S_n.$$

Now, the fact that  $V$  contains all order two even permutations in  $S_4$ , so every conjugate of  $V$  is contained in  $V$ , so normal.  $\downarrow$  5m

b). Consider  $S_3$  is a subgroup of  $S_4$ , the number 4 fixed. Then,

$$\varphi: S_3 \rightarrow S_4/V: \sigma \mapsto \sigma V$$

is well-defined, and given  $\sigma_1, \sigma_2 \in S_3 \leq S_4$ ,

$$\varphi(\sigma_1 \sigma_2) = \sigma_1 \sigma_2 V = \sigma_1 V \sigma_2 V = \varphi(\sigma_1) \varphi(\sigma_2)$$

so it's homomorphism. And, since all elements in  $S_3$  fix 4,  $S_3 \cap V = \{\text{id}\}$ , so  $\ker \varphi = \{\text{id}\}$ . Now, since  $|S_3| = 6 = |\ker \varphi| \cdot |\text{Im } \varphi|$ ,  $|\text{Im } \varphi| = 6$ , so  $\text{Im } \varphi = S_4/V$ , since  $|S_4/V| = 6$ , so  $\varphi$  is onto.

Now, the first isomorphism theorem gives  $S_3 \cong S_4/V$ .  $\square$

2023 Spring

5. Let  $I = (2, x^2 + x + 1) \subseteq \mathbb{Z}[x]$  be an ideal

show that  $I$  is maximal ideal and give a multiplication table for  $\mathbb{Z}[x]/I$ .

Proof. Define a map

$$\pi: \mathbb{Z}[x] \rightarrow \mathbb{Z}/2\mathbb{Z}[x]; \sum_{n=0}^{\infty} a_n x^n \mapsto \sum_{n=0}^{\infty} (a_n \bmod 2) x^n$$

Then,  $\pi$  is an onto Homomorphism, ( $\pi(0) = 0$ ,  $\pi(1) = 1$ ) with

$\ker \pi = 2\mathbb{Z}[x] = (2)$ , so  $\mathbb{Z}[x]/\ker \pi = \mathbb{Z}[x]/2\mathbb{Z}[x] \cong (\mathbb{Z}/2\mathbb{Z})[x]$ .

Meanwhile,  $x^2 + x + 1$  is irreducible over  $\mathbb{Z}/2\mathbb{Z}$ , because it is of degree 2 and  $0^2 + 0 + 1 = 1 \neq 0$ ,  $1^2 + 1 + 1 = 1 \neq 0$  in  $\mathbb{Z}/2\mathbb{Z}$ .

Now, Consider the natural Homomorphism

$$\phi: \mathbb{Z}/2\mathbb{Z}[x] \rightarrow (\mathbb{Z}/2\mathbb{Z}[x])/(x^2 + x + 1); p(x) \mapsto p(x) + (x^2 + x + 1)$$

It is trivially onto, so  $\phi \circ \pi$  is onto Homomorphism.

Moreover,

$$\ker(\phi \circ \pi) = (\phi \circ \pi)^{-1}[\{0\}] = \pi^{-1}[\phi^{-1}[\{0\}]] = \pi^{-1}[\ker \phi] = \pi^{-1}[(x^2 + x + 1)]$$

Now,

$$f \in \pi^{-1}[(x^2 + x + 1)] \Leftrightarrow \pi(f) \in (x^2 + x + 1)$$

$$\Leftrightarrow \pi(f) = (x^2 + x + 1) \cdot \pi(g) \text{ for some } g(x) \in \mathbb{Z}[x]$$

$$\Leftrightarrow \pi(f - (x^2 + x + 1)g) = 0$$

$$\Leftrightarrow f - (x^2 + x + 1)g \in (2)$$

$$\Leftrightarrow f \in (2) + (x^2 + x + 1) = (2, x^2 + x + 1).$$

So,  $\mathbb{Z}[x]/\ker(\phi \circ \pi) = \mathbb{Z}[x]/(2, x^2 + x + 1) \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$ .

And, since  $\mathbb{Z}/2\mathbb{Z}$  is field and  $x^2 + x + 1$  is irreducible over  $\mathbb{Z}/2\mathbb{Z}$ , so  $(\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$  is a field, so, the quotient is field if and only if  $(x^2 + x + 1)$  is maximal, so proved.

Since  $\mathbb{Z}[x]/I \cong (\mathbb{Z}/2\mathbb{Z})[x]/(x^2 + x + 1)$ , Let  $\alpha = x + (x^2 + x + 1)$ .

Then  $\alpha^2 = x^2 + (x^2 + x + 1) = x + 1 + (x^2 + x + 1) = x + 1$ , so

|          |          |          |
|----------|----------|----------|
|          | 1        | $\alpha$ |
| 1        | 1        | $\alpha$ |
| $\alpha$ | $\alpha$ | $x + 1$  |

is the table,

Generalized in the next page.

Let  $I = (a, f) \subseteq R[x]$  be an ideal generated by  $a, f \in R[x]$ .

Consider the diagram:

$$\begin{array}{ccccc} R[x] & \xrightarrow{\mathcal{Q}} & (R/aR)[x] & \xrightarrow{\pi_f} & (R/aR)[x]/(\overline{f}) \\ & \searrow \pi & \cong \nearrow & & \\ & & R[x]/aR[x] & & \end{array}$$

$$\begin{array}{ll} R[x] \rightarrow R[x]/aR[x] & R[x]/aR[x] \rightarrow (R/aR)[x] \\ p(x) \mapsto p(x) + aR[x] & p(x) + aR[x] \mapsto \overline{p(x)}, \text{ where } \overline{\sum a_n \cdot x^n} = \sum (a_n + aR) \cdot x^n. \\ \text{Onto Homomorphism.} & \text{Isomorphism.} \end{array}$$

$\Rightarrow \mathcal{Q}: R[x] \rightarrow (R/aR)[x]$  is Onto Homomorphism, so is  $\pi_f \circ \mathcal{Q}$  onto Homomorphism. Moreover,

$$\begin{aligned} \ker(\pi_f \circ \mathcal{Q}) &= (\pi_f \circ \mathcal{Q})^{-1}[\{0\}] = \mathcal{Q}^{-1}[\ker \pi_f] \\ &= \{p(x) \in R[x] \mid \overline{p(x)} \in \ker \pi_f\} = \{p(x) \in R[x] \mid \overline{p(x)} \in (\overline{f(x)})\} \\ &\stackrel{(*)}{=} \{p(x) \in aR[x] + f(x)R[x]\} = (a) + (f) = (a, f) = I. \end{aligned}$$

Assume  $a \in R$  and  $(a) \subseteq R$  is maximal, and  $\overline{f} \in (R/aR)[x]$  is irreducible.

Then,  $R/aR$  is a field, and therefore  $(R/aR)[x]$  is P.I.D.

So,  $(\overline{f})$  is maximal, it implies that  $(R/aR)[x]/(\overline{f})$  is a field.

Now, by the first-isomorphism theorem,

$$R[x]/(aR[x] + fR[x]) \cong (R/aR)[x]/\overline{f} \cdot (R/aR)[x]$$

Note: (\*) is obtained by: given  $p(x) \in R[x]$ ,

$$\overline{p(x)} \in (\overline{f(x)}) \Leftrightarrow \overline{p(x)} = \overline{g(x) \cdot f(x)} \text{ for some } g(x) \in R[x]$$

$$\Leftrightarrow p(x) - g(x)f(x) \in aR[x] \text{ for some } g(x) \in R[x]$$

Another Solution.

Observe that  $\mathbb{Z}[x]/(2, x^2+x+1)$ .

Given  $f(x) \in \mathbb{Z}[x]$ , since the leading coefficient of  $x^2+x+1$  is a unit, so we can apply the division Algorithm.

$$f(x) = g(x) \cdot (x^2+x+1) + r(x) \text{ for some } g, r \in \mathbb{Z}[x] \text{ with } \deg r \leq 1.$$

Therefore,  $\mathbb{Z}[x]/(2, x^2+x+1) = \{ax+b + (2, x^2+x+1) \mid a, b \in \mathbb{Z}\}$

And, for any  $a, b \in \mathbb{Z}$ ,  $ax+b = (a \bmod 2) \cdot x + (b \bmod 2) + 2nx + 2m$  for some  $n, m \in \mathbb{Z}$ , so

$$= \{ax+b + (2, x^2+x+1) \mid a, b \in \{0, 1\}\}$$

And,  $1 \notin (2, x^2+x+1)$  and  $x \notin (2, x^2+x+1)$  and  $x+1 \notin (2, x^2+x+1)$  implies these are distinct with zero, and also these are mutually distinct, just simple calculation. Let  $\alpha = x+1$ .

Now,  $\mathbb{Z}[x]/I$  has four elements,  $\{0, 1, \alpha, \alpha+1\}$  where  $\alpha^2+\alpha+1=0$ , so the table

|            | 0 | 1          | $\alpha$   | $\alpha+1$ |
|------------|---|------------|------------|------------|
| 0          | 0 | 0          | 0          | 0          |
| 1          | 0 | 1          | $\alpha$   | $\alpha+1$ |
| $\alpha$   | 0 | $\alpha$   | $\alpha+1$ | 1          |
| $\alpha+1$ | 0 | $\alpha+1$ | 1          | $\alpha$   |

'is obtained. Since every non-zero element has an inverse, it is a field.

Comment. Multiplication table 요구한 부분에서 이게 다 풀지 키(오)기 가깝습니다?

2023, Spring.

7. Let  $f: \mathbb{C} \rightarrow \mathbb{C}$  be an entire satisfying 10m

$$U^2(z) - 2 \cdot V^2(z) = 1 \text{ for all } z \in \mathbb{C}$$

Prove that  $f$  is constant.

Proof. Observe that:  $f(z) = 0 \Leftrightarrow U(z) = V(z) = 0$

But,  $U^2 - 2V^2 = 1$ , so  $f(z) \neq 0$  for all  $z \in \mathbb{C}$ . Now,

$$\frac{1}{f^2(z)} = \frac{1}{U^2(z) + V^2(z)} = \frac{1}{1 + 3V^2(z)} \leq 1,$$

therefore by the Liouville's theorem, the bounded analytic function  $\frac{1}{f^2}$  is constant, so  $f$  is also constant function.  $\square$

$$\text{Or, } |f^2| = U^2 + V^2 = 1 + 3V^2 \Rightarrow \frac{1}{f^2} \leq \frac{1}{1 + 3V^2} \leq 1.$$

observe that ~~differentiating gives~~ that

$$2U U_x - 2V V_x = 0 \Rightarrow U U_x = V V_x = V \cdot U_y$$

2023 Spring.

8. (10 pts) Let  $C$  be a closed subset of  $\mathbb{R}$  and let  $F: C \rightarrow C$ . Suppose that there exists  $\lambda \in (0, 1)$  such that

$$|F(x) - F(y)| \leq \lambda |x - y| \text{ for all } x, y \in C.$$

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We say that  $x_0 \in C$  is a fixed point of  $F$  if it satisfies  $F(x_0) = x_0$ .

- (a) Let  $(x_n)_{n=1}^{\infty}$  be a sequence in  $C$  such that  $x_{n+1} = F(x_n)$  for all  $n \in \mathbb{N}$ . Prove that the sequence  $(x_n)_{n=1}^{\infty}$  is convergent and the limit is a fixed point of  $F$ .  
(b) Prove that  $F$  has a unique fixed point.

Proof. a). Given  $n \in \mathbb{N}$ ,

$$\begin{aligned} |x_{n+1} - x_n| &= |F(x_n) - F(x_{n-1})| \leq \lambda |x_n - x_{n-1}| = \lambda |F(x_{n-1}) - F(x_{n-2})| \leq \dots \\ &\leq \lambda^n |x_1 - x_0| \end{aligned}$$

Now, for any fixed  $N \in \mathbb{N}$ ,  $m > n \geq N$  implies that

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \dots + |x_{n+1} - x_n| \\ &\leq (\lambda^{m-1} + \lambda^{m-2} + \dots + \lambda^n) \cdot |x_1 - x_0| \\ &\leq \sum_{n=N}^{\infty} \lambda^n \cdot |x_1 - x_0| \rightarrow 0 \text{ as } N \rightarrow \infty \end{aligned}$$

Therefore,  $\{x_n\}$  is a Cauchy sequence in  $\mathbb{R}$ , so  $x_n$  converges, and since  $C$  is closed,  $x_n \rightarrow x$  for some  $x \in C$ . Moreover, given  $\varepsilon > 0$ ,

$$|x - y| < \frac{\varepsilon}{\lambda} \Rightarrow |F(x) - F(y)| < \frac{\varepsilon}{\lambda} \cdot \lambda = \varepsilon, \text{ so } F \text{ is continuous, thus}$$

$$F(y) = F\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} F(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x$$

So, the limit  $x$  is the fixed point we desired.  $\square$  3m

b). Let  $x, y \in C$  be fixed points of  $F$ . Then

$$|F(x) - F(y)| = |x - y| \leq \lambda |x - y|$$

Since  $\lambda < 1$ , so  $x = y$ . Thus the fixed point is unique.  $\square$  3m

2023 Spring

10. Let  $f: [0, \infty) \rightarrow [0, \infty)$  be a continuous function satisfying

$$\int_0^{\infty} f(x) dx < \infty.$$

a). Prove or disprove that  $\lim_{x \rightarrow \infty} f(x) = 0$ .

b). If  $f$  is continuous uniformly then  $\lim_{x \rightarrow \infty} f(x) = 0$ ? Prove or disprove.  
Proof.

a). Define a map on  $[0, \infty)$

$$f_n(x) = \begin{cases} 0 & \text{otherwise} \\ n^2(x-n) & \text{if } x \in [n, n + \frac{1}{n^2}] \\ n^2(n + \frac{2}{n^2} - x) & \text{if } x \in [n + \frac{1}{n^2}, n + \frac{2}{n^2}] \end{cases}$$

Then  $\int_0^{\infty} f_n = \frac{1}{n}$  and  $f_n$  is continuous, for each  $n \in \mathbb{N}$ .

Let  $f = f_1 + f_2 + \dots + f_n + \dots$ . Then  $f$  is continuous and

$$\int_0^{\infty} f = \sum_{n=1}^{\infty} \frac{1}{n} < \infty.$$

(The continuity is obtained just investigate each  $[n, n + \frac{2}{n^2}]$ )

But, for  $\epsilon = \frac{1}{2}$ , for any  $\delta > 0$ , there is  $x > \delta$  such that  $|f(x) - 0| = 1 \geq \frac{1}{2}$ .  
So  $\lim_{x \rightarrow \infty} f(x) \neq 0$ .

b). Suppose  $\lim_{x \rightarrow \infty} f(x) = 0$  is false but  $f$  is continuous uniformly.

Let  $\epsilon = 2$ . Then, for any  $n \in \mathbb{N}$ , there is  $x_n > n$  such that  $f(x_n) \geq 2$ .

Since  $f$  is uniformly continuous, take  $\delta > 0$  such that  $|x - y| < \delta$  implies  $|f(x) - f(y)| < 1$ . Then, for each  $n \in \mathbb{N}$ ,

$$\int_{x_n}^{x_n + \delta} f(x) dx \geq \delta \cdot 1$$

So,  $\int_0^{\infty} f \geq \sum_{n=1}^N \int_{x_n}^{x_n + \delta} f \geq \delta \cdot N$  diverges, So contradicts.

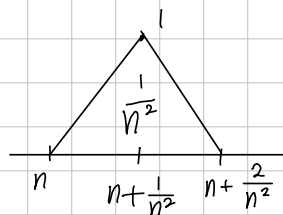
Thus  $\lim_{x \rightarrow \infty} f = 0$ .  $\square$

Revised in the Next.

Let  $f: [0, \infty) \rightarrow \mathbb{R}$  be non-negative valued continuous function,  
does not Converges to Zero,  
but the 'improper integral' Converges.

Find and Convince Concretely, and change the Condition as  $f$  Continuous Uniformly.

Define  $f_n(x) = \begin{cases} n^2(x-n) & \text{if } x \in [n, n + \frac{1}{n^2}] \\ n^2(n + \frac{2}{n^2} - x) & \text{if } x \in [n + \frac{1}{n^2}, n + \frac{2}{n^2}] \\ 0 & \text{otherwise.} \end{cases}$



Then,  $f_n$  is continuous, with  $\int_0^\infty f_n = \frac{1}{n^2}$ .

Now, let  $f = \sum_{n=2}^\infty f_n$ . Since the interval  $[n, n + \frac{2}{n^2}]$  are disjoint mutually,  $n \geq 2$ ,  
so the  $f$  is well-defined, but it does not uniformly converges, because

$$\sup_{x \geq 0} \left| f(x) - \sum_{n=2}^N f_n(x) \right| = 1 \text{ for any } N \in \mathbb{N}.$$

Nevertheless,  $\int_0^\infty f = \int_0^\infty \sum_{n=2}^\infty f_n = \sum_{n=2}^\infty \int_0^\infty f_n = \sum_{n=2}^\infty \frac{1}{n^2}$ , the integral Converges.

But, for each  $n \in \mathbb{N}$ ,  $f(n + \frac{1}{n^2}) = 1$ , so  $f(x)$  does not tends to zero as  $x \rightarrow \infty$ .

Assume that  $f(x)$  is Continuous Uniformly

If  $f$  does not Converges to Zero, then there is  $\varepsilon > 0$  such that  
for each  $R > 0$ , there is  $x \geq R$  such that  $f(x) \geq \varepsilon$ .

Now, let  $\delta > 0$  such that  $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \frac{\varepsilon}{2}$ .

And, let  $\{x_n\}$  be a Positive Sequence such that for all  $n \in \mathbb{N}$ ,

$$x_{n+1} > x_n + 2\delta \text{ and } f(x_n) \geq \varepsilon.$$

Then,  $\int_0^{x_n + \delta} f \geq \sum_{k=2}^n \frac{\varepsilon}{2} \cdot 2\delta = n\varepsilon\delta \rightarrow \infty$  as  $n \rightarrow \infty$ ,

so it Contradicts to  $\int_0^\infty f < \infty$ .  $\square$

2023, Spring

11. (10 pts) Prove or disprove the following statements.

- (a) For a metric space  $X$ , a subspace  $A \subset X$  and a point  $x \in X$ , there is a sequence of points of  $A$  converging to  $x$  if and only if  $x$  is in the closure  $\bar{A}$  of  $A$ .
- (b) Any two metrics on  $\mathbb{R}^n$  induce the same metric topology.

Proof.

a). It is true, because that:

Suppose that there is a sequence  $\{a_n\}$  in  $A$  converges to  $x \in X$ .

Then, for any  $\epsilon > 0$ ,  $B_\epsilon(x) \cap A \neq \emptyset$ , because since  $a_n \rightarrow x$ , so for some  $N \in \mathbb{N}$ ,  $n \geq N \Rightarrow a_n \in B_\epsilon(x)$ . Thus  $x \in \bar{A}$ . Conversely, let  $x \in \bar{A}$ .

Then, for each  $n \in \mathbb{N}$ ,  $B_{1/n}(x) \cap A \neq \emptyset$ , so we can construct a sequence  $\{a_n\}$  such that  $a_n \in B_{1/n}(x) \cap A$  for each  $n \in \mathbb{N}$ . Now, the  $\{a_n\}$  is a sequence in  $A$  and it converges to  $x$ .

b). Define  $d: \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}: (x, y) \mapsto \begin{cases} 1 & x \neq y \\ 0 & x = y. \end{cases}$

Then,  $d(x, x) = 0 \Leftrightarrow x = x$  and  $d(x, y) = d(y, x)$  for all  $x, y \in \mathbb{R}^n$ , moreover given  $x, y, z \in \mathbb{R}^n$ ,

$$d(x, z) \leq d(x, y) + d(y, z), \text{ by just casework}$$

(If  $d(x, z) = 0$ , then we are done, and if  $d(x, z) = 1$ , then either  $d(x, y) = 1$  or  $d(y, z) = 1$ , so we are done)

This  $d$  generates the discrete topology, but

the euclidean metric is not discrete, so it is false.  $\square$

2023, Spring

12. (15 pts) Let  $X_n$  be the real projective space  $\mathbb{R}P^n$ , i.e. the quotient space of  $\mathbb{R}^{n+1} \setminus \{(0, \dots, 0)\}$  where the equivalence relation is defined by

$$(x_1, \dots, x_{n+1}) \sim (y_1, \dots, y_{n+1}) \Leftrightarrow \exists t \in \mathbb{R} \setminus \{0\} \text{ such that } (x_1, \dots, x_{n+1}) = t(y_1, \dots, y_{n+1}).$$

(a) Show that  $X_1$  is homeomorphic to the circle  $S^1$ .

(b) For a topological space  $X$  and  $A \subset X$ , a retraction of  $X$  to  $A$  is a continuous map  $r: X \rightarrow A$  such that the restriction of  $r$  to  $A$  is the identity. Remove a point from  $X_2$  and consider its retraction to a circle in  $X_2$ . (You may assume that such a retraction exists.) Using the retraction, show that  $X_2$  is not homeomorphic to the 2-sphere  $S^2$ .

Proof. a). In  $X_1$ , that is, we are in the  $\mathbb{R}^2$ -plane. In  $\mathbb{R}^2 \setminus \{(0,0)\}$ , we can correspond  $(x_1, x_2) = r \cdot e^{i\theta}$ , so that for  $(x_1, x_2) = r_1 e^{i\theta_1}$ ,  $(y_1, y_2) = r_2 e^{i\theta_2}$ ,  
 $(x_1, x_2) \sim (y_1, y_2) \Leftrightarrow e^{i(\theta_1 - \theta_2)} = \pm 1 \Leftrightarrow \theta_1 - \theta_2 = k\pi$  for some  $k \in \mathbb{Z}$ .

Define a map

$$f: \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow S^1, r e^{i\theta} \mapsto e^{2i\theta}$$

Then,  $f$  is continuous onto map, being composition of continuous maps and for  $r=1$ ,  $0 \leq \theta \leq \pi$  fills the codomain. Moreover,

$$\theta_1 = \theta_2 + k\pi \text{ for some } k \in \mathbb{Z} \Leftrightarrow e^{2i\theta_1} = e^{2i\theta_2 + 2ik\pi} \text{ for some } k \in \mathbb{Z} \\ \Leftrightarrow f(r_1 e^{i\theta_1}) = f(r_2 e^{i\theta_2})$$

therefore,  $f$  induces a homeomorphism,

$$\tilde{f}: \mathbb{R}P^1 \rightarrow S^1, [(x_1, x_2)] \mapsto f(x_1, x_2).$$

Which is well-defined and injective by (a), onto since  $f$  is onto, and  $(\tilde{f})^{-1} \circ f = \pi: \mathbb{R}^2 \setminus \{(0,0)\} \rightarrow X_1$  which is a continuous map, so  $\tilde{f}^{-1}$  is conti. Consequently  $\mathbb{R}P^1 \cong S^1$ .

b). Let  $r: X_2 \setminus \{p\} \rightarrow A \subset X_2 \setminus \{p\}$  be the restriction.

2023 Spring.

13. Let  $\alpha$  be a geodesic on the unit sphere  $S^2 = \{(x, y, z) \mid x^2 + y^2 + z^2 = 1\}$ . Prove that  $\alpha$  is contained in a plane.

Proof. Let  $\alpha: I \rightarrow S^2$  be the geodesic. Then, the normal vector of  $S^2$  at  $\alpha(t)$  is  $\alpha(t)$ . And, since  $\alpha$  is geodesic, by definition,

$$[N, T, T'] = [\alpha(t), \alpha'(t), \alpha''(t)/|\alpha'(t)|] = \left\langle \underbrace{\alpha \times \alpha'}_{\vec{T} \times \vec{N}}, \underbrace{\alpha''}_{\vec{N}} \right\rangle / |\alpha'(t)| = 0$$

Meanwhile, the torsion of  $\alpha$  is

$$\tau(t) = \langle B', N \rangle = \left\langle \alpha' \times \alpha'' \cdot \frac{1}{|\alpha'|}, \alpha'' \cdot \frac{1}{|\alpha'|} \right\rangle = [\alpha']$$

$$\begin{matrix} T \times N \\ (\alpha' \times \alpha'') \cdot \frac{1}{|\alpha'|} \end{matrix}$$

$$\begin{pmatrix} & \kappa & \\ -\kappa & & \tau \\ & -\tau & \end{pmatrix} \quad B',$$

$$B' = -\tau N$$

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1. Suppose that  $\{a_n\}$  and  $\{b_n\}$  are bounded sequences of real numbers.

a). prove that

$$\limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n.$$

b). Give an example for the equality does not hold, prove

(a). Since for each  $n \in \mathbb{N}$ ,

$$\sup_{k \geq n} (a_k + b_k) \leq \sup_{k \geq n} a_k + \sup_{k \geq n} b_k,$$

taking limit to the both sides we obtain the result.

b). Let  $a_n = (-1)^n$  and  $b_n = (-1)^{n+1}$ . Then,

$$\limsup (a_n + b_n) = 0$$

but

$$\limsup a_n = \limsup b_n = 1.$$

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2. Find all solutions to the ODE

$$xy' + y = x^2 - y^2 \quad \text{and} \quad y(1) = 1.$$

Proof. Since  $y(1) = 1 \neq 0$ ,  $y > 0$  for some neighbor of  $x=1$ . In this neighbor,

$$xy' + y = x^2 - y^2 \Rightarrow \frac{1}{x} \cdot \frac{1}{y^2} \cdot y' + \frac{1}{x^2} \cdot \frac{1}{y} = 1$$

$$\Rightarrow \left( \frac{1}{x} \cdot \frac{1}{y} \right)' = -1$$

$$\Rightarrow \frac{1}{xy} = -x + C$$

$$\Rightarrow \frac{-1}{x(x-C)} = y$$

$$\text{Since } y(1) = 1, \quad \frac{1}{C-1} = 1 \Rightarrow C = 2.$$

$$\text{Finally, } y(x) = \frac{1}{x(2-x)} \quad \text{for some neighbor of } x=1.$$

is the solution of the initial value problem, we desired.

Moreover, by the uniqueness theorem, it is the unique solution through  $(1, 1) \in \mathbb{R}^2$ .

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4. Find all  $\alpha > 0$  such that

$$\int_0^{\infty} \cos(x^\alpha) dx$$

exists as a real number.

Proof.

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6. Let  $A \in M_{n \times n}(\mathbb{R})$  be given. Put  $m_A(t) \in \mathbb{R}[t]$  be the minimal polynomial and  $\chi_A(t) \in \mathbb{R}[t]$  be the characteristic polynomial.

a). Does there exist a 3 by 3 matrix  $A$  such that

$$m_A(t) = t^2 - t \quad \text{and} \quad \chi_A(t) = t^3 - t^2 ?$$

b). If  $C \in M_{5 \times 5}(\mathbb{R})$  such that  $C^5 = I_5$ , is it necessarily true that

$$m_C(t) = t^5 - 1 ?$$

Proof.

a). Let  $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ . Then,  $\chi_A(t) = t^2(t-1) = t^3 - t^2$ .  
And,  $A \cdot (A - I) = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \cdot \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ ,  
so  $m_A(t) = t(t-1)$ .

Therefore, exists.

b). if  $C = I_5$ , then  $C^5 = I_5$  but the minimal polynomial is  $t - 1$ .  
So, not necessary.  $\square$

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7. Let  $K = \mathbb{F}_{p^2}$  be a finite field of order of  $p^2$ .

a), Find the number of all  $3 \times 3$  invertible matrices over  $K$ .

b), Let  $V$  be a 5-dimensional vector space over  $K$  and  $W \subset V$  be a 2-dimensional subspace. Find the number of all  $K$ -linear maps

$$T: V \rightarrow V \text{ such that } T[W] \subseteq W.$$

Proof a), Since  $A \in M_{3 \times 3}(K)$  is invertible if and only if the rows of  $A$  are linearly independent, so the possible cases are

$$\underbrace{[(p^2)^3 - 1]}_{\text{Choose first row except zero}} \cdot \underbrace{[(p^2)^3 - p^2]}_{\text{Choose second row except multiple of the first}} \cdot \underbrace{[(p^2)^3 - p^2 \cdot p^2]}_{\text{similar}}$$

$$\text{so, } |GL_3(K)| = (p^6 - 1)(p^6 - p^2)(p^6 - p^4).$$

b), Let  $\{\alpha_1, \alpha_2\}$  be the basis for  $W$ .

Since  $T[W] \subseteq W \Leftrightarrow T(\alpha_1), T(\alpha_2) \in W$ , so  $T[W] \subseteq W$  iff

$$T(\alpha_1) = a \cdot \alpha_1 + b \cdot \alpha_2 \text{ and } T(\alpha_2) = c \cdot \alpha_1 + d \cdot \alpha_2 \text{ for some } a, b, c, d \in K$$

And, let  $\{\alpha_1, \alpha_2, \dots, \alpha_5\}$  be the extended basis for  $V$ .

$T(\alpha_i) = \sum_{j=1}^5 a_{ij} \alpha_j$  for  $i \geq 3$ , so the number of such a  $T$  is

$$p^2 \cdot p^2 \cdot p^2 \cdot p^2 \cdot ((p^2)^5)^3.$$

2023, Fall.

14. If  $X$  is Lindelöf and  $Y$  is compact, then  $X \times Y$  is Lindelöf. Prove.

Proof. Let  $\mathcal{O}$  be the open cover for  $X \times Y$ . Let  $x_0 \in X$  be fixed. Since  $\{x_0\} \times Y \rightarrow Y : (x_0, y) \mapsto y$  is homeomorphism,  $\{x_0\} \times Y \cong Y$  is compact.

Since  $\mathcal{O}$  covers  $X \times Y$ , it also covers  $\{x_0\} \times Y$ , so there is a finite subcover  $F_0 \subset \mathcal{O}$ , which satisfies  $\{x_0\} \times Y \subset \bigcup F_0 \subset X \times Y$ .

Given  $y \in Y$ ,  $(x_0, y) \in \bigcup F$ , open, so there exist open sets  $U_y \times V_y \subset X \times Y$  such that  $(x_0, y) \in U_y \times V_y \subset \bigcup F$ . The compactness gives that the open cover  $\{U_y \times V_y \mid y \in Y\}$  of  $\{x_0\} \times Y$  has a finite subcover  $\{U_{y_1} \times V_{y_1}, \dots, U_{y_n} \times V_{y_n}\}$ . Let  $W_0 = U_{y_1} \cap \dots \cap U_{y_n}$ .

Then,  $x_0 \in W_0$  since  $x_0 \in U_{y_i}$  for all  $1 \leq i \leq n$ , so that  $\{x_0\} \times Y \subset W_0 \times Y$ , and  $W_0$  is open, being finite intersection of open sets. Moreover,

$$\begin{aligned} W_0 \times Y &\subset \bigcup F, \text{ because } (x, y) \in W_0 \times Y \Leftrightarrow x \in U_{y_1} \cap \dots \cap U_{y_n} \text{ and } y \in V_{y_k} \exists k, \\ &\Rightarrow (x, y) \in U_{y_k} \times V_{y_k} \exists k, \\ &\Rightarrow (x, y) \in \bigcup F. \end{aligned}$$

Since  $x_0 \in X$  was arbitrary, we can consider open cover  $\{W_x \mid x \in X\}$  for  $X$ , and since  $X$  is Lindelöf, it has a countable subcover  $\{W_{x_k} \mid k \in \mathbb{N}\}$ .

$$\text{So, } X \times Y = \left( \bigcup_{k \in \mathbb{N}} W_{x_k} \right) \times Y = \bigcup_{k \in \mathbb{N}} W_{x_k} \times Y \subset \bigcup_{k \in \mathbb{N}} \bigcup F_k,$$

where  $F_k \subset \mathcal{O}$  is the finite subcover determined when  $x_k \in X$  chosen, thus  $\mathcal{O}$  has a countable subcover.  $\square$

Debris

Let  $\mathcal{O} = \{U_\lambda \mid \lambda \in \Lambda\}$  be an open cover of  $X \times Y$ .

Given  $x_0 \in X$ , since the projection  $\{x_0\} \times Y \rightarrow Y : (x_0, y) \mapsto y$  is homeomorphism, so  $\{x_0\} \times Y \cong Y$  is compact. Therefore, there is a finite subcollection

$$\mathcal{O}_{x_0} \subset \mathcal{O} \text{ satisfying } \{x_0\} \times Y \subset \bigcup \mathcal{O}_{x_0}.$$

For each  $x \in X$ , define  $W_x = \bigcup \mathcal{O}_x$ . Let  $x_0 \in X$  be fixed.

Since  $W_{x_0}$  is open, being union of open sets, so given  $(x, y) \in W_{x_0}$ , there exist open sets  $U_x \subset X$  and  $V_x \subset Y$  such that

$$(x, y) \in U_x \times V_x \subset W_{x_0}$$

Then,  $\{U_x \times V_x \mid x \in W_{x_0}\}$  is an open cover of  $\{x_0\} \times Y$ , so there is finite subcover  $\{U_{x'_0} \times V_{x'_0}, \dots, U_{x''_0} \times V_{x''_0}\}$ . Take  $A_{x_0} = U_{x'_0} \cap \dots \cap U_{x''_0}$ .

Then, since  $x_0 \in A_{x_0}$ , it is non-empty open set satisfying

$$\{x_0\} \times Y \subset A_{x_0} \times Y \subset W_{x_0}$$

2024, Spring.

1. Let  $A, B \in M_{m \times n}(F)$ . Show that  $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$ . Sym

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given  $\alpha \in \text{Im}(A+B)$ ,  $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$  for some  $\beta \in F^n$ , therefore  $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$ . Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that  $P$  and  $Q$  are positive definite such that  $P^2 = Q^2$ . Show  $P = Q$ .

Proof. Since  $P$  is Hermitian, there exists an orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  such that  $P\alpha_i = C_i\alpha_i$  for some  $C_i > 0$ , for each  $i$ .

Let  $1 \leq i \leq n$  be fixed. Then, by the assumption,

Sym

$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2 I)\alpha_i = (Q - C_i I)(Q + C_i I)\alpha_i = 0.$$

Since  $Q$  is positive definite,  $(Q - C_i I)\alpha_i = 0$ , because: if it is not zero, then  $(Q - C_i I)(Q + C_i I)\alpha_i = (Q + C_i I)(Q - C_i I)\alpha_i = 0$ ,

so  $-C_i$  is negative eigenvalue of  $Q$ , it contradicts to  $Q$  is positive definite. Therefore, for each  $1 \leq i \leq n$ ,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since  $\mathcal{B}$  is basis, therefore  $P = Q$ .  $\square$

2024, Spring

3. Let  $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$  where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists  $T$ -invariant subspace  $V \subseteq \mathbb{R}^5$  s.t.  $\dim V = 4$ ?

b). "

s.t.  $\dim V = 2$ ,  $T|_V$  is diagonalizable?

c). "

$\dim V = 3$ , "

Proof.

a). Let  $V = \langle e_1, e_2, e_3, e_4 \rangle$ . Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in  $V$ , so  $T[V] \subseteq V$ . (clearly  $\dim V = 4$ )

b). Let  $V = \langle e_1, e_3 \rangle$ . Then  $Ae_1 = e_1$ ,  $Ae_3 = e_3$  are contained in  $V$ , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of  $T$  is  $(t-1)^5$ .

If such a subspace  $V$  exists, then the characteristic polynomial of  $T|_V$  is

$(t-1)^3$ . Since  $T|_V$  is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of  $T|_V$  must be  $(t-1)$ .

That is,  $T|_V = I_V$  but it implies that  $\ker(T-I)$  has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

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4. (10 pts.) Let  $R$  be a ring. Denote by  $GL_3(R)$  the set of all  $3 \times 3$  invertible matrices with entries in  $R$ . Let  $G$  be the subset of  $GL_3(R)$  consisting of upper-triangular matrices.

(a) Show that  $G$  is a subgroup of  $GL_3(R)$ .

(b) Show that  $G$  is not abelian.

(c) Show that  $G$  is solvable.

Proof. a). Given  $A, B \in G$ ,

$$(AB)_{ij} = \sum_{k=1}^3 A_{ik} B_{kj}. \text{ If } i > j, \text{ then } (i, j) \in \{(2, 1), (3, 1), (3, 2)\}, \text{ so}$$

$$= A_{i1} B_{1j} + A_{i2} B_{2j} + A_{i3} B_{3j}$$

$$= 0 + 0 + 0$$

$$= 0, \text{ by just casework, so } AB \text{ is also upper-triangular.}$$

Moreover, since  $A$  and  $B$  are invertible, so  $AB \cdot B^{-1} A^{-1} = I$ , thus

$AB$  is invertible with the inverse  $B^{-1} A^{-1}$ . Thus,  $AB \in G$ .

Clearly  $I \in G$ , and, given  $A \in G$ , there is  $A^{-1} \in GL_3(R)$ . Hence  $B = A^{-1}$ , then

$$\begin{pmatrix} A_{11} & A_{12} & A_{13} \\ 0 & A_{22} & A_{23} \\ 0 & 0 & A_{33} \end{pmatrix} \cdot \begin{pmatrix} B_{11} & B_{12} & B_{13} \\ B_{21} & B_{22} & B_{23} \\ B_{31} & B_{32} & B_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then,  $A_{33} B_{31} = 0$  and  $A_{33} B_{33} = 1$ , but every unit cannot be a zero divisor,

so  $B_{31} = 0$ . Similarly  $B_{32} = 0$ . And, now,  $A_{22} B_{22} = 1$  but  $A_{22} B_{21} = 1$ ,

so  $B_{21} = 0$ . Therefore,  $B = A^{-1}$  is upper-triangular, so  $A^{-1} \in G$ .

It proves  $G$  is a group.

$$\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

b). Define a 3 by 3 matrix over  $\mathbb{Z}$  as:

$$(E_{ab})_{ij} = \begin{cases} 0 & (a, b) \neq (i, j) \\ 1 & (a, b) = (i, j) \end{cases}$$

Let  $A = I + E_{12}$ . Then,  $A \cdot (I - E_{12}) = I^2 + E_{12} - E_{12} - E_{12} E_{12} = I$ .

and let  $B = I + E_{23}$ . Then similarly  $B \cdot (I - E_{23}) = I$ , so  $A, B \in G$ .

Moreover,  $AB = I + E_{12} + E_{23} + E_{12} E_{23} = I + E_{12} + E_{23} + E_{13}$

$$BA = I + E_{12} + E_{23} + E_{23} E_{12} = I + E_{12} + E_{23} + 0$$

So  $AB \neq BA$ , thus  $G$  is non-abelian.

$$\text{Observe that } \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \text{ but}$$

$$\begin{pmatrix} -1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \text{ so}$$

these are not commute. And, these have the determinant  $\pm 1$ , units in  $R$ , so contained in  $GL$ , and clearly upper-triangles, so  $G$ .

c). Let  $GL(R) \rightarrow \mathbb{Z} : A \mapsto \det A$

Then,  $\det$  is a homomorphism with kernel  $SL(R) \triangleleft GL(R)$ .

Now  $G / (SL(R) \cap G)$

2024, Spring.

5. Let  $K$  be a field, and  $L_1, L_2 \subseteq \overline{K}$  be finite extensions of  $K$ . 41

a). Prove  $L = L_1 L_2$  is finite extension.

b). If  $L_1$  and  $L_2$  are Galois over  $K$ , then  $L/K$  is Galois.

a). Since  $L_1$  and  $L_2$  both are finite, denote

$$L_1 = K(\alpha_1, \dots, \alpha_n) \quad \text{and} \quad L_2 = K(\beta_1, \dots, \beta_r) \quad \text{where } \alpha_i, \beta_j \in \overline{K} \text{ are algebraic over } K.$$

$$\text{Now, } L = L_1 L_2 = \bigcap_{\substack{E \subseteq \overline{K} \\ L_1, L_2 \subseteq E}} E = K(\alpha_1, \dots, \alpha_n, \beta_1, \dots, \beta_r)$$

is finitely generated by algebraic elements  $\alpha_i, \beta_j$ , so  $L$  is finite. 14m.

b). Since  $L_1$  and  $L_2$  are splitting field of  $K$ , let  $f_1$  and  $f_2$  be polynomials over  $K$  such that  $L_i$  is the splitting field of  $f_i$ ,  $i=1,2$ . It is possible, because  $L_1, L_2$  both are finite extension.

Now let  $E$  be the splitting field of  $f_1, f_2$ . Then,  $E \subseteq L_1 L_2$ , because  $E$  is the smallest field containing all roots of  $f_1, f_2$ ,  $L_1 L_2 \subseteq E$  because  $E$  contains all roots of  $f_1$  and  $f_2$ . So,  $L = E$ . Therefore  $L$  is splitting field over  $K$ . 17m.

Since  $L_1$  and  $L_2$  both are Galois, for some separable  $f_1$  and  $f_2$  over  $K$ ,  $L_1$  and  $L_2$  are splitting field of  $f_1$  and  $f_2$  respectively.

$L_1 L_2$  is  $f_1, f_2$  splitting field  $\Rightarrow OK$

즉  $f_1, f_2$  is separable 이란 보장 X.

중복 linear factor 소거  $\Rightarrow f_1, f_2$ 가 irreducible over  $K$  인가?

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7. Let  $f: (-1, 1) \rightarrow \mathbb{R}$  be a differentiable function satisfying  $f(0) = 0$  and

$$\text{let } g: (-1, 1) \rightarrow \mathbb{R} : \begin{cases} f(x)/x & \text{if } x \neq 0 \\ f'(0) & \text{if } x = 0. \end{cases}$$

a). If  $f \in C^2$ , then  $g \in C^1$  (prove)

b). If  $f \in C^1$  but  $\notin C^2$ , then  $g \in C^1$  may not hold.

Proof. Notice that  $g$  is continuous, since

$$\lim_{x \rightarrow 0} g(x) = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0).$$

a) For  $x \neq 0$ ,

$$g'(x) = \left( \frac{f(x)}{x} \right)' = \frac{x f'(x) - f(x)}{x^2} \quad \text{exists, By the L'Hospital Theorem,}$$

$$\lim_{x \rightarrow 0} g'(x) = \lim_{x \rightarrow 0} \frac{f'(0) + x f''(x) - f'(0)}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} f''(x) = \frac{1}{2} f''(0).$$

(Since  $x^2 \rightarrow 0$  and  $x f'(x) - f(x) \rightarrow 0$  as  $x \rightarrow 0$ , we can use the theorem).

$$\text{And, } g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)/x - f'(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x) - x f'(0)}{x^2} = \frac{1}{2} f''(0).$$

Since the L'Hopital theorem again, so  $g \in C^1$ .

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8. Let  $C_b(\mathbb{R})$  be the set of all real-valued bounded continuous functions on  $\mathbb{R}$ . For two functions  $f, g \in C_b(\mathbb{R})$ , define

$$d(f, g) := \sup_{x \in \mathbb{R}} |f(x) - g(x)|.$$

Let  $\{f_n\}_{n=1}^{\infty}$  be a sequence in  $C_b(\mathbb{R})$  such that:

$\forall \varepsilon > 0, \exists N \in \mathbb{N}$  such that  $m, n \geq N \Rightarrow d(f_m, f_n) < \varepsilon$ .

a). Prove that there exists  $f \in C_b(\mathbb{R})$  such that  $\lim_{n \rightarrow \infty} d(f, f_n) = 0$

b). Prove that  $\lim_{n \rightarrow \infty} \left( \sup_{x \in \mathbb{R}} |f_n(x)| \right) = \sup_{x \in \mathbb{R}} |f(x)|$ .

[Proof. For each  $x \in \mathbb{R}$ , a sequence  $\{f_n(x)\}$  in  $\mathbb{R}$  is Cauchy sequence by hypothesis. Since  $\mathbb{R}$  is complete,  $\lim_{n \rightarrow \infty} f_n(x)$  exists, put this  $f(x) \in \mathbb{R}$ .

a).

Since  $x \in \mathbb{R}$  was arbitrary,  $f(x)$  is well-defined on  $\mathbb{R}$ , but still not necessarily  $f \in C_b(\mathbb{R})$ , and  $d(f, f_n) \rightarrow 0$ .

Given  $\varepsilon > 0$ , there exists  $N \in \mathbb{N}$  such that  $m > n \geq N$  implies that  $\sup_{x \in \mathbb{R}} |f_m(x) - f_n(x)| \leq \varepsilon$ .

So, for any  $x \in \mathbb{R}$ ,  $|f_m(x) - f_n(x)| \leq \varepsilon$ . Letting  $m \rightarrow \infty$ , we obtain  $|f(x) - f_n(x)| \leq \varepsilon$ .

Since  $x$  was arbitrary,  $\sup_{x \in \mathbb{R}} |f(x) - f_n(x)| \leq \varepsilon$ , for  $n \geq N$ .

Therefore, the triangle inequality gives  $|f(x)| \leq |f_n(x)| + \varepsilon$  for all  $x \in \mathbb{R}$

and since  $f_n$  was bounded,  $f$  is bounded. And, the argument above

proves  $\sup_{x \in \mathbb{R}} |f - f_n| \rightarrow 0$  as  $n \rightarrow \infty$ , so since  $f_n$  is continuous and

$f_n \rightarrow f$  uniformly, so  $f$  is also continuous, so  $f \in C_b(\mathbb{R})$  and a) is proved.

b). Since  $\left| \sup_{x \in \mathbb{R}} |f_n(x)| - \sup_{x \in \mathbb{R}} |f(x)| \right| \leq \sup_{x \in \mathbb{R}} |f_n(x) - f(x)| \rightarrow 0$  as  $n \rightarrow \infty$

Prove directly.

2024 Spring

12. Compute the Gauss curvature at  $p=(0,0,0)$  of the smooth surface given by  $z=y^2-x^2$ .

Proof. Consider the Monge patch of the given surface

$$X: \mathbb{D} \rightarrow \mathbb{R}^3; (u,v) \mapsto (u, v, v^2 - u^2)$$

Then,  $X_u = (1, 0, -2u)$  and  $X_v = (0, 1, 2v)$ , so the first fundamental form

$$E = 1 + 4u^2, \quad F = -2uv, \quad G = 1 + 4v^2$$

$$\text{so } \det \begin{pmatrix} E & F \\ F & G \end{pmatrix} = 1 + 4u^2 + 4v^2 + 16u^2v^2 - 4u^2v^2 = 1 \quad \text{at } (u,v) = (0,0)$$

$$\text{and } \begin{pmatrix} E & F \\ F & G \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{at } (u,v) = (0,0).$$

Meanwhile, the normal of the given surface is

$$X_u \times X_v = (2u, -2v, 1) \text{ gives } n = (0, 0, 1) \text{ at } (u,v) = (0,0).$$

Meanwhile,  $X_{uu} = (0, 0, -2)$ ,  $X_{uv} = (0, 0, 0)$ ,  $X_{vv} = (0, 0, 2)$  so

$$L_{uu} = -2, \quad L_{uv} = 0, \quad L_{vv} = 2.$$

$$L_u^u = L_{uu} g^{uu} + L_{vu} \cdot g^{vu} = 2$$

$$L_u^v = L_v^u = L_{uv} g^{uv} + L_{vu} \cdot g^{vv} = -4$$

$$L_v^v = L_{vv} g^{vv} + L_{vv} \cdot g^{vv} = 4.$$

Therefore, the Gaussian curvature is

$$\det \begin{pmatrix} 2 & -4 \\ -4 & 4 \end{pmatrix} = 24.$$

Note: Gaussian Curvature is  $\det II / \det I$ .

2024, Spring.

14. Let  $X$  be a compact Hausdorff space. Prove that

$X$  is metrizable if and only if  $X$  has a countable basis.

Proof Suppose that  $X$  is metrizable by the metric  $d: X \times X \rightarrow \mathbb{R}$ .

Given  $n \in \mathbb{N}$ ,  $\{B_{\frac{1}{n}}(x) \mid x \in X\}$  is an open cover of  $X$ , so the compactness gives that  $\mathcal{B}_n = \{B_{\frac{1}{n}}(x_i) \mid i \in F_n\}$ , the finite subcover exists

with finite index set  $F_n$ . Now, put  $\mathcal{B} = \bigcup_{n \in \mathbb{N}} \mathcal{B}_n$ .

Let  $x \in X$  be given and let  $B_r(y)$  be an open ball containing  $x$ . Then, for  $\varepsilon = r - d(x, y)$ ,  $x \in B_\varepsilon(y) \subseteq B_r(y)$ , because

$$a \in B_\varepsilon(x) \Leftrightarrow d(a, x) < \varepsilon = r - d(x, y) \Rightarrow d(a, y) = d(a, x) + d(x, y) < r.$$

Now, let  $N \in \mathbb{N}$  such that  $\frac{1}{N} < \frac{\varepsilon}{2}$ . Then, since each  $\mathcal{B}_n$  covers  $X$ , for some  $x_i$ ,  $x \in B_{\frac{1}{N}}(x_i)$ , and

$$z \in B_{\frac{1}{N}}(x_i) \Leftrightarrow d(z, x_i) < \frac{1}{N} \Rightarrow d(z, x) \leq d(z, x_i) + d(x_i, x) < \frac{1}{N} + \frac{1}{N} < \varepsilon.$$

So,  $x \in B_{\frac{1}{N}}(x_i) \subseteq B_\varepsilon(x)$ , it proves  $\mathcal{B}$  is a basis for  $(X, d)$ .

and since  $\mathcal{B}$  is countable union of finite sets, so it is countable.

Conversely, since compact Hausdorff space is normal, so it is regular. By the Urysohn metrization theorem, second countable regular space is metrizable, so proved.

2024, Fall.

1. Let  $V$  be a finite-dimensional vector space over a field  $F$ .  
Let  $T: V \rightarrow V$  be a linear operator such that  $T^2 = T$ . Prove

$\eta_m$

a).  $V = \ker T \oplus \operatorname{Im} T$

b).  $T$  is diagonalizable.

Proof. a). Given  $\alpha \in V$ ,  $\alpha = \alpha - T(\alpha) + T(\alpha)$ . And,  $T(\alpha - T(\alpha)) = T(\alpha) - T^2(\alpha) = 0$  since  $T^2 = T$ , so  $\alpha - T(\alpha) \in \ker T$ . Clearly  $T(\alpha) \in \operatorname{Im} T$ , so

$V = \ker T + \operatorname{Im} T$ . Meanwhile, let  $\alpha \in \ker T \cap \operatorname{Im} T$ . Then,  $\alpha = T(\beta)$  for some  $\beta \in V$ , and  $\alpha \in \ker T$  so that

$$0 = T(\alpha) = T(T(\beta)) = T(\beta) = \alpha, \text{ thus } \alpha = 0, \ker T \cap \operatorname{Im} T = \{0\}.$$

It proves a).

b). Since for any  $\alpha \in \operatorname{Im} T$ ,  $\alpha = T(\beta)$  for some  $\beta \in V$  and so that  $T(\alpha) = T^2(\beta) = T(\beta) = \alpha$ , thus  $T(\alpha) - \alpha = 0$ . Therefore,  $\operatorname{Im} T \subseteq \ker(T - I)$ , and given  $\alpha \in \ker(T - I)$ ,  $T(\alpha) = \alpha$ , so  $\alpha \in \operatorname{Im} T$ .  
Now,  $V = \ker T \oplus \ker(T - I)$ ,

therefore  $T$  is diagonalizable.  $\square$

2024 Fall.

2. (10 pts) Find all  $2 \times 2$  matrices  $A$  with entries in  $\mathbb{R}$  such that  $AA^t = I_2$  and that  $A$  is diagonalizable over  $\mathbb{R}$ , that is,  $Q^{-1}AQ$  is a diagonal matrix for some invertible  $2 \times 2$  matrix  $Q$  with real entries.

Proof. Let  $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ . Then,

22m

$$A^t = A^{-1} \Leftrightarrow \begin{pmatrix} a & c \\ b & d \end{pmatrix} = \frac{1}{\det A} \cdot \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} \text{ with } \det A \neq 0.$$

So,  $\det A \cdot a = d$  and  $\det A \cdot d = a$ , so that  $(\det A)^2 = 1$ .

If  $\det A = 1$ , then  $a = d$  and  $b = -c$ , so that

$$A = \begin{pmatrix} a & -b \\ b & a \end{pmatrix}, \text{ so } A = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \text{ for some } \theta.$$

And, since a matrix is diagonalizable if and only if the minimal polynomial splits and separable,

$$\det(A - tI) = t^2 - 2\cos \theta \cdot t + 1 = (t - (\cos \theta + \sqrt{-\sin^2 \theta}))(t + (\cos \theta - \sqrt{-\sin^2 \theta}))$$

so that the characteristic splits only when  $\theta = 0$  or  $\pi$ , so the minimal polynomial splits only  $\theta = 0$  or  $\pi$ , and in such a case, it is separable,

$$\text{so } A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ or } \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

If  $\det A = -1$ , then  $a = -d$  and  $b = c$ , so that

$$A = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix},$$

$$\text{and } \det(tI - A) = t^2 - \cos^2 \theta - \sin^2 \theta = (t-1)(t+1),$$

so  $A$  is always diagonalizable, so consequently

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \text{ or } \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix} \text{ or } \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix} \quad (\theta \in [0, 2\pi))$$

are we desired.

2024 Fall

3. (a) An element  $a$  of a ring  $R$  is nilpotent if  $a^n = 0$  for some integer  $n > 0$ . Show that the collection of all nilpotent elements in a commutative ring  $R$  is an ideal. Note that this ideal is called the nilradical of  $R$ .

(b) Find the nilradical of the ring  $\mathbb{Z}_{12}$ .

Proof. a). Let  $N$  be the nilradical of  $R$ . Let  $a, b \in N$  be given. Then,  $a^n = b^m = 0$  for some  $n, m \in \mathbb{N}$ . Now,

$$(a-b)^{n+m} = \sum_{k=0}^{n+m} \binom{n+m}{k} a^{n+m-k} b^k = 0, \text{ so } a-b \in N.$$

And,  $(ab)^{nm} = a^{nm} b^{nm} = 0$ , so  $ab \in N$ . Therefore,  $N$  is a subring. Moreover, given  $r \in R$ ,  $(ra)^n = r^n a^n = r^n 0 = 0$ , so  $ra \in N$ .

Thus,  $N \subset R$  is an ideal.

$2^2 \cdot 3$

b). The nilradical of  $\mathbb{Z}_{12}$  is  $\{0, 6\}$ , because

$k \in \mathbb{Z}_{12}$  satisfies  $k^n = 0$  for some  $n \in \mathbb{N}$

$\Leftrightarrow$  for some  $n \in \mathbb{N}$ ,  $12 \mid k^n$ .

So  $12 = 2^2 \cdot 3$ ,  $k = 0$  or  $6$ .

2024. Fall

4. (10 pts) Let  $G$  be a finite group. Show that if the number of solutions  $x$  of the equation  $x^m = e$  in  $G$  is at most  $m$  for each positive integer  $m$ , then  $G$  is cyclic. (Hint: you can use the fact that  $n = \sum_{d|n} \phi(d)$ , where  $\phi(d)$  is the Euler-phi function.)

Proof. Denote that  $G$  as a Partition

$$G = \bigcup_{C \leq G \text{ cyclic}} \mathcal{G}(C) \text{ where } \mathcal{G}(C) = \{x \in G \mid \langle x \rangle = C\}.$$

It is actually a partition, because: given  $x \in G$ ,  $x \in \langle x \rangle$  and so  $x \in \mathcal{G}(C)$ . And for cyclic subgroups  $C_1, C_2 \leq G$ ,

$$x \in \mathcal{G}(C_1) \cap \mathcal{G}(C_2) \Rightarrow \langle x \rangle = C_1 = C_2, \text{ so the claim is proved.}$$

Meanwhile, by the Lagrange's theorem, for each subgroup of  $G$  has order of factor of order of  $G$ . Put  $n = |G|$ . For each divisor  $d$  of  $n$ , if there is an order  $d$  element  $r \in G$ , then  $|\langle r \rangle| = d$ , and clearly

$$\langle r \rangle \subseteq \{x \in G \mid x^d = e\}$$

But, by the assumption,

$$d = |\langle r \rangle| \leq |\{x \in G \mid x^d = e\}| \leq d$$

therefore  $\langle r \rangle = \{x \in G \mid x^d = e\}$ , it contains all order  $d$  elements, and so that it is the unique cyclic subgroup of order  $d$ .

Now, by the Lagrange's theorem again, every element of  $G$  has order of divisor for  $|G|$ , therefore for each  $d|n$ ,  $\#\{x \in G \mid |x| = d\} \in \{0, \phi(d)\}$  so

$$n = |G| = \sum_{d|n} \#\{x \in G \mid |x| = d\} \leq \sum_{d|n} \phi(d) = n$$

forces for all  $d|n$ ,  $\#\{x \in G \mid |x| = d\} = \phi(d)$ , so  $G$  has  $\phi(n)$ -numbers order  $n$  elements, so cyclic.  $\square$

Additionally,  $\#\{x \in G \mid |x| = d\} \in \{0, \phi(d)\}$  is given by:

if there is no such element  $|x| = d$ , then  $\#\{x \in G \mid |x| = d\} = 0$ ,

if such an element exists, as  $x' \in G$ ,

$$\#\{x \in G \mid |x| = d\} = \mathcal{G}(\langle x' \rangle) = \phi(d).$$

2024 Fall.

5. Let  $K = \mathbb{Q}(\sqrt[4]{2}, i)$  be the splitting field in  $\mathbb{C}$  of  $x^4 - 2$  over  $\mathbb{Q}$ .

Describe all subgroups  $H$  of  $\text{Gal}(K/\mathbb{Q})$  of order 2 and corresponding subfields  $K_H$ .

Proof. Since  $i \notin \mathbb{Q}(\sqrt[4]{2})$  and the minimal polynomial of  $i$  over  $\mathbb{Q}$  is  $x^2 + 1$ ,

$$[K:\mathbb{Q}] = [K:\mathbb{Q}(\sqrt[4]{2})] \cdot [\mathbb{Q}(\sqrt[4]{2}):\mathbb{Q}] = 2 \cdot 4 = 8$$

So  $|\text{Gal}(K/\mathbb{Q})| = 8$ . And,  $x^4 - 2$  is the minimal polynomial of  $\sqrt[4]{2}$ ,  
so that  $x^4 - 2 = (x^2 - \sqrt{2})(x^2 + \sqrt{2}) = (x - \sqrt[4]{2})(x + \sqrt[4]{2})(x - \sqrt[4]{2}i)(x + \sqrt[4]{2}i)$

and the  $\mathbb{Q}$ -automorphism maps an element to the conjugate,  
the  $\sigma \in \text{Aut}(K/\mathbb{Q})$  is determined by

$$\sigma_{s,k}: \begin{cases} i \mapsto \pm i \quad (= i^{2s+1}, \quad s=0,1) \\ \sqrt[4]{2} \mapsto \sqrt[4]{2} \cdot i^k \quad (k=0,1,2,3) \end{cases}$$

Since  $(\sigma_{1,0})^2 = \text{id}$  and  $|\langle \sigma_{0,1} \rangle| = 4$ , and

$|\langle \sigma_{1,0} \circ \sigma_{0,1} \rangle| = 2$ , because  $\sigma_{1,0} \circ \sigma_{0,1} \neq \text{id}$  but  $(\sigma_{1,0} \circ \sigma_{0,1})^2 = \text{id}$ ,  
therefore  $\text{Gal}(K/\mathbb{Q}) \cong D_8$ , and all order 2 elements are

$\sigma_{1,0}, \sigma_{0,2}, \sigma_{1,2}, \sigma_{1,3}, \sigma_{1,4}$  so  $H = \langle \sigma_{i,2} \rangle, \langle \sigma_{1,i} \rangle$   $i=0,1,2,3$ .

And, observe that: since the trace is invariant value, so

$$\text{Tr}_{\langle \sigma_{1,0} \rangle}(\sqrt[4]{2}) = \sqrt[4]{2} + \sqrt[4]{2} = 2 \cdot \sqrt[4]{2}, \text{ so } K_{\langle \sigma_{1,0} \rangle} = \mathbb{Q}(\sqrt[4]{2})$$

$$\text{Tr}_{\langle \sigma_{0,2} \rangle}(\sqrt[4]{2}i) = i + \sqrt[4]{2} + i - \sqrt[4]{2} = 2i, \text{ and}$$

$$\text{Tr}_{\langle \sigma_{0,2} \rangle}(\sqrt{2}) = \sqrt{2} + \sqrt{2} = 2\sqrt{2}, \text{ so } K_{\langle \sigma_{0,2} \rangle} = \mathbb{Q}(i, \sqrt{2})$$

$$\text{Tr}_{\langle \sigma_{1,2} \rangle}(\sqrt[4]{2}i) = \sqrt[4]{2} \cdot i + (-\sqrt[4]{2}) \cdot -i = \sqrt[4]{2}i, \text{ so } K_{\langle \sigma_{1,2} \rangle} = \mathbb{Q}(\sqrt[4]{2}i).$$

$$\text{Tr}_{\langle \sigma_{1,1} \rangle}(\sqrt[4]{2}) = \sqrt[4]{2} + \sqrt[4]{2}i, \text{ so } K_{\langle \sigma_{1,1} \rangle} = \mathbb{Q}(\sqrt[4]{2}(1+i))$$

$$\text{Tr}_{\langle \sigma_{1,3} \rangle}(\sqrt[4]{2}) = \sqrt[4]{2} - \sqrt[4]{2}i, \text{ so } K_{\langle \sigma_{1,3} \rangle} = \mathbb{Q}(\sqrt[4]{2}(1-i))$$

(The equality is obtained by the Galois theorem gives that

$$|H|=2 \Rightarrow [K:H] = 2).$$

2024 Fall

7. (10 pts) Let  $(f_n(x))_{n \in \mathbb{N}}$  be a sequence of continuously differentiable functions on  $[0, 1]$ .

(a) Suppose that the sequence  $(f_n(x))_{n \in \mathbb{N}}$  converges pointwise to a function  $f$ . Does  $(f_n(x))_{n \in \mathbb{N}}$  converge uniformly to  $f$ ? Justify your answer.

(b) Suppose that there is a uniform constant  $C > 0$  such that, for all  $n = 1, 2, \dots$ ,

$$\sup_{x \in [0, 1]} (|f_n(x)| + |f'_n(x)|) < C.$$

Show that  $(f_n(x))_{n \in \mathbb{N}}$  has a uniformly convergent subsequence in  $[0, 1]$ .

Proof. a). Define for each  $n \in \mathbb{N}$ ,

$$f_n: [0, 1] \rightarrow \mathbb{R}; x \mapsto x^n.$$

Then, clearly  $f_n \in C^1$ , and  $f_n \rightarrow f$  pointwise where

$$f(x) = \begin{cases} 1 & \text{if } x=1 \\ 0 & \text{if } 0 \leq x < 1 \end{cases}$$

But for each  $n \in \mathbb{N}$ ,  $\lim_{x \rightarrow 1^-} f_n(x) = 1$  so that  $\sup_{x \in [0, 1]} |f(x) - f_n(x)| = 1$ ,

therefore  $f_n$  does not uniformly converge.

b). Let  $n \in \mathbb{N}$  be given. Given  $\varepsilon > 0$ , take  $\delta = \frac{\varepsilon}{C}$ . Then, for any  $x, y \in [0, 1]$ ,

$$|x - y| < \delta \Rightarrow |f_n(x) - f_n(y)| = |f'_n(t)| \cdot |x - y| \leq C \cdot \frac{\varepsilon}{C} < \varepsilon$$

for some  $t$  between  $x$  and  $y$ , by the Mean-Value theorem  
therefore the  $\{f_n\}$  is equicontinuous, and pointwise-bounded by

$$\sup_{x \in [0, 1]} |f_n(x)| \leq \sup_{x \in [0, 1]} (|f_n(x)| + |f'_n(x)|) < C.$$

Finally, by the Arzela-Ascoli theorem, the sequence of functions  $\{f_n\}$  which is pointwise-bounded, equicontinuous on the compact set  $[0, 1]$  has a uniformly convergent subsequence.  $\square$

Similar Problem. Rudin, PMA, Chapter 7.

18. Let  $\{f_n\}$  be a uniformly bounded sequence of functions which are Riemann-integrable on  $[a, b]$ , and put

$$F_n(x) = \int_a^x f_n(t) dt \quad (a \leq x \leq b).$$

Prove that there exists a subsequence  $\{F_{n_k}\}$  which converges uniformly on  $[a, b]$ .

2024 Fall.

8. Solve the followings:  $x \in \mathbb{R}$ ,

a).  $x''(t) - 2x'(t) + x(t) = 0$

b).  $x''(t) - 2x'(t) + x(t) = te^t + 4$ .

Proof.

a). Since the characteristic polynomial of the equation is

$$r^2 - 2r + 1 = (r-1)^2,$$

the general solution is

$$x_1(t) = C_1 e^t + C_2 \cdot t e^t \quad \text{where } C_1, C_2 \in \mathbb{R}.$$

b). Calculate the Wronskian:

$$W(e^t, t e^t) = \begin{vmatrix} e^t & t e^t \\ e^t & e^t + t e^t \end{vmatrix} = e^{2t} + t e^{2t} - t e^{2t} = e^{2t}.$$

Therefore, the particular solution is

$$\begin{aligned} Y(t) &= \int_{t_0}^t \begin{vmatrix} e^u & e^t \\ u e^u & t e^t \end{vmatrix} \cdot b(u) / W(e^u, u e^u) du \\ &= \int_{t_0}^t (e^u t e^t - u e^u e^t) \cdot (u e^u + 4) / e^{2u} du \\ &= \int_{t_0}^t (u e^{2u} t e^t + 4 e^u t e^t - u^2 e^{2u} e^t - 4 u e^u e^t) / e^{2u} du \\ &= \int_{t_0}^t u \cdot t e^t + 4 \cdot t e^t \cdot e^{-u} - u^2 \cdot e^t - 4 e^t \cdot u e^{-u} du \end{aligned}$$

Now,  $x_2 + Y$  is the general solution.

2024, Fall,

1. For a fixed  $z \in \mathbb{R}$ , let

$$f(z) = \frac{e^{-2\pi i z \tau}}{\cosh(\pi z)} \quad (z \in \mathbb{C}).$$

a). Show that  $\alpha = i/2$  and  $\beta = 3i/2$  are simple poles of  $f$ .

b). Compute the residues of  $f$  at  $\alpha$  and  $\beta$ .

c). Evaluate  $\int_{-\infty}^{\infty} \frac{e^{-2\pi i x \tau}}{\cosh(\pi x)} dx$ .

Proof. a). Notice that  $e^{-2\pi i z \tau}$  is entire and has no zero, we need to investigate pole of  $1/\cosh(\pi z)$ .

Since  $\cosh(\pi z) = 0$  if  $z_0 = \alpha$  or  $\beta$ , and

$$\lim_{z \rightarrow z_0} \frac{\cosh(\pi z) - \cosh(\pi z_0)}{z - z_0} = \sinh(\pi z_0) \cdot \pi \neq 0, \quad \text{so}$$

$$\lim_{z \rightarrow z_0} (z - z_0)' \cdot \frac{1}{\cosh(\pi z)} = \frac{1}{\pi \sinh(\pi z_0)} \text{ exists, so } \alpha \text{ and } \beta \text{ are simple poles.}$$

b). Since  $\alpha$  and  $\beta$  both are simple pole,

$$\text{Res}(f, \alpha) = \lim_{z \rightarrow \alpha} (z - \alpha) \cdot f(z) = e^{-2\pi i \alpha \tau} \cdot \frac{1}{\pi \cdot \sinh(\pi \alpha)} = \frac{e^{\pi \tau}}{\pi \cdot \sinh(\pi i/2)} = \frac{e^{\pi \tau}}{\pi i}.$$

$$\text{Res}(f, \beta) = e^{-2\pi i \beta \tau} \cdot \frac{1}{\pi \cdot \sinh(\pi \beta)} = \frac{e^{3\pi \tau}}{\pi \cdot (-i)} = -\frac{e^{3\pi \tau}}{\pi i}.$$

c). Let  $C_R$  be the rectangle  $C_R = C_1 * C_2 * C_3 * C_4$  along four line segments  $-R$  to  $R$ ,  $R$  to  $R+i$ ,  $R+i$  to  $-R+i$ ,  $-R+i$  to  $-R$ .

And, observe that

$$f(x+i) = \frac{e^{-2\pi i x \tau + 2\pi \tau}}{\cosh(\pi x + \pi i)} = e^{2\pi \tau} \cdot -f(x)$$

$$\text{Therefore, } \int_{C_1 + C_3} f = \left( \int_{-R}^R f(x) dx \right) \cdot (1 + e^{2\pi \tau}).$$

And, since  $|f(R+it)| = \left| \frac{e^{-2\pi i R \tau + 2\pi \tau}}{\cosh(\pi R + \pi it)} \right| \rightarrow 0$  as  $|\cosh(\pi R + \pi it)| \rightarrow \infty$  as  $R \rightarrow \infty$ .

Therefore,  $\int_{C_2} f, \int_{C_4} f \rightarrow 0$ , so, by the Residue theorem,

$$\int_{-\infty}^{\infty} f(x) dx = \frac{1}{1 + e^{2\pi \tau}} \cdot \lim_{R \rightarrow \infty} \int_{C_R} f = \frac{1}{1 + e^{2\pi \tau}} \cdot 2e^{\pi \tau}.$$

12/24 Fall.

10. (10 pts) Let  $f, f_n : [a, b] \rightarrow \mathbb{R}$ ,  $n \in \mathbb{N}$ , be functions such that  $(f_n)_{n \in \mathbb{N}}$  converges uniformly to  $f$ . Prove or disprove the following.

(a) If  $f_n$  is Riemann integrable for all  $n \in \mathbb{N}$ , then  $f$  is Riemann integrable.

(b) If  $f$  and  $f_n$ ,  $n \in \mathbb{N}$ , are all differentiable, then  $(f'_n)_{n \in \mathbb{N}}$  converges to  $f'$ .

Proof. (a). For each  $n \in \mathbb{N}$ , define  $\varepsilon_n = \sup\{|f_n(x) - f(x)| \mid x \in [a, b]\}$ .

Then,  $\varepsilon_n \rightarrow 0$  as  $n \rightarrow \infty$  and for all  $x \in [a, b]$

$$f_n - \varepsilon_n < f(x) < f_n + \varepsilon_n, \quad \text{so that}$$

$$\int_a^b (f_n - \varepsilon_n) < \int_a^b f < \int_a^b (f_n + \varepsilon_n) \quad \text{and}$$

$$\int_a^b (f_n - \varepsilon_n) < \int_a^b f < \int_a^b (f_n + \varepsilon_n), \quad \text{and since } f_n \text{ is integrable,}$$

$$\text{so } \int_a^b f_n = \int_a^b f_n = \int_a^b f_n, \quad \text{therefore}$$

$$\int_a^b f - \int_a^b f < 2\varepsilon_n \cdot [b-a] \rightarrow 0 \quad \text{as } n \rightarrow \infty,$$

so  $f$  is Riemann integrable.

b) Let  $f_n : \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sin(nx)$ .

Then,  $f_n \rightarrow 0$  uniformly, but,  $f'_n = \cos(nx)$  does not converge to 0, so it disproves b).



2024 Fall,

12. Let  $p: X \rightarrow Y$  be a quotient map.

Show that if  $Y$  is connected and  $p^{-1}[\{y\}]$  is connected for each  $y \in Y$ , then  $X$  is connected.

Proof. Suppose that  $X$  is disconnected, with a separation  $\{U, V\}$ . Given  $y \in Y$ ,  $p^{-1}[\{y\}] \subseteq X$  is connected, so either

$$p^{-1}[\{y\}] \subseteq U \text{ or } p^{-1}[\{y\}] \subseteq V.$$

[If not, then  $\{U, V\}$  is a separation of  $p^{-1}[\{y\}]$

Meanwhile, observe that

$$Y = p[X] = p[U \cup V] = p[U] \cup p[V]$$

Let  $a \in p[U]$  be given. Then,  $a = p(b)$  for some  $b \in U$ .

And,  $p^{-1}[\{a\}]$  contains  $b$ , so  $p^{-1}[\{a\}] \cap U \neq \emptyset$ , therefore  $p^{-1}[\{a\}] \subseteq U$

Thus,  $p^{-1}[p[U]] \subseteq U$  and  $U \subseteq p^{-1}[p[U]]$  generally, so

$U = p^{-1}[p[U]]$  is open. Since  $p$  is quotient map,  $p^{-1}[p[U]] = U$  is open implies that  $p[U]$  is also open set, similarly  $p[V]$  is open.

Moreover,  $p[U] \cap p[V] = \emptyset$ , because suppose that

for some  $a \in Y$ ,  $a = p(b) = p(c)$  for some  $b \in U$  and  $c \in V$ .

Then,  $p^{-1}[\{a\}] = p^{-1}[p[b]] = p^{-1}[p[c]]$ , but

since  $b \in p^{-1}[p[b]] \cap U$  and  $c \in p^{-1}[p[c]] \cap V$ ,

$p^{-1}[\{a\}] \subseteq U$  and  $V$ , but it contradicts to  $U \cap V = \emptyset$ .

So,  $\{p[U], p[V]\}$  is a separation of  $Y$ , it is contradiction.  $\square$

Similar problem.

Thm. Let  $p: X \rightarrow Y$  be a continuous, onto, closed map such that:

for all  $y \in Y$ ,  $p^{-1}[\{y\}] \subset X$  is compact set. (such  $p$  is called perfect)

If  $Y$  is compact, then  $X$  is compact.

Proof. Let  $\mathcal{O} = \{\mathcal{U}_\alpha \mid \alpha \in I\}$  be an open cover of  $X$ . For each  $y \in Y$ ,

$$p^{-1}[\{y\}] \subset p^{-1}[Y] = X = \bigcup_{\alpha \in I} \mathcal{U}_\alpha.$$

By the hypothesis,  $p^{-1}[\{y\}]$  is compact, so that there is a finite index set

$$I_y \subset I \text{ such that } p^{-1}[\{y\}] \subset \bigcup_{\alpha \in I_y} \mathcal{U}_\alpha.$$

Now, since  $p$  is closed map,

$$V_y = Y \setminus p[X \setminus \bigcup_{\alpha \in I_y} \mathcal{U}_\alpha]$$

is open set containing  $y$ .

12. Let  $p: X \rightarrow Y$  be a closed continuous onto map s.t. for all  $y \in Y$ ,  $p^{-1}[y]$  is compact.

If  $Y$  is compact, then  $X$  is compact. (Such  $p$  is called perfect map.)

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{y \in Y} \{y\}\right] = \bigcup_{y \in Y} p^{-1}[\{y\}]$$

Let  $\mathcal{O} = \{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$  be an open cover of  $X$ . For each  $y \in Y$ ,

$$p^{-1}[\{y\}] \subset \bigcup_{y \in Y} p^{-1}[\{y\}] \subset \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha$$

And by the assumption,  $p^{-1}[\{y\}]$  is compact, therefore there is a finite index  $\Lambda_y \subset \Lambda$  s.t.

$$p^{-1}[\{y\}] \subset \bigcup_{i \in \Lambda_y} \mathcal{U}_i.$$

Now, since  $p$  is closed map,

$$V_y = Y \setminus p[X \setminus (\bigcup_{i \in \Lambda_y} \mathcal{U}_i)]$$

is an open set of  $Y$ , containing  $y$ . Now, since  $Y$  is compact, for some  $y_1, \dots, y_n$ ,

$$Y = \bigcup_{y \in Y} V_y = \bigcup_{j=1}^n V_{y_j} \Rightarrow X = p^{-1}[Y] = p^{-1}\left[\bigcup_{j=1}^n V_{y_j}\right] = \bigcup_{j=1}^n p^{-1}[V_{y_j}]$$

Meanwhile,  $p^{-1}[V_{y_j}] = p^{-1}\left[Y \setminus p\left[X \setminus \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i\right]\right] = X \setminus p^{-1}\left[p\left[X \setminus \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i\right]\right] \subset \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i,$

therefore 
$$Y \subset \bigcup_{j=1}^n \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i$$

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13. Let  $X$  be any space and  $Y$  be Compact Hausdorff space.

$f: X \rightarrow Y$  is Continuous map if and only if the graph  $G_f = \{(x, f(x)) \mid x \in X\}$  is closed.

Proof. Let  $f: X \rightarrow Y$  be a Continuous map.

Let  $(x, y) \in X \times Y \setminus G_f$  be given. Then,  $y \neq f(x)$ , so there exist disjoint open sets  $U, V$

$$U \cap V = \emptyset, \quad y \in U, \quad f(x) \in V$$

Since  $f(x) \in V \stackrel{\text{def}}{\Leftrightarrow} x \in f^{-1}[V]$  and  $f$  is Continuous,  $f^{-1}[V]$  is an open set in  $X$ .

Now,  $(x, y) \in f^{-1}[V] \times U$

And,  $(f^{-1}[V] \times U) \cap G_f = \emptyset$ , because if  $(x, f(x)) \in f^{-1}[V] \times U$ , then

$f(x) \in V$  and  $f(x) \in U$ , it is contradiction to disjointness.

Therefore,  $(x, y) \in f^{-1}[V] \times U \subseteq X \times Y \setminus G_f$ , so proved.  $\Rightarrow$

Conversely, let  $G_f$  is closed in  $X \times Y$ , under the Product Topology.

Let  $C \subseteq Y$  be a closed set in  $Y$ . Then,  $G_f \cap (X \times C)$  is closed. Moreover,

$$P_X[G_f \cap (X \times C)] = \{x \in X \mid f(x) \in C\} = f^{-1}[C]$$

is closed, because  $P_X: X \times Y \rightarrow X; (x, y) \mapsto x$  is closed map if  $Y$  is Compact.  $\square$

Note: for  $g = f \times \text{id}: X \times Y \rightarrow Y \times Y; (x, y) \mapsto (f(x), y)$ ,

$$g^{-1}[\Delta(Y)] = \{(x, y) \in X \times Y \mid (f(x), y) \in \Delta(Y)\} = G_f \quad \leftarrow$$

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14, Let  $S \subset \mathbb{P}^3$  be a smooth surface and  $p \in S$  be a point.

2025. Spring.

1. Let  $F$  be any field, and let  $V = M_{n \times n}(F)$ , and let

$$T: V \rightarrow V: A \mapsto A + A^t$$

a). Show that  $V = \ker T \oplus \operatorname{Im} T$ .

b). Is  $T$  diagonalizable? Find all eigenvalues of  $T$  with multiplicities.

Proof. Let  $\operatorname{ch}(F) \neq 2$ .

a). Given  $A \in V$ ,

$$A = \frac{1}{2}(A - A^t) + \frac{1}{2}(A + A^t)$$

And, since  $-\left(\frac{1}{2}(A - A^t)\right)^t = -\frac{1}{2}(A^t - A) = \frac{1}{2}(A - A^t)$  implies  $\frac{1}{2}(A - A^t) \in \ker T$ ,  
and  $\frac{1}{2}(A + A^t) = T\left(\frac{1}{2}A\right) \in \operatorname{Im} T$ , so  $V = \ker T + \operatorname{Im} T$ .

Meanwhile, given  $A \in \ker T \cap \operatorname{Im} T$ ,

$$A + A^t = 0 \quad \text{and} \quad A = B + B^t \quad \text{for some } B \in V.$$

$$\text{But, } 0 = A + A^t = B + B^t + B^t + B = 2 \cdot (B + B^t) = 2 \cdot A,$$

so  $A = 0$ . It proves  $\ker T \cap \operatorname{Im} T = \{0\}$ , thus  $V = \ker T \oplus \operatorname{Im} T$ .

b). Since  $A \in \ker T \Leftrightarrow A = -A^t$ , so, all diagonal entries are zero and if the upper entries are chosen, then the lower entries are determined. So  $(n^2 - n)/2$  is the dimension of  $\ker A$ .

Meanwhile, given  $A \in \operatorname{Im} T$ ,  $A = B + B^t$  for some  $B \in V$  and

$$T(A) = T(B) + T(B^t) = B + B^t + B + B^t = 2 \cdot A.$$

So,  $\operatorname{Im} T \subseteq \ker(T - 2I)$ . Conversely, given  $A \in \ker(T - 2I)$ ,  
 $A + A^t - 2A = A^t - A = 0$ , so  $A = A^t$ . Meanwhile,

$$T\left(\frac{1}{2}A\right) = \frac{1}{2}A + \frac{1}{2}A^t = A, \quad \text{so } A \in \operatorname{Im} T. \quad \text{Thus, } \operatorname{Im} T = \ker(T - 2I).$$

By the dimension theorem,

$$\dim \operatorname{Im} T = \frac{n^2 + n}{2}.$$

Therefore, 0 and 2 are eigenvalues of  $T$  with multiplicities  $\frac{n^2 - n}{2}$ ,  $\frac{n^2 + n}{2}$  respectively, and diagonalizable.  $\square$

3.

b) Find  $f(x) \in \mathbb{C}[x]$  such that  $f(x) = (x^5 + 2x^4 + 3x^3 + 2x^2 + x, x^6 - 1)$ .

a). Let  $I \subseteq F[x]$  be an ideal. If  $I = F[x]$  or  $I = \{0\}$ , then  $I = (1)$  or  $(0)$ .

$$\deg: F[x] \rightarrow \mathbb{N} \cup \{0\}; f(x) \mapsto \deg f(x)$$

and  $m \geq 1$ , because  $I$  cannot contain a unit, by the assumption.

$$f(x) = p(x)q(x) + r(x) \text{ for some } q, r \in \mathbb{C}[x], \text{ with } r=0 \text{ or } \deg r < \deg p$$

A degree of  $p$  gives that  $r=0$ , so  $f=pq \in \mathbb{Z}$ . Therefore,  $I \subseteq (\mathbb{Z}\omega)$ .

b). Since  $x^6 - 1 = (x-1)(x^5 + x^4 + x^3 + x^2 + x + 1) = (x-1)(x^2 + x + 1)(x^3 + 1)$

So, the greatest common divisor is  $(x^2+x+1)$ , so  $x^2+x+1$  is the generating

polynomial we desired, because  $(g, h) = (\gcd(g, h))$  in P.I.D. 

b). By the division,

$$x^6 - 1 = (x^5 + 2x^4 + 3x^3 + 2x^2 + x)(x - 2) + x^4 + 4x^3 + 3x^2 + 2x$$

$$x^5 + 2x^4 + 3x^3 + 2x^2 + x = (x^3 + 4x^2 + 3x + 1)(x - 2) + 8x^3 + 6x^2 + 5x$$

\* 생각을 해보자,  $\mathbb{C}$  는 Algebraic Closure 이고, 즉 각 polynomial 들은

linear factor 로 존재한다. 즉, 나머지 2차항의 근은 5, 62 번은  $\pm i\sqrt{13}$ 이다.

과다리 공통 근무 가능한지 더 여성적.

Let  $I \subseteq F[x]$  be an ideal. If  $I = F[x]$ , then  $(1) = I$ .

If  $I = \{0\}$ , then  $I = (0)$ . Now, assume  $I$  is non-trivial proper Ideal.

Let  $p(x) \in \mathbb{Z}[x]$  be given with  $\deg p(x) \leq \deg h(x)$  for all  $h(x) \in F[x] \setminus \{0\}$ .

(It is Possible by well-ordering Principle)

Now, let  $f(x) \in I$  be given. Then, by the division algorithm,

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4. (10 pts)

- (a) Show that a finite abelian group is not cyclic if and only if it contains a subgroup isomorphic to  $\mathbb{Z}_p \times \mathbb{Z}_p$  for some prime  $p$ .
- (b) For positive integers  $d_1, d_2, d_3, d_4$  such that  $d_1$  divides  $d_2$  and  $d_3$  divides  $d_4$ , assume that  $\mathbb{Z}_{d_1} \times \mathbb{Z}_{d_2}$  is isomorphic to  $\mathbb{Z}_{d_3} \times \mathbb{Z}_{d_4}$ . Show that  $d_1 = d_3$  and  $d_2 = d_4$ .

Proof.

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5. Consider  $K = \mathbb{Q}(\sqrt[4]{2}, i)$ . Find the Norm

$$N_{K/\mathbb{Q}}(1 + \sqrt[4]{2} + i).$$

Proof Since  $[\mathbb{Q}(\sqrt[4]{2}) : \mathbb{Q}] = 4$  and  $[K : \mathbb{Q}(\sqrt[4]{2})] = 2$ , by the first is obtained by the Eisenstein and the second is given by  $\pm i \notin \mathbb{Q}(\sqrt[4]{2})$ , thus  $[K : \mathbb{Q}] = 8$ .

Moreover, the minimal polynomial of  $\sqrt[4]{2}$  over  $\mathbb{Q}$  is  $x^4 - 2$  and

$$\text{“ } i \text{ over } \mathbb{Q} \text{ is } x^2 + 1$$

$$\text{which split as } x^4 - 2 = (x - \sqrt[4]{2})(x - \sqrt[4]{2} \cdot \zeta)(x - \sqrt[4]{2} \cdot \zeta^2)(x - \sqrt[4]{2} \cdot \zeta^3) \\ \text{and } x^2 + 1 = (x + i)(x - i)$$

where  $\zeta = i$ . Since the  $\mathbb{Q}$ -Automorphisms of  $K$  must map a root to the conjugate, and the  $K$  contains all roots of  $x^4 - 2$  and  $x^2 + 1$ , thus the  $\sigma \in \text{Aut}(K/\mathbb{Q})$  is determined by

$$\sigma(\sqrt[4]{2}) = \sqrt[4]{2} \cdot i^k \quad (k = 0, 1, 2, 3) \quad \text{and} \quad \sigma(i) = \pm i.$$

And, the  $K$  is the splitting field of  $x^4 - 2$ ,  $K/\mathbb{Q}$  is Galois. Since  $[K : \mathbb{Q}] = 8$ , these are all elements of  $\text{Gal}(K/\mathbb{Q})$ . Now,

$$N_{K/\mathbb{Q}}(1 + \sqrt[4]{2} + i)$$

$$\begin{aligned} &= (1 + \sqrt[4]{2} + i)(1 + \sqrt[4]{2}i + i)(1 - \sqrt[4]{2} + i)(1 - \sqrt[4]{2}i + i)(1 + \sqrt[4]{2} - i)(1 + \sqrt[4]{2}i - i)(1 - \sqrt[4]{2} - i)(1 - \sqrt[4]{2}i - i) \\ &= (2i - \sqrt{2})(2i + \sqrt{2})(-2i - \sqrt{2})(-2i + \sqrt{2}) \\ &= (-4 - 2)(-4 - 2) \\ &= 36. \end{aligned}$$

$$(1 + i)^2 = 1 + 2i - 1 = 2i$$

$$(1 - i)^2 = -2i$$

2025 Spring.

6. Prove or disprove that

If  $f: [0,1] \rightarrow \mathbb{R}$  is uniformly continuous, then it is Lipschitz continuous.

Proof. Let  $f: [0,1] \rightarrow \mathbb{R}$ ;  $x \mapsto \sqrt{x}$ . Since  $f$  is continuous and  $[0,1]$  is compact, thus  $f$  is continuously uniformly. But, given  $L > 0$ ,

$$|f(0) - f(x)| = |\sqrt{x}| \geq L \cdot |x| \text{ when } x < \frac{1}{L^2}$$

Therefore,  $f$  does not satisfies the Lipschitz condition, so the given statement is false.  $\square$

Note: Given  $p > 0$ , Consider  $f(x) = x^p$ . ( $x \geq 0$ )

If  $0 < p \leq 1$ , assume that  $x \geq y \geq 0$ . Put  $h = x - y \geq 0$ . Then, we want to show

$$x^p - y^p = (y+h)^p - y^p \leq h^p = (x-y)^p.$$

Given  $\varepsilon > 0$ , take  $\delta =$  . Then, for all  $x, y \in [0,1]$ ,

$$|x-y| < \delta = \varepsilon \Rightarrow |f(x) - f(y)| \leq |\sqrt{x} - \sqrt{y}| = \frac{|x-y|}{\sqrt{x} + \sqrt{y}}$$

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8. Let  $U = \{z \in \mathbb{C} \mid |z| = 1\}$  be the unit circle in the complex plane. Prove or disprove that:

Every continuous function on  $U$  can be approximated uniformly by polynomials in the variable  $z \in \mathbb{C}$ .

Proof. Let  $f: U \rightarrow \mathbb{C}; z \mapsto 1/z$  be a continuous map.

Directly, for a curve  $C: t \mapsto e^{it}$  ( $0 \leq t \leq 2\pi$ ),

$$\oint_C \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{it}} i e^{it} dt = 2\pi i.$$

Now, suppose that there is a sequence of polynomials  $\{P_n\}$  converges uniformly to  $f$ . Then, for each  $n \in \mathbb{N}$ ,  $P_n$  is entire and so

$$\int_C P_n = 0 \quad \text{by the Cauchy-Goursat theorem,}$$

therefore

$$\int_C f = \int_C \lim_{n \rightarrow \infty} P_n = \lim_{n \rightarrow \infty} \int_C P_n = 0$$

But it contradicts to  $\int_C f = 2\pi i$ .  $\square$

2025. Spring.

9. (10 pts) Find general solutions  $x = x(t)$  to the following second-order ODEs:

(a)  $t^2 x'' + tx' + x = 0, \quad t > 0;$

(b)  $t^2 x'' + 5tx' + 4x = 0, \quad t > 0.$

Justify your answers to each problem.

Proof.

a). Put  $x = t^r$ . Then,

$$\begin{aligned} t^2 \cdot x'' + t \cdot x' + x &= 0 \Rightarrow t^2 \cdot (r \cdot (r-1) \cdot t^{r-2}) + t \cdot r \cdot t^{r-1} + t^r \\ &= r(r-1)t^r + r \cdot t^r + t^r \\ &= t^r (r^2 + 1). \end{aligned}$$

Therefore, if  $r = \pm i$ ,  $x = t^{\pm i}$  satisfies a). If  $r = i$ , then

$$x = t^i = (e^{\log t})^i = e^{i \log t} = \cos(\log t) + i \cdot \sin(\log t).$$

And if  $r = -i$ , similarly,

$$x = \cos(\log t) - i \cdot \sin(\log t)$$

therefore, the general solution is

$$x(t) = C_1 \cdot \cos(\log t) + C_2 \cdot \sin(\log t), \quad C_1, C_2 \in \mathbb{R}.$$

b)

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13. Let  $D^2$  be a closed unit disc in  $\mathbb{R}^2$  centered at the origin.

a). Show that the quotient space  $\mathbb{R}^2/D^2$  is homeomorphic to  $\mathbb{R}^2$ .

b). Let  $U = \text{int } D^2$ . Determine  $\mathbb{R}^2/U \cong \mathbb{R}^2$ .

Proof. Let  $\sim$  be a relation defined on  $\mathbb{R}^2$  generated by:

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow (x_1, y_1), (x_2, y_2) \in D^2.$$

Then, it is clearly an equivalence relation, and  $\mathbb{R}^2/\sim = \mathbb{R}^2/D^2$ , by definition.

a). Define a map

$$f: \mathbb{R}^2 \rightarrow \mathbb{R}^2; (x, y) \mapsto \begin{cases} (0, 0) & \text{if } x^2 + y^2 \leq 1 \\ \left( \left(1 - \frac{1}{\sqrt{x^2 + y^2}}\right) \cdot (x, y) \right) & \text{if } x^2 + y^2 > 1 \end{cases}$$

Then, given  $(x_1, y_1), (x_2, y_2) \in \mathbb{R}^2$ ,

$$(x_1, y_1) \sim (x_2, y_2) \Leftrightarrow (x_1, y_1), (x_2, y_2) \in D^2 \text{ or } (x_1, y_1) = (x_2, y_2)$$

$$\Leftrightarrow f(x_1, y_1) = f(x_2, y_2)$$

and, it is onto continuous map, so  $f$  induces a homeomorphism

$$\tilde{f}: \mathbb{R}^2/D^2 \rightarrow \mathbb{R}^2; [(x, y)] \mapsto f(x, y).$$

2025. Spring.

14. For  $a, b, c > 0$ , consider the ellipsoid

$$E_{a,b,c} = \left\{ (x, y, z) \in \mathbb{R}^3 \mid \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 + \left(\frac{z}{c}\right)^2 = 1 \right\}.$$

- a). If  $a=b$ , then  $E_{a,c}$  is a surface of revolution (Pravel.)  
b) Give a parametrization of  $E_{a,c}$ . Compute the first fundamental form of this parametrization and the Christoffel symbols  
c). Show that the curve  $E_{a,b,c} \cap \{x=0\}$  is the image of a geodesic.

a). Let

$$X: (0, 2\pi) \times (0, 2\pi) \rightarrow \mathbb{R}^3; (t, \theta) \mapsto (a \cdot \cos t \cdot \cos \theta, a \cdot \cos t \cdot \sin \theta, c \cdot \sin t)$$

It covers  $E_{a,c}$  excepts a line, so  $E_{a,c}$  is a surface of revolution.

b). The parametrization  $X$  is given in a). And,

$$X_t = (-a \cdot \sin t \cdot \cos \theta, -a \cdot \sin t \cdot \sin \theta, c \cdot \cos t)$$

$$X_\theta = (-a \cdot \cos t \cdot \sin \theta, a \cdot \cos t \cdot \cos \theta, 0). \text{ So the first fundamental form is}$$

$$E = a^2 \sin^2 t + c^2 \cos^2 t, \quad F = 0, \quad G = a^2 \cos^2 t. \quad \begin{bmatrix} E & F \\ F & G \end{bmatrix}^{-1} = \begin{bmatrix} E^{-1} & 0 \\ 0 & G^{-1} \end{bmatrix}$$

$$X_{tt} = (-a \cos t \cdot \cos \theta, -a \cos t \cdot \sin \theta, -c \cdot \sin t)$$

$$X_{t\theta} = (+a \sin t \cdot \sin \theta, -a \sin t \cdot \cos \theta, 0)$$

$$X_{\theta\theta} = (-a \cdot \cos t \cdot \cos \theta, -a \cos t \cdot \sin \theta, 0). \text{ Now,}$$

$$\Gamma_{tt}^t = \langle X_{tt}, X_t \rangle g^{tt} + \langle X_{tt}, X_\theta \rangle g^{t\theta} = (a^2 - c^2) \cdot \sin t \cdot \cos t / (a^2 \sin^2 t + c^2 \cos^2 t)$$

$$\Gamma_{\theta\theta}^t = \langle X_{\theta\theta}, X_t \rangle \cdot g^{tt} + \langle X_{\theta\theta}, X_\theta \rangle \cdot g^{t\theta} = a^2 \sin t \cos t / (a^2 \sin^2 t + c^2 \cos^2 t)$$

$$\Gamma_{t\theta}^\theta = \langle X_{t\theta}, X_t \rangle g^{t\theta} + \langle X_{t\theta}, X_\theta \rangle \cdot g^{\theta\theta} = -a^2 \sin t \cos t / a^2 \cos t = -\tan t.$$

$$\text{And, } \Gamma_{t\theta}^t = \Gamma_{t\theta}^\theta = \Gamma_{\theta\theta}^\theta = 0.$$

2025, Fall

1. a). Show that  $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$  is linear functional.

8m

b). Find a dual basis corresponding to  $\{1, t, \dots, t^n\}$  of  $P_n(\mathbb{R})$ .

Proof. Let  $f(x) = \sum_{i=0}^n a_i x^i$  and  $g(x) = \sum_{i=0}^n b_i x^i$  and  $C \in \mathbb{R}$  be given.

Then,  $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each  $k=0, 1, \dots, n$ ,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in  $\mathcal{B}^*$  are all linear functional, so the  $\mathcal{B}^*$  is dual basis.

2025. Fall

2. Let  $A \in M_{\text{max}}(1\mathbb{R})$ .

15m:  $A^n$ 의 rank,  $n=2$ 일 때부터 시작

a). For  $n \in \mathbb{N}$ ,  $\text{rank } A^n \leq \text{rank } A$ .

b). Let  $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$ .

Show that for  $n \geq 2$ ,

$$\text{rank } A^n = \text{rank } A \text{ implies } \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given  $x \in \text{Im } A^n$ ,  $x = A^n y$  for some  $y \in \mathbb{R}^m$ .

So,  $x = A^n y = A^{n-1} Ay \in \text{Im } A$ , for  $n \geq 2$ . Thus  $\text{Im } A^n \subseteq \text{Im } A$ , it follows  $\text{rank } A^n \leq \text{rank } A$ . If  $n=1$ , clear.

b). Given  $x \in \ker A$ ,  $A^n x = A^{n-1} Ax = A^{n-1} 0 = 0$ , so  $x \in \ker A^n$ , so  $\ker A \subseteq \ker A^n$ . Since  $\text{rank } A^n = \text{rank } A$ , the dimension theorem gives  $\text{nullity } A^n = \text{nullity } A$ .

Therefore, since  $\mathbb{R}^m$  is finite dimensional, so is  $\ker A^n$ , thus  $\ker A = \ker A^n$ .

Now, let  $x \in \ker L_A \cap \text{Im } L_A$  be given. Then,  $Ax = 0$  and  $x = Ay$  for some  $y \in \mathbb{R}^m$ .

Then,  $A^{n-1}x = A^n y = 0$ , because  $Ax = 0$ , so  $A^{n-2}Ax = 0$ , and this implies  $y \in \ker A^n$ .

Since  $\ker A^n = \ker A$ ,  $Ay = 0$ , so  $x = Ay = 0$ , so proved.  $\square$

2025. Fall

3. Find all  $a, b, c \in \mathbb{R}$  such that

6m

$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of  $A$  is regardless of  $a, b, c$ ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the  $A$  is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore,  $a=0$ ,  $b \in \mathbb{R}$ ,  $c \in \mathbb{R}$  are all possible  $(a, b, c)$ .

2025. Fall,

4. Let  $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$  be the set of all pairs  $(A, B)$  such that  $\det A = \pm \det B$ .

20m

Show that  $N$  is normal and determine the factor group  $(G \times G)/N$ .

Sol). Let  $(A, B) \in N$  and  $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$  be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

So, since  $(A, B) \in N$ ,  $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore  $(C, D)(A, B)(C, D)^{-1} \in N$ . And, to show  $N$  is subgroup,

first,  $(I, I) \in N$  trivially, so  $N \neq \emptyset$ . Let  $(A, B), (C, D) \in N$  be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies  $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$ .

therefore, by the subgroup criterion,  $N$  is subgroup, and normal.  $\perp$  hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B^{-1})^2$$

Then,  $\ker \mathcal{Q} = N$  and  $\mathcal{Q}$  is homomorphism, onto  $(0, \infty)$ , because

the det preserves multiplication, so is  $\mathcal{Q}$ , and onto is given by: fix  $B = I$ , and  $A = \text{diag}(r, 1, \dots, 1)$  where  $r > 0$  is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \xrightarrow{\perp} \text{hom.} \quad \mathcal{Q}$$

Comment:  $\mathcal{Q}$  은 먼저  $\mathbb{R}^{\times}$  구조 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

12-25. Fall

10m

5. Let  $G$  be an order 21 subgroup of  $\mathbb{C}^\times$ .

a)  $G$  is cyclic.

b). For any  $g \in G$ ,  $x^{12} = g^{15}$  has 3 distinct solutions in  $G$ .

Proof. a). Since  $\mathbb{C}$  is abelian, so  $G$  is also abelian.

By the Cauchy's theorem,  $G$  has order 3 element  $x$  and order 7 element  $y$ .  
Now, since 3 and 7 are relatively prime,  $|xy| = 21$ , so  $G = \langle xy \rangle$ .

b). By a),  $G \cong \mathbb{Z}/21\mathbb{Z}$ . So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of  $g$ ,  $\gcd(12, 21) = 3$  divides  $15g$ , so it has three distinct solutions.

Note:  $ax \equiv b \pmod{n}$  has a solution iff  $\gcd(a, n) \mid b$ .

And in such a case, it has  $\gcd(a, n)$  solutions.

2025. Fall.

6. Let  $\{a_n\}$  be a bounded sequence in  $\mathbb{R}$ . Define

22m

$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that  $\{a_n\}^*$  is closed.

b). Show  $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence  $\{b_n\}$  such that  $\{b_n\}^* = [0, 1]$ ?

Proof a). If  $\{a_n\}^*$  is finite, then clearly closed in  $\mathbb{R}$ .

Assume that  $\{a_n\}^*$  is infinite. Since  $\{a_n\}$  is bounded, so the set of subsequential limits  $\{a_n\}^*$  is bounded set. So it has a limit point.

Let  $\alpha$  be a limit point of  $\{a_n\}^*$ . Then, given  $\epsilon \in \mathbb{R}$ , there exists a

subsequential limit  $p$  of  $\{a_n\}$  such that  $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ .

For  $k=1$ , choose such a  $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ , and since  $p$  is a subsequential limit, choose  $a_{p_1} \in B_{\frac{1}{k}}(p)$ , then  $a_{p_1} \in B_{\frac{\epsilon}{2}}(\alpha)$ .

For chosen  $a_{p_1}, \dots, a_{p_{k-1}}$ , choose  $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ .

Then, similarly, there is  $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$  such that  $p_k > p_{k-1}$ .

because no such  $p_k$  exists, then it is contradiction to  $a_{p_k}$  is a subsequential limit. Therefore, the arbitrary limit point  $\alpha$  is contained in  $\{a_n\}^*$ ,

so it is closed.

b). Since a),  $\{a_n\}^*$  is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  such that  $n \geq N \Rightarrow a_n < \alpha + \epsilon$  (since  $\alpha = \max \{a_n\}^*$ )

and there exists infinitely many terms in  $\{a_n\}$  such that  $\alpha - \epsilon < a_n$  (since  $a_{p_k} \rightarrow \alpha$ )

Thus  $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let  $b_n$  be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given  $b \in [0, 1]$ , there exists a subsequence in  $\{b_n\}$  converges to  $b$ , because: for any  $\epsilon > 0$ , there are infinitely many terms in  $b_n$  such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let  $E$  be a nonempty bounded subset of a Metric space  $(X, d)$ .

10m

Define  $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove  $\text{diam } E = \text{diam } \bar{E}$

b). If  $E$  is compact, then  $d(x, y) = \text{diam } E$  for some  $x, y \in E$ .

Proof.

a). Since  $E \subseteq \bar{E}$ ,  $\text{diam } E \leq \text{diam } \bar{E}$  is clear.

To show the inverse inequality, using the property of supremum.

Given  $\varepsilon > 0$ , there exists  $x, y \in \bar{E}$  such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since  $\bar{E}$  is closure, there are  $x', y' \in E$  such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now,  $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$ .

Since  $\varepsilon > 0$  was arbitrary,  $\text{diam } \bar{E} \leq \text{diam } E$ , so proved. 7m.

b). Since  $E$  is compact set, so  $E \times E$  is also compact set.

Since the metric function  $d$  is continuous, it preserves compactness, so  $d[E \times E] \subseteq \mathbb{R}$  is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is  $(x', y') \in E \times E$  such that  $d(x', y') = \max d[E \times E]$  3m.

2025 Fall.

8. Let  $a > 1$ . Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore  $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

$$\begin{aligned} \text{Now, } \int_t^{at} \frac{\cos x}{x} dx &= \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx \\ &= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} \end{aligned}$$

(It is possible, because  $\sum (-1)^n x^{2n-1}/(2n)!$  converges for all  $x \in \mathbb{R}$ )

$$\text{Now, } \lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0, \text{ so}$$

the given limit is  $\ln a$ .

Revised in next

Observation for  $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$ .

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all  $x \neq 0$ . Therefore, for any  $t > 0$ ,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \end{aligned}$$

Meanwhile, the series converges uniformly in  $[t, a]$  for small enough  $t > 0$ , specifically,  $t < \frac{1}{a}$ , then for each  $n \in \mathbb{N}$ ,

$$\left| (-1)^n \frac{x^{2n-1}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left( \frac{1}{2n \cdot (2n)!} \cdot (a^{2n} t^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note:  $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$ ,  $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} \rightarrow 0$  as  $t \rightarrow 0$ .

2025. Fall,

9. (10 pts) Consider the following differential equation with independent variable  $t$ :

$$(1+t^2)y'' + ty' + y = 0, \quad \begin{cases} y(0) = 0, \\ y'(0) = 1. \end{cases}$$

Assume that a solution  $y$  has a power series expansion of the form

$$y(t) = \sum_{n=0}^{\infty} a_n t^n.$$

- (a) Determine the first five coefficients:  $a_0, a_1, a_2, a_3, a_4$ .
- (b) Prove that the power series  $y(t)$  converges for  $|t| < 1$ .

Proof.

2025. Fall.

12. Define an equi. rel on  $\mathbb{R}$  as  $x \sim y \Leftrightarrow x - y \in \mathbb{Q}$ .

" as  $x \sim y \Leftrightarrow x = y$  or  $x - y \in \mathbb{Q}$ .

a). Describe the quotient topology on  $\mathbb{R}/\sim$ .

b). " on  $\mathbb{R}/\sim$ .

c). In  $\mathbb{R}/\sim$ , determine if every point is closed.

d). " the image of the set of irrationals under quotient map is compact.

Proof. Since  $\mathbb{Q}$  is closed under the operation  $-$ , so these relations are equi. rel.

a). First,  $\mathbb{R}/\sim = \{a + \mathbb{Q} \mid a \in \mathbb{R}\}$ , where  $a + \mathbb{Q} = \{a + q \mid q \in \mathbb{Q}\}$ . And, given a basis element  $(a, b) \in \mathbb{R}$ , for the projection  $\pi: \mathbb{R} \rightarrow \mathbb{R}/\sim: a \mapsto a + \mathbb{Q}$ .

$$\pi^{-1}[\pi[(a, b)]] = \bigcup_{r \in (a, b)} r + \mathbb{Q} = \bigcup_{q \in \mathbb{Q}} (a + q, b + q)$$

is open and since  $\pi$  is quotient map, so  $\pi[(a, b)]$  is open, so  $\pi$  is open. Therefore, the topology on  $\mathbb{R}/\sim$  is generated by

$$\mathcal{B}_\sim = \{\pi[(a, b)] \mid a < b\} = \{(a, b) + \mathbb{Q} \mid a < b\}.$$

But, by the density of rationals, for any  $a < b$ ,  $(a, b) + \mathbb{Q} = \mathbb{R}/\mathbb{Q}$ . Therefore the topology on  $\mathbb{R}/\sim$  is indiscrete.

2025, Fall.

13. Consider a regular plane curve  $C: I \rightarrow \mathbb{R}^2$ .

a). Give a formula for the normal  $n$  to  $C$ , and define the signed curvature.

b). Explain the relation between the total angle change and the signed curvature.

c). Assume  $C$  is a closed plane curve parametrized by arc-length, and  $I = [0, 2\pi]$ .

Prove or disprove that: there are  $t, t' \in I$  such that  $C'(t) = -C'(t')$ .

and if  $t=0$ , there is such a  $t' \in I$ ?

Moreover, show that  $\int_0^{2\pi} |K(s)| ds \geq 2\pi$  where  $K(s)$  is the signed curvature.

Proof. Denote  $C(t) = (x(t), y(t))$ .

a). The tangent vector is

$$T(t) = \frac{C'(t)}{|C'(t)|} = \frac{(x'(t), y'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

So, the normal vector is

$$N(t) = \frac{(-y'(t), x'(t))}{\sqrt{x'(t)^2 + y'(t)^2}}$$

And, since the curvature  $K(s(t))$  satisfies  $T'(t) = K(t) \cdot N(t)$ ,

$$K(t) = K(t) \langle N(t), N(t) \rangle = \langle T'(t), N(t) \rangle$$

Pebris,

Denote  $C = (x, y)$ . Then,

$$T = (x', y') / |(x', y')| = \frac{(x', y')}{\sqrt{x'^2 + y'^2}}$$

is the tangent vector, and the normal vector is

$$N = \frac{(-y', x')}{\sqrt{x'^2 + y'^2}}$$

Moreover, the signed curvature  $K$  satisfies  $K N = T'$ ,

so  $K = K \cdot \langle N, N \rangle = \langle T', N \rangle \cdot |C'(t)|^{-1}$  Meanwhile,

$$T' = \frac{(x'', y'')}{\sqrt{x'^2 + y'^2}} + (x', y') \cdot \left( -\frac{1}{2} \cdot \frac{2x'x'' + 2y'y''}{(x'^2 + y'^2)^{\frac{3}{2}}} \right) \text{ so}$$

$$K = \frac{1}{x'^2 + y'^2} \cdot (x'y'' - x''y') \cdot \frac{1}{\sqrt{x'^2 + y'^2}}$$

TSNU. Q. 16.

Prove that  $\lim_{b \rightarrow \infty} \int_0^b \frac{\sin x}{x} dx$  exists.

Proof. First, let  $a_n = n\pi$  ( $n \in \mathbb{N}$ ). Then, a sequence  $\left\{ \int_0^{a_n} \frac{\sin x}{x} dx \right\}$  converges, because:

$\left| \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx \right|$  is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx$  is positive if  $n$  is even and is negative if  $n$  is odd.

That is,  $\int_0^{n\pi} \frac{\sin x}{x} dx = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} \frac{\sin x}{x} dx$  is alternating series, so converges.

2026. Spring.

1. Find a  $2 \times 2$  matrix  $A$  over  $\mathbb{C}$  such that

$$A^{2025} = \begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix}$$

Proof. Since  $\det(tI - A) = (t-2)(t-5)$ ,  $\begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix}$  is diagonalizable to

$$\begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix} = P \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} P^{-1} \text{ where } P = \begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix}$$

Therefore,  $A = P \begin{bmatrix} 2^{1/2025} & 0 \\ 0 & 5^{1/2025} \end{bmatrix} P^{-1}$  is the matrix we desired.  $\square$

$$\begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \cdot \frac{1}{3} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 6 & 0 \\ -4 & 5 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 6 & 0 \\ 6 & 15 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 2 & 5 \end{bmatrix}.$$

$$\text{Ker} \begin{bmatrix} 0 & 0 \\ 2 & 3 \end{bmatrix} = \text{span} \left\{ \begin{pmatrix} 3 \\ -2 \end{pmatrix} \right\}, \quad \text{Ker} \begin{bmatrix} -3 & 0 \\ 2 & 0 \end{bmatrix} = \text{span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

2026, Spring.

2. Let  $W$  be the subspace of  $\mathbb{R}^4$  spanned by all solutions of  
 $x_1 + x_2 - x_3 - x_4 = 0$ .

a). Find a basis for  $W^\perp$ .

b). Find  $V_1 \in W$  and  $V_2 \in W^\perp$  such that  $(1, 1, 0, 1) = V_1 + V_2$ .

Proof Let  $T: \mathbb{R}^4 \rightarrow \mathbb{R}; \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \mapsto (1 \ 1 \ -1 \ -1) \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$ . Then,  $W = \ker T$ .

Since  $(\ker T)^\perp = \text{Im } T^*$ , because: given  $\alpha \in \text{Im } T^*$ , and  $\beta \in \ker T$ ,

$$\langle \alpha, \beta \rangle = \langle T(\gamma), \beta \rangle = \langle \gamma, T(\beta) \rangle = \langle \gamma, 0 \rangle = 0 \text{ for some } \gamma,$$

so that  $\text{Im } T^* \subseteq (\ker T)^\perp$ , and since  $\text{rank } T^* = \text{rank } T$  and  
 $\dim(\ker T)^\perp = \text{rank } T$ , so  $\text{Im } T^* = (\ker T)^\perp$ . Now, for

$$T^*: \mathbb{R} \rightarrow \mathbb{R}^4; x \mapsto \begin{pmatrix} x \\ x \\ -x \\ -x \end{pmatrix}.$$

$\text{Im } T^* = \text{span} \left( \begin{pmatrix} 1 \\ 1 \\ -1 \\ -1 \end{pmatrix} \right) = (\ker T)^\perp$ , so  $\{(1, 1, -1, -1)\}$  is the basis we desired,

b). Since  $\{(1, 1, -1, -1)\}$  spans  $(\ker T)^\perp$ , so orthogonal projection of  $(1, 1, 0, 1)$  onto  $(\ker T)$  is

$$(1, 1, 0, 1) - \frac{1}{4} \cdot 1 \cdot (1, 1, -1, -1) = \left( \frac{3}{4}, \frac{3}{4}, \frac{1}{4}, \frac{5}{4} \right) \in (\ker T) = W$$

Therefore,  $V_1 = \frac{1}{4}(3, 3, 1, 5)$  and

$$V_2 = \left( \frac{1}{4}, \frac{1}{4}, -\frac{1}{4}, -\frac{1}{4} \right)$$

are we desired.  $\square$

7226. Prima.

3. Let  $V$  be an  $n$ -dimensional vector space over  $\mathbb{C}$ ,  
Let  $T: V \rightarrow V$  be a linear operator such that  $T^3 = T$ .

27m

a). Find all possible characteristic polynomials and corresponding minimal polynomials for  $T$ .

b). Prove  $V = \ker T \oplus \operatorname{Im} T$ .

Proof.

a). Since  $T^3 - T = 0$ , the minimal polynomial  $m(t)$  for  $T$  divides  $t^3 - t = 0$ .  
 $t^3 - t = t(t-1)(t+1)$ , there are possibilities:

$$m(t) = t \text{ or } t-1 \text{ or } t+1 \text{ or } t(t-1) \text{ or } t(t+1) \text{ or } (t-1)(t+1) \text{ or } t(t-1)(t+1).$$

Since  $m(t)$  divides the characteristic polynomial and they have same characteristic values, the possible characteristic polynomial is:

$$t^{r_1} \cdot (t-1)^{r_2} \cdot (t+1)^{r_3} \text{ where } r_1, r_2, r_3 \text{ are non-negative integers and } r_1 + r_2 + r_3 = n.$$

If  $r_1 = 0$  and  $r_2, r_3 \neq 0$ , then the corresponding minimal polynomial is  
 $(t-1)(t+1)$

and other cases are similar.

b). ~~Given  $\alpha \in \ker T \cap \operatorname{Im} T$ ,  $T\alpha = 0$  and  $\alpha = T\beta$  for some  $\beta \in V$ .~~

Given  $\alpha \in V$ ,  $\alpha = T^2\alpha + \alpha - T^2\alpha$  and since  $T(\alpha - T^2\alpha) = (T - T^3)\alpha = 0\alpha = 0$ ,  
so  $\alpha - T^2\alpha \in \ker T$  and trivially  $T^2\alpha \in \operatorname{Im} T$ . Moreover, if  $T\alpha = 0$  and  $\alpha \in \operatorname{Im} T$ ,  
Then  $\alpha = T\beta = T^3\beta$  for some  $\beta \in V$  and  $T\alpha = T^2\beta = 0$  so  $T^3\beta = TT^2\beta = \alpha = 0$ ,  
so  $V = \ker T \oplus \operatorname{Im} T$ .

Note: Consider a case of  $T^2 + T + I = 0$ . Does still hold  $V = \ker T \oplus \operatorname{Im} T$ ?

Modified.

Let  $T$  be a linear operator on the fin. dim space  $V$  over  $\mathbb{C}$  satisfying  $T^4 = T^2$ .

a). Prove that  $V = \ker T^2 \oplus \operatorname{Im} T^2$ .

b). Describe  $T|_{\operatorname{Im} T^2}$ .

c). Prove that  $T$  is diagonalizable  $\Leftrightarrow \ker T = \ker T^2$ .

observe:  $T^4 - T^2 = 0$ , so the minimal polynomial divides  $t^2(t-1)(t+1)$ .

a). Given  $\alpha \in V$ ,  $\alpha = T^2\alpha + \alpha - T^2\alpha$  and  $T^2\alpha \in \operatorname{Im} T^2$  clearly, and  $T^2(I - T^2)\alpha = (T^2 - T^4)\alpha = 0 \cdot \alpha = 0$ , so  $\alpha - T^2\alpha$ , which proves  $\alpha - T^2\alpha \in \ker T^2$ , so  $V = \ker T^2 + \operatorname{Im} T^2$ . Moreover, given  $\alpha \in \ker T^2 \cap \operatorname{Im} T^2$ ,  $T^2\alpha = 0$  and  $\alpha = T^2\beta$  for some  $\beta \in V$ , so

$$\alpha = T^2\beta = T^4\beta = T^2T^2\beta = T^2\alpha = 0$$

so  $V = \ker T^2 \oplus \operatorname{Im} T^2$ .

b). Given  $\alpha \in \operatorname{Im} T^2$ ,  $\alpha = T^2\beta$  for some  $\beta \in V$ . so,  $T\alpha = T^3\beta$ .

Multiplying  $T$ , then  $T^2\alpha = T^4\beta = T^2\beta$ , so  $T^2(\alpha - \beta) = 0$ .

Therefore,  $\alpha - \beta \in \ker T^2$ , but since  $\alpha \in \operatorname{Im} T^2$  and  $\operatorname{Im} T^2 \cap \ker T^2 = 0$ ,

$\beta = \alpha + \gamma$ , where  $\gamma \in \ker T^2$ . Now,  $T\alpha = T^3\beta = T^3\alpha + T^3\gamma = T^3\alpha$ ,

so,  $T|_{\operatorname{Im} T^2} = T^3|_{\operatorname{Im} T^2}$ , moreover, since  $T^2 = T^4$ ,  $T^2|_{\operatorname{Im} T^2} = I$ .

c). Since  $T$  is diagonalizable  $\Leftrightarrow$  minimal polynomial of  $T$  splits and separable.

That is, the minimal polynomial is  $t^a(t-1)^b(t+1)^c$ ,  $a, b, c \in \{0, 1\}$ .

$$x^4 + 2x + 3 \text{ over } \mathbb{F}_7$$

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5. Let  $f(x) = x^4 + 2x + 3 \in \mathbb{F}_7[x]$  and  $K = \mathbb{F}_7[x]/(f)$ .

Let  $\mathcal{Q}: K \rightarrow K: a \mapsto a^7$ .

a). Prove  $K$  is a field and  $\mathcal{Q}$  is Automorphism.

b). Let  $G = \langle \mathcal{Q} \rangle \leq \text{Aut}(K)$ . Determine the number of  $a \in K$  such that  $|\text{Orb}(a)| = 2$ .

Proof

a). Since  $f(0)=3, f(1)=6, f(2)=2, f(3)=6, f(4)=1, f(5)=1, f(6)=2$ , so  $f(x)$  has no linear factor. So, if  $f(x)$  is reducible,

$$f(x) = (x^2 + ax + b)(x^2 + cx + d) \text{ for some } a, b, c, d \in \mathbb{F}_7.$$

Then, 
$$\begin{cases} a+c=0 & \text{If } a=0 \text{ or } c=0, \text{ then } a=c=0, \text{ so } 0=2. \\ ac+b+d=0 & \text{If } b=0 \text{ or } d=0, \text{ then } 0=3, \\ ad+bc=2 & \text{Therefore, } a, b, c, d \text{ are non-zero.} \\ bd=3 & \text{Reduce the system:} \end{cases}$$

$$\Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases}$$

If  $b=d$ , then  $b^2=3$ , but  $\sqrt{3} \notin \mathbb{F}_7$ , so  $b \neq d$ .

And,  $3 = 1 \cdot 3$  or  $2 \cdot 5$  or  $4 \cdot 6$ , so

$a^2 = 4$  or  $0$  or  $3$ . Since  $\sqrt{3} \notin \mathbb{F}_7$  and  $a \neq 0$ ,  $a^2 = 4$ , so  $a = 2$  or  $5$ ,

with  $\{b, d\} = \{1, 3\}$ . But,  $2 \cdot (1-3) = 3 \neq 2$ ,  
 $2 \cdot (3-1) = 4 \neq 2$ ,  
 $5 \cdot (1-3) = 4 \neq 2$ ,  
 $5 \cdot (3-1) = 3 \neq 2$

So contradiction. Therefore,  $f$  must be an irreducible.

Since  $\mathbb{F}_7$  is a field,  $\mathbb{F}_7[x]$  is p.z.d., so  $(f)$  is maximal, so  $\mathbb{F}_7[x]/(f)$  is a field. And, since the characteristic of  $K$  is 7, for any  $a, b \in K$ ,

$$(a+b)^7 = \sum_{k=0}^7 \binom{7}{k} a^{7-k} b^k = a^7 + b^7$$

and  $(ab)^7 = a^7 b^7$ , so  $\mathcal{Q}$  is Homomorphism. Moreover,

$\mathcal{Q}(a) = 0 \Leftrightarrow a^7 = 0 \Leftrightarrow a = 0$ , so  $\ker \mathcal{Q}$  is trivial, so  $\mathcal{Q}$  is 1-1.

Now, since  $[K: \mathbb{F}_7] = \deg f = 4$ , so  $\#K = 7^4$ , finite, so  $\mathcal{Q}$  is bijective.

b). Since every finite field is perfect, so  $K$  is Galois over  $\mathbb{F}_7$ ,  
so  $|\text{Aut}(K)| = \deg f = 4$ ,

Consider an Action

$$G \times K \rightarrow K : (\mathbb{Q}^n, \alpha) \mapsto \mathbb{Q}^n(\alpha).$$

So  $|\text{Orb}(\alpha)| = 2 \iff \mathbb{Q}^2(\alpha) = \mathbb{Q}$  and  $\alpha \neq 0 \iff \alpha^{4q} = \alpha$  and  $\alpha \neq 0$   
So, the fixed field by  $\langle \mathbb{Q}^2 \rangle$ ,  $\langle \mathbb{Q}^2 \rangle \setminus \{0\}$  is we desired.

Revised in the next

Revised.

Prove that  $f(x) = x^4 + 2x + 3$  is irreducible over  $\mathbb{F}_7$ .

And describe the Galois group of  $f(x)$ .

1).  $f(x)$  has no linear factor.

$f(0) = 3 \neq 0$ . And, since  $\mathbb{F}_7^\times$  is a group of order 6 under the multiplication. Therefore, for any  $\alpha \in \mathbb{F}_7^\times$ ,  $\alpha^6 = (\alpha^3)^2 = 1$  implies  $\alpha^3 = \pm 1$ .

If  $\alpha^3 = 1$ , then  $\alpha^4 = \alpha$ , so  $f(\alpha) = \alpha^4 + 2\alpha + 3 = 3(\alpha + 1) = 0$  implies  $\alpha = -1$  but  $(-1)^3 = -1 = 6 \neq 1$ , so  $\alpha^3 \neq 1$ .

If  $\alpha^3 = -1$ , then  $\alpha^4 = -\alpha$ , so  $f(\alpha) = \alpha + 3 = 0$  implies  $\alpha = 4$ , but  $4^3 = 64 = 1 \neq -1$ , so  $\alpha^3 \neq -1$ . Now,  $f(x)$  has no root in  $\mathbb{F}_7^\times$ .

2).  $f(x)$  has no factor of degree 2.

Suppose that  $f(x) = (x^2 + ax + b)(x^2 + cx + d)$ . Then, we have a system:

$$\begin{cases} a+c=0 \\ ac+b+d=0 \\ ad+bc=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} b+d=a^2 \\ a(d-b)=2 \\ bd=3 \end{cases} \Rightarrow \begin{cases} (d-b)^2 = (b+d)^2 - 4bd = a^4 - 12 = a^4 + 2 \\ a^2(d-b)^2 = 4 \end{cases}$$

$\alpha \neq 0$ , so  $\alpha^6 = 1$

$\Rightarrow \alpha^6 + 2\alpha^2 = 1 + 2\alpha^2 = 4 \Rightarrow \alpha^2 = 5$ . But,  $\{x^2 | x \in \mathbb{F}_7\} = \{0, 1, 4, 2\}$ , so  $\sqrt{\alpha} \notin \mathbb{F}_7$ .

Now,  $f$  is irreducible  $\Rightarrow \mathbb{F}_7[x]/(f)$  is a field.

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7. Let  $f: (a-R, a+R) \rightarrow \mathbb{R}$  be an analytic function given by

$$f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n \quad (a, C_n \in \mathbb{R}).$$

with radius of convergence  $R > 0$ .

a). Derive the power series for

$$F(x) = \int_a^x f(t) dt.$$

b). Prove that  $F$  is analytic on  $(a-R, a+R)$  and determine its radius of conver.

Proof. Let  $f_N = \sum_{n=0}^N C_n (x-a)^n$ .

a). Given  $x \in (a-R, a+R)$ ,  $f_N(x)$  converges and for all  $|a-t| \leq |a-x|$ ,

$$|C_n| \cdot |t-a|^n \leq |C_n| \cdot |x-a|^n \text{ and } \sum_{n=0}^{\infty} |C_n| \cdot |x-a|^n \text{ converges}$$

So the Weierstrass M-test gives  $f_N \rightarrow f$  uniformly on  $|a-t| \leq |a-x|$ .

Therefore, by the theorem for integrability of uniformly convergence functions,

$$\int_a^x f(x) dx = \int_a^x \lim_{N \rightarrow \infty} f_N(x) dx = \lim_{N \rightarrow \infty} \int_a^x f_N(x) dx = \sum_{n=0}^{\infty} \frac{C_n}{n+1} \cdot (x-a)^{n+1}$$

$$b). \text{ Since } \limsup_{n \rightarrow \infty} \left( \frac{|C_n|}{n+1} \right)^{\frac{1}{n+1}} = \limsup_{n \rightarrow \infty} |C_n|^{\frac{1}{n}},$$

the Radius of Convergence of  $F$  equals to  $f$ , and thus it is also analytic on  $(a-R, a+R)$ .

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8. Let  $\mathbb{D}$  be the unit open disc in  $\mathbb{C}$ . Prove or disprove:

Every holomorphic bijection  $f: \mathbb{D} \rightarrow \mathbb{D}$  has a point  $w \in \mathbb{D}$  such that  $f(w) = w$ .

Proof. Let  $f: \mathbb{D} \rightarrow \mathbb{D}$  be a holomorphic bijection.

Suppose that for all  $w \in \mathbb{D}$ ,  $f(w) \neq w$ . Then, a map

$$z \mapsto \frac{1}{f(z) - z} \quad (z \in \mathbb{D}) \text{ is well-defined,}$$

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13. (10 pts.) Let  $S^1 = \{z \in \mathbb{C} \mid |z| = 1\}$  and  $X$  a topological space.

(a) Let  $f, g : S^1 \rightarrow X$  be homotopic maps.

Show that the spaces  $X \cup_f D^2$  and  $X \cup_g D^2$  obtained by attaching a disc  $D^2$  to  $X$  using  $f$  and  $g$ , respectively, are homotopy equivalent.

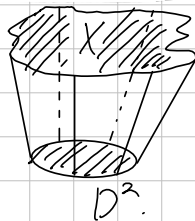
(b) Let  $h : S^1 \rightarrow S^1$  be given by  $h(e^{2\pi i s}) = e^{8\pi i s}$  for  $0 \leq s \leq 1/2$  and  $h(e^{2\pi i s}) = e^{-4\pi i s}$  for  $1/2 \leq s \leq 1$ . Show that  $S^1 \cup_h D^2$  is contractible.

Proof. Define a map

$h : X \cup_f D^2 \rightarrow X \cup_g D^2$ :

$$[\cdot] \mapsto \begin{cases} [x]_g & \text{if } [\cdot] = [x]_f \text{ where } x \in X \\ [2r \cdot e^{i\theta}]_g & \text{if } 0 \leq r \leq \frac{1}{2} \\ [F(e^{i\theta}, 2-2r)]_g & \text{if } \frac{1}{2} \leq r \leq 1 \end{cases}$$

if  $[\cdot] = [r \cdot e^{i\theta}]_f$



Observe:  $i : S^1(CD^2) \rightarrow D^2 : z \mapsto z$  is an inclusion.

$f \sim g \Leftrightarrow$  A Homotopy  $F : S^1 \times I \rightarrow X$  st  $F(z, 0) = f(z)$  and  $F(z, 1) = g(z)$  exists.

$S^1 \neq D^2$

$$\begin{array}{ccccc} X & \xleftarrow{g} & S^1 & \xrightarrow{f} & X \\ \downarrow \iota_X & & \downarrow i & & \downarrow \iota_X \\ X \cup_g D^2 & \xleftarrow{\quad} & D^2 & \xrightarrow{\quad} & X \cup_f D^2 \end{array}$$

⤵

2026 Fall.

2. Let  $A \in \text{Mat}_4(\mathbb{R})$  be a symmetric with eigenvalues  $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \lambda_4$ .

Let 
$$P = \frac{1}{2} \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}$$

a) Prove that  $P$  is an orthogonal projection.  
 b) prove  $\text{tr}(AP) \leq \lambda_1 + \lambda_2$   
 c)  $\max_{\substack{x, y \in \mathbb{R}^4 \\ \|x\| = \|y\| = 1 \\ \langle x, y \rangle = 0}} x^T A x + y^T A y = \lambda_1 + \lambda_2$ .

a). By just calculation,  $P^2 = P = P^T$ , and in  $\mathbb{R}^4$ ,  $P^T = P^*$ , so  $P$  is an orthogonal projection,  $\text{rank } P = 2$ , because the row vectors space has dimension 2.

b). Since  $P$  is orthogonal projection,  $\mathbb{R}^4 = \text{Im } P \oplus \text{Ker } P$ , with  $(\text{Ker } P)^\perp = \text{Im } P$ , and  $\text{rank } P = \text{nullity } P = 2$ . Since  $A$  is symmetric, there are four eigenvectors  $\{\alpha_1, \dots, \alpha_4\}$  corresponding to  $A\alpha_i = \lambda_i \alpha_i$  ( $i=1, \dots, 4$ ) and  $\{\alpha_1, \dots, \alpha_4\}$  is orthonormal.

And, let  $\{v_1, v_2\}$  be a basis for  $\text{Im } P$  and  $\{v_3, v_4\}$  be a basis for  $\text{Ker } P$ , satisfying  $\{v_1, \dots, v_4\}$  is orthonormal.

Since the trace is invariant under the similar, for  $B = (v_1 \dots v_4)$ ,

$$\begin{aligned} \text{tr}(AP) &= \text{tr}(B^T A P B) = v_1^T A P v_1 + \dots + v_4^T A P v_4 = v_1^T A v_1 + v_2^T A v_2 \\ &= \langle A v_1, v_1 \rangle + \langle A v_2, v_2 \rangle \end{aligned}$$

Now, since  $v_1, v_2 \in \langle \alpha_1, \dots, \alpha_4 \rangle$  and  $\|v_1\| = \|v_2\| = 1$  and  $\langle v_1, v_2 \rangle = 0$ ,  $v_1 = a_1 \alpha_1 + \dots + a_4 \alpha_4$  and  $v_2 = b_1 \alpha_1 + \dots + b_4 \alpha_4$  such that  $a_1^2 + \dots + a_4^2 = b_1^2 + \dots + b_4^2 = 1$  and  $a_1 b_1 + \dots + a_4 b_4 = 0$ . Now,

$$\begin{aligned} \langle A v_1, v_1 \rangle + \langle A v_2, v_2 \rangle &= a_1^2 \lambda_1 + \dots + a_4^2 \lambda_4 + b_1^2 \lambda_1 + \dots + b_4^2 \lambda_4 \\ &= \lambda_1 (a_1^2 + b_1^2) + \dots + \lambda_4 (a_4^2 + b_4^2) \end{aligned}$$

But, for each  $1 \leq i \leq 4$ ,  $a_i^2 + b_i^2 \leq 1$  and  $a_1^2 + \dots + a_4^2 + b_1^2 + \dots + b_4^2 = 2$ , therefore  $\leq \lambda_1 \cdot 1 + \lambda_2 \cdot 1 = \lambda_1 + \lambda_2$ . It proves b)

c) Given  $x, y \in \mathbb{R}^4$  such that  $\langle x, y \rangle = 0$ , with  $\|x\| = \|y\| = 1$ ,  $x^T A x + y^T A y = \langle A x, x \rangle + \langle A y, y \rangle \leq \lambda_1 + \lambda_2$ , by the argument in b). And, take  $x = \alpha_1$  and  $y = \alpha_2$ , we obtain c).

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3. Find all prime and Maximal ideals of  $\mathbb{Z} \times \mathbb{Z}$

Proof. Notice that: an ideal is Prime if and only if the quotient is integral domain.

2026 Fall.

4. Let  $G$  be a group and  $H_1, H_2 \leq G$  be subgroups of finite index.

a). Prove that the map

$$T: G \rightarrow S_{G/H_1 \times G/H_2} : x \mapsto T_x \text{ where } T_x(aH_1, bH_2) = (xaH_1, xbH_2)$$

is Homomorphism and  $\text{Ker } T \subseteq H_1 \cap H_2$ .

b). Prove  $H_1 \cap H_2$  has finite index.

Proof

$$\begin{aligned} \text{a). Given } g_1, g_2 \in G, T(g_1 g_2)(aH_1, bH_2) &= (g_1 g_2 aH_1, g_1 g_2 bH_2) \\ &= T(g_1) \circ T(g_2)(aH_1, bH_2) \end{aligned}$$

for each  $a, b \in G$ , so it preserves the operation.

$$\text{And } g \in \text{Ker } T \Leftrightarrow T_g = \text{id} \Leftrightarrow \text{for all } a, b \in G, T_g(aH_1, bH_2) = (gaH_1, gbH_2) = (aH_1, bH_2)$$

$$\begin{aligned} &\Leftrightarrow \text{for all } a, b \in G, a^{-1}ga \in H_1 \text{ and } b^{-1}gb \in H_2 \\ &\Rightarrow g \in H_1 \cap H_2 \text{ (take } a=b=1_G) \end{aligned}$$

b). By the first-isomorphism theorem,

$$G/\text{Ker } T \cong \text{Im } T \leq S_{G/H_1 \times G/H_2}$$

and since  $H_1, H_2$  have finite index, so the Codomain of  $T$  is finite set.

$$\text{Now, } |G : H_1 \cap H_2| \leq |G : \text{Ker } T| \leq \# S_{G/H_1 \times G/H_2}$$

because  $\text{Ker } T \subseteq H_1 \cap H_2$ , so proved.

Note: If  $A \leq B \leq G$  and  $|G:A|$  is finite, then

$$G/A \rightarrow G/B : gA \mapsto gB$$

is well-defined and onto, because: Since  $A \leq B$ ,

$$g_1 A = g_2 A \Leftrightarrow g_1 g_2^{-1} \in A \leq B \Rightarrow g_1 g_2^{-1} \in B \Leftrightarrow g_1 B = g_2 B$$

and surjectiveness is clear.

12026 Fall,

5. Let  $F = \mathbb{Q}(\sqrt{2+\sqrt{2}})$ .

a). Prove that  $F = \mathbb{Q}(\cos \frac{\pi}{8})$

b) Prove that  $F$  is normal extension of  $\mathbb{Q}$  and find  $\text{Gal}(F/\mathbb{Q})$ .  
Proof.

a). Observe that  $\cos \frac{\pi}{4} = \cos^2 \frac{\pi}{8} - \sin^2 \frac{\pi}{8} = 2\cos^2 \frac{\pi}{8} - 1$ , so that

$$\cos \frac{\pi}{8} = \frac{1}{\sqrt{2}} \cdot \sqrt{\frac{\sqrt{2}}{2} + 1} = \sqrt{\frac{1}{2} + \frac{1}{\sqrt{2}}} = \frac{1}{2} \cdot \sqrt{2+\sqrt{2}}$$

Therefore,  $F = \mathbb{Q}(\cos \frac{\pi}{8})$ .

b). Let  $\zeta = \exp(2\pi i/16)$ . Then,  $\cos \frac{\pi}{8} = (\zeta + \zeta^{-1})/2$ . Let  $E = \mathbb{Q}(\zeta)$ .

Since  $\zeta$  is 16th primitive root of unity,  $E$  is the splitting field over  $\mathbb{Q}$  of

$$(x - \zeta)(x - \zeta^3) \cdots (x - \zeta^{15}) \in \mathbb{Q}[x]$$

and  $\text{Gal}(E/\mathbb{Q}) \cong (\mathbb{Z}/16\mathbb{Z})^\times$ , abelian group.

Since  $(\zeta + \zeta^{-1})/2 \in \mathbb{Q}(\zeta)$ , so that  $\mathbb{Q} \subset F \subset E$ . Meanwhile, observe

$$x^2 - (\zeta + \zeta^{-1})x + \zeta \cdot \zeta^{-1} = x^2 - (\zeta + \zeta^{-1})x + 1 = (x - \zeta)(x - \zeta^{-1}) \in F[x]$$

is irreducible over  $F$  because  $\zeta, \zeta^{-1} \notin F$ , so  $E$  is the splitting field of  $x^2 - 2\cos \frac{\pi}{8}x + 1$  over  $F$ , so that  $[E:F] = 2$ , thus  $[F:\mathbb{Q}] = 4$ .

And, since  $\text{Gal}(E/\mathbb{Q}) = \{\sigma_a \mid 1 \leq a \leq 16, \gcd(a, 16) = 1\}$

where  $\sigma_a(\zeta) = \zeta^a$ , and  $\sigma_{15}(\zeta + \zeta^{-1}) = \zeta^{-1} + \zeta$ , therefore

$$F = \mathbb{Q}(\zeta + \zeta^{-1}) = E^{\langle \sigma_{15} \rangle}$$

Now, since  $\text{Gal}(E/F)$  is abelian,  $F/\mathbb{Q}$  is normal extension and

$$\text{Gal}(F/\mathbb{Q}) = \text{Gal}(E^{\langle \sigma_{15} \rangle}/\mathbb{Q}) \cong \text{Gal}(E/\mathbb{Q}) / \langle \sigma_{15} \rangle \cong \mathbb{Z}/4\mathbb{Z}. \quad \square$$

Note:  $\Phi_n(x) = \prod_{\substack{1 \leq a \leq n \\ \gcd(a, n) = 1}} (x - \zeta^a)$  so that  $x^n - 1 = \prod_{d|n} \Phi_d(x)$ .

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8. For  $f \in C([0,1])$ , define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x < 0 \\ f(x) & \text{if } 0 \leq x \leq 1 \\ f(1) & \text{if } x > 1 \end{cases}$$

For each  $n \geq 1$ , define

$$p_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define  $T_n f(x) := \int_{\mathbb{R}} p_n(x-y) \tilde{f}(y) dy$ ,  $x \in \mathbb{R}$ .

a). Show that  $T_n f \rightarrow f$  on  $[0,1]$ .

b). Determine  $T_n f \rightarrow f$  uniformly on  $[0,1]$  or not.

Proof. Given  $x \in \mathbb{R}$  and  $n \in \mathbb{N}$ ,  $x-y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n} + x \leq y \leq \frac{1}{n} + x$ . Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y) dy.$$

Since  $\tilde{f}$  is continuous on  $[-1,2]$ , so it is continuously uniformly.

Given  $\varepsilon > 0$ , take  $\delta > 0$  such that  $\forall x, t \in [0,1]$ ,  $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$ .

Now, given  $x \in [0,1]$ , and  $n > \frac{1}{\delta}$ ,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y) dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} f(x) dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} [\tilde{f}(y) - f(x)] dy \right| \\ &\leq \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)| dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore,  $\sup_{x \in [0,1]} |T_n f - f| \rightarrow 0$  as  $n \rightarrow \infty$ , so converges uniformly.

2026 Fall

a. Let  $f \in C([0, \pi])$  and define

$$(Tf)(x) := \int_0^\pi \min\{x, y\} \cdot f(y) \, dy, \quad 0 \leq x \leq \pi$$

a). Show that  $Tf$  is twice differentiable on  $(0, \pi)$  and satisfies

$$(Tf)'(x) = \int_x^\pi f(y) \, dy, \quad (Tf)''(x) = -f(x)$$

b). Suppose that  $f \neq 0$  and  $\lambda \neq 0$  satisfy  $Tf = \lambda f$ .

Show that  $\lambda > 0$  and that  $f$  satisfies

$$-f''(x) = \frac{1}{\lambda} \cdot f(x), \quad 0 < x < \pi$$

With the boundary conditions  $f(0) = 0$ ,  $f'(\pi) = 0$ .

c). Determine all  $\lambda > 0$  and all non-trivial functions  $f$  satisfying the differential equation and the boundary conditions in (b).

Proof. Observe that: given  $0 \leq x \leq \pi$ ,

$$\begin{aligned} \int_0^\pi \min\{x, y\} \cdot f(y) \, dy &= \int_0^x \min\{x, y\} \cdot f(y) \, dy + \int_x^\pi \min\{x, y\} \cdot f(y) \, dy \\ &= \int_0^x y f(y) \, dy + x \cdot \int_x^\pi f(y) \, dy \end{aligned}$$

a). By the Fundamental Theorem of Calculus and  $f$  is continuous, both  $\int_0^x y f(y) \, dy$  and  $x \cdot \int_x^\pi f(y) \, dy$  are differentiable, so

$$(Tf)'(x) = x f(x) + \int_x^\pi f(y) \, dy - x f(x) = \int_x^\pi f(y) \, dy$$

And, FTC again,  $(Tf)'$  is differentiable, and  $(Tf)''(x) = -f(x)$ .

b). To show  $\lambda > 0$ , observe that

$$(\lambda f)'' = (Tf)'' = -f(x), \text{ so } -f'' = \frac{1}{\lambda} f, \quad x \in (0, \pi).$$

Moreover, since the observation above,  $(Tf)(0) = 0$ , so  $\lambda f(0) = 0$ , so  $f(0) = 0$ .

And, since the formula in a),

$$f'(\pi) = \frac{1}{\lambda} (Tf)'(\pi) = \frac{1}{\lambda} \cdot \left( \int_\pi^\pi \cdot \right) = 0.$$

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9. Let  $f \in C([0, \pi])$  and define

$$(Tf)(x) := \int_0^\pi \min\{x, y\} \cdot f(y) \, dy, \quad 0 \leq x \leq \pi$$

a). Show that  $Tf$  is twice differentiable on  $(0, \pi)$  and satisfies

$$(Tf)'(x) = \int_x^\pi f(y) \, dy, \quad (Tf)''(x) = -f(x)$$

b). Suppose that  $f \neq 0$  and  $\lambda \neq 0$  satisfy  $Tf = \lambda f$ .

Show that  $\lambda > 0$  and that  $f$  satisfies

$$-f''(x) = \frac{1}{\lambda} \cdot f(x), \quad 0 < x < \pi$$

With the boundary conditions  $f(0) = 0$ ,  $f'(\pi) = 0$ .

c) Determine all  $\lambda > 0$  and all non-trivial functions  $f$  satisfying the differential equation and the boundary conditions in (b).

Proof of c).



Another Problems. 26/09/08

1. Let  $T \in \text{End}(V)$  where  $V$  is fin. dim v.s over  $F$  with  $\text{ch}(F) \neq 2$ .

a) Prove that  $V = \ker T \oplus \text{Im } T$ .

b) Prove that  $T|_{\text{Im } T}: \text{Im } T \rightarrow \text{Im } T; \alpha \mapsto T(\alpha)$  is an Automorphism.

c) In addition, if  $T^3 = T$ , prove  $T$  is diagonalizable and find all possible eigenvalues. Proof.

a). Given  $\alpha \in \ker T$ ,  $T^2(\alpha) = T(T(\alpha)) = T(0) = 0$ , so  $\ker T \subseteq \ker T^2$ .

Meanwhile, by the dimension theorem and by the assumption,

$$\dim V = \text{rank } T + \text{nullity } T = \text{rank } T^2 + \text{nullity } T = \text{rank } T^2 + \text{nullity } T^2$$

So,  $\dim \ker T = \dim \ker T^2$ , and  $\dim \ker T \leq \dim V < \infty$  gives  $\ker T = \ker T^2$ . Let  $\alpha \in \ker T \cap \text{Im } T$ . Then,  $\alpha = T(\beta)$  for some  $\beta \in V$  and  $T(\alpha) = T^2(\beta) = 0$ . So,  $\beta \in \ker T^2 = \ker T$ , it gives  $\alpha = T(\beta) = 0$ . And, let  $\{\alpha_1, \dots, \alpha_k\}$  and  $\{\alpha_{k+1}, \dots, \alpha_n\}$  be the basis for  $\ker T$  and  $\text{Im } T$ . Then,  $\{\alpha_1, \dots, \alpha_n\}$  is linearly independent, because  $\ker T \cap \text{Im } T = \{0\}$ , which proves  $V = \ker T \oplus \text{Im } T$ .

b) Since  $\text{Im } T \supseteq \text{Im } T^2$  and  $\text{rank } T = \text{rank } T^2$ ,  $\text{Im } T = \text{Im } T^2$  and so

$$T[\text{Im } T] \subseteq \text{Im } T$$

And, since  $V = \text{Im } T \oplus \ker T$ ,  $\ker T|_{\text{Im } T} = \{0\}$ , so the restriction is injective, and also surjective by  $\dim V < \infty$ . So it is automorphism.

c).  $T^3 = T$  means  $\chi(x^2 - 1)$  is a multiple of the minimal polynomial, but  $\chi(x-1)(x+1)$  is separable and splits over  $F$ , so the minimal polynomial also splits and is separable, thus  $T$  is diagonalizable. And the possible eigenvalues are  $0, \pm 1$ .

Note: Given  $T: V \rightarrow V$ , (not necessary finite-dimensional)

$V = \ker T \oplus \text{Im } T$  if and only if  $T|_{\text{Im } T}$  is an Automorphism.

In such the case, given  $\alpha \in V$ ,

$$\alpha = \underbrace{[\alpha - (T|_{\text{Im } T})^{-1}(T(\alpha))]}_{\ker T} + \underbrace{(T|_{\text{Im } T})^{-1}(T(\alpha))}_{\text{Im } T}$$

2. Let  $V, W$  be fin. dim real inn. prod space, and let  $T: V \rightarrow W$  be injection.  
 a). Prove that  $T^* \circ T: V \rightarrow V$  is invertible.

b). Prove that  $T \circ (T^* \circ T)^{-1} \circ T^*: W \rightarrow W$  is an orthogonal proj of  $W$  onto  $\text{Im } T$ .  
 Proof.

a).  $T$  is 1-1  $\Leftrightarrow \ker T = \{0\}$ , since  $\dim V < \infty$ .

Meanwhile, observe that given  $\alpha \in V$ ,

$$T^* T(\alpha) = 0 \Rightarrow \langle T^* T(\alpha), \alpha \rangle = 0$$

$$\Rightarrow \langle T(\alpha), T(\alpha) \rangle = 0$$

$$\Rightarrow T(\alpha) = 0$$

So  $\ker T^* T \subset \ker T = \{0\}$ , thus  $\ker T^* T = \{0\}$ .

Since  $\dim V < \infty$ ,  $T^* T$  is 1-1 and onto.

b) observe that

$$P^2 = T(T^* T)^{-1} T^* T(T^* T)^{-1} T^* = T(T^* T)^{-1} T^* = P$$

$$P^* = P, \text{ since } (ABC)^* = (C^* B^* A^* \text{ and } (T^{-1})^* = (T^*)^{-1}$$

So,  $P^2 = P^* = P$ ,  $P$  is orthogonal projection.

And, since  $T$  is 1-1, so is  $T^*$ , thus  $T^*$  is onto, so

$$\text{Im } P = T(T^* T)^{-1} T^*[W] = T[W] = \text{Im } T,$$

So proved

1. Com.  
11.2

3. Let  $G$  be a group of order 56.  
Prove that  $G$  is non-simple.

ans: 50

Proof. Notice that  $56 = 2^3 \cdot 7$ , and denote  $n_p$  is number of Sylow  $p$ -subgroups.  
By the Sylow theorem,  $n_7 \in \{1, 8\}$  and  $n_2 \in \{1, 7\}$ .

Either  $n_7 = 1$  or  $n_2 = 1$ , then the Sylow subgroup is normal, so we are done. Suppose that  $n_7 = 8$  and  $n_2 = 7$ .

By the Lagrange's theorem, all non-identity elements in Sylow 7-subgroup has order of 7. Since  $n_7 = 8$ , there are  $8 \cdot 6 = 48$  distinct elements of order 7 in  $G$ . Meanwhile, since  $n_2 = 7$ , let  $P, Q \in \text{Syl}_2(G)$ . Then  $|P \cap Q| \leq 4$ , so there are  $7 + 4 = 11$  distinct elements whose order is multiple of 2, but  $48 + 11 = 59 > 56$ , it contradicts.  $\square$

1 cm

4. Let  $R = \mathbb{Z}[x]/(6, x^2+x+1)$ .

a) Decompose  $R$  as a product over  $\mathbb{F}_2$  and  $\mathbb{F}_3$ .

b) Identify the two factors up to isomorphism as explicitly as possible.

c) Determine the number of units, nilpotent, and idempotent.

Proof: a), Notice that  $x^2+x+1$  is irreducible, because  $0, \pm 1$  are only possibilities for root, but  $0^2+0+1 \neq 0$ ,  $1^2+1+1=3 \neq 0$ ,  $1+(-1)+1=1 \neq 0$ . So, the degree 2 polynomial  $x^2+x+1$  has no root in  $\mathbb{Z}$ , so irreducible. Denote  $f = x^2+x+1$ . Meanwhile, observe that

$$(6, f) = (2, f) \cap (3, f) \quad \text{and} \quad (2, f) + (3, f) = (1) = R, \quad \text{since } 3-2=1.$$

Now, by the Chinese remainder theorem,

$$\mathbb{Z}[x]/(6, f) \cong \mathbb{Z}[x]/(2, f) \times \mathbb{Z}[x]/(3, f)$$

Since 2 is prime and  $f$  is irreducible, so  $\mathbb{Z}[x]/(2, f) \cong \mathbb{F}_{2^2}$ .

But,  $f$  is not irreducible over  $\mathbb{Z}/3\mathbb{Z}$ , as  $f = (x+2)^2$ , so

$$\mathbb{Z}[x]/(3, f) \cong (\mathbb{Z}/3\mathbb{Z})[x]/(\overline{x+2})^2.$$

$$\text{Consequently, } \mathbb{Z}[x]/(6, f) \cong \mathbb{F}_{2^2} \times (\mathbb{F}_3[x]/(\overline{x+2})^2)$$

Since  $\mathbb{F}_{2^2}^\times$  is a cyclic group, thus only nilpotent element in  $\mathbb{F}_{2^2}$  is 0.

And, since  $(\overline{x+2})^2 = 0$  in  $\mathbb{F}_3[x]/(\overline{x+2})^2$ , so

$$(\overline{x+2}) \cdot a \quad (a=0, 1, 2) \text{ are nilpotent elements in } \mathbb{F}_3[x]/(\overline{x+2})^2.$$

Now, all nilpotent in  $R$  is  $(0, a\overline{x+2})$ ,  $a \in \mathbb{F}_3$ .

35. Let  $I, J, K$  be ideals of  $R$ . (with 1)

(a) Prove that  $I(J+K) = IJ + IK$  and  $(I+J)K = IK + JK$ .

(b) Prove that if  $J \subseteq I$  then  $I \cap (J+K) = J + (I \cap K)$ .

$R$  is commutative, with 1

If  $I+J=R$ , then  $IJ = I \cap J$ .

$$(3, f) + (2, f) = \mathbb{Z}[x], \quad \text{since } 3-2=1.$$

$$\begin{aligned} (3, f) \cap (2, f) &= (3, f)(2, f) = [(3) + (f)] \cdot [(2) + (f)] \\ &= (3)(2) + (3)(f) + (2)(f) + (f)(f) \\ &= (6) + (3f, 2f, f^2) = (3, 2, f)(f) \\ &= (6) + (f) \\ &= (6, f). \end{aligned}$$

$$(6, f) = (6) + (f) = (3)(2) + (f)$$

5. Let  $\alpha = \sqrt[4]{3}$  and let  $K$  be the splitting field of  $x^4 - 3$  over  $\mathbb{Q}$ .

a). Prove  $K = \mathbb{Q}(\alpha, i)$  and show  $[K:\mathbb{Q}] = 8$ .

b). Define  $r, s \in \text{Gal}(K/\mathbb{Q})$  as

$r(\alpha) = i\alpha, r(i) = i, s(\alpha) = \alpha, s(i) = -i$ . Prove  $\text{Gal}(K/\mathbb{Q}) \cong D_8$ .

c). Determine  $\langle r \rangle$  and  $\langle s \rangle$

Proof. a).

6. Let  $(f_n)$  be a sequence in  $C^1([0,1])$  such that  
 $\sup_n \|f_n'\|_\infty \leq 1$  and  $\int_0^1 |f_n(x)| dx \rightarrow 0$ . prove  $f_n \rightarrow 0$  uniformly.

Proof.

7.

a)

b)

c)

$a,$

Proof,

l).

a)

b)

c).

2. **Linear Algebra - commuting projections.** Let  $P, Q : V \rightarrow V$  be linear maps on a finite-dimensional vector space over a field  $F$ . Assume

$$P^2 = P, \quad Q^2 = Q, \quad PQ = QP.$$

Prove the direct-sum decomposition

$$V = (\operatorname{im} P \cap \operatorname{im} Q) \oplus (\operatorname{im} P \cap \ker Q) \oplus (\ker P \cap \operatorname{im} Q) \oplus (\ker P \cap \ker Q),$$

and deduce that  $P$  and  $Q$  are simultaneously diagonalizable.

3. **Finite Abelian Groups.** Let  $G$  be a finite abelian group. Suppose that for every prime  $p$  dividing  $|G|$ , the equation  $x^p = e$  has exactly  $p$  solutions in  $G$ . Prove that  $G$  is cyclic.

6. **Real Analysis - Dini type argument.** Let  $X$  be a compact metric space and  $f_n, f \in C(X)$  satisfy

$$f_1(x) \geq f_2(x) \geq \cdots \geq f(x) \quad \text{for every } x \in X,$$

and  $f_n(x) \rightarrow f(x)$  pointwise. Prove that  $f_n \rightarrow f$  uniformly.

7. **Bounded Variation.** Define on  $[-1, 1]$

$$f(0) = g(0) = 0, \quad f(x) = x^2 \sin \frac{1}{x^2}, \quad g(x) = x^3 \sin \frac{1}{x^2} \quad (x \neq 0).$$

Determine which of  $f, g$  are of bounded variation on  $[-1, 1]$ . Justify both answers.

